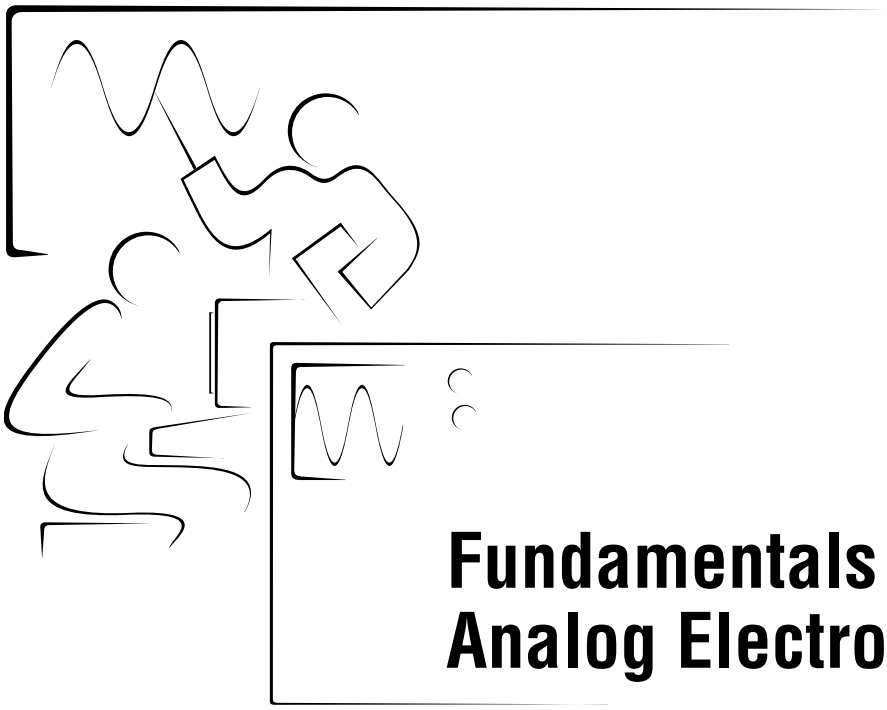


Fundamentals of Analog Electronics

by Professor Barry Paton
Dalhousie University

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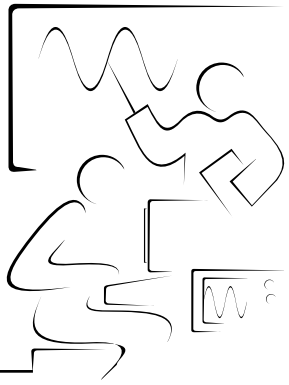
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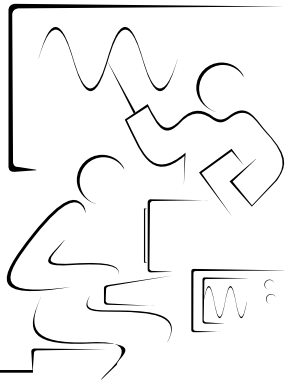
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Introduction

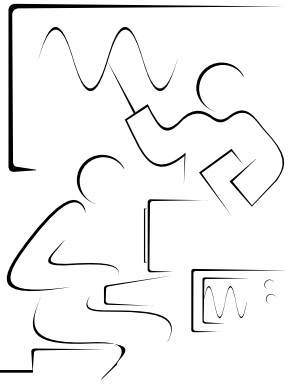
Analog Electronics is one of the fundamental courses found in all Electrical Engineering and most science programs. The great variety of LabVIEW Boolean and numeric controls/indicators, together with the wealth of programming structures and functions make LabVIEW an excellent tool to visualize and demonstrate many of the fundamental concepts of analog electronics. The inherent modularity of LabVIEW is exploited in the same way that complex analog integrated circuits are built from circuits of less complexity which in turn are built from fundamental amplifiers. This project is designed as a teaching resource to be used in the classroom, in tutorial sessions or in the laboratory.

Operational Amplifiers are the heart and soul of all modern electronic instruments. Their flexibility, stability and ability to execute many functions make op-amps the ideal choice for analog circuits. Historically, op-amps evolved from the field of analog computation where circuits were designed to add, subtract, multiply, integrate, differentiate etc. in order to solve differential equations found in many engineering applications. Today analog computers op-amps are found in countless electronic circuits and instruments. This project focuses on op-amps as the soul and heart of all analog electronic instruments.

The labs cover op-amp basics including AC and DC characteristics, filters, monostables, astable and log amp circuits. Electronic labs (eLabs) using real components are found at the end of each lab. They are designed to demonstrate an electronic principle but can be used as a template for more complex real op-amp circuits. The 741 and 555 chips are studied and used to build more complex circuits such as a voltage-to-frequency converter. Sensors including photodiodes and thermistors are used with op-amps to build a photometer and a temperature transducer. All eLabs are described in detail and simulated in the text. Computer Automation labs also found at the end of the lab, employ a DAQ card to show how LabVIEW can be used for automated testing and analysis of the eLab circuits.

For engineers, students and instructors, this project provides a dynamic settings to display analog characteristics in the classroom or your home computer. In tutorial sessions, the analog VIs can provide a template to build better simulations and demonstrations. In the lab, the eLabs can provide a template to build real analog circuits, to better understand analog principles and to design more complex circuits. LabVIEW is used throughout the course for calculations, simulations and data collection. Readers wishing to learn LabVIEW should look behind the front panel onto the diagram page where many unique LabVIEW constructs are used to generate the analog simulations and measurements. Enjoy!

Lab 1
Operational Amplifiers: The Basics



Operational Amplifiers or op-amps are the heart and soul of all modern electronic instruments. Their flexibility, stability and ability to execute many functions make op-amps the ideal choice for analog circuits. Historically, op-amps evolved from the field of analog computation where circuits were designed to add, subtract, multiply, integrate, differentiate etc. in order to solve differential equations found in many engineering applications. Today analog computers have been mostly replaced by digital computers; however the high functionality of op-amp circuits remains its legacy and op-amps are found in countless electronic circuits and instruments.

The op-amp is basically a very high gain differential amplifier with bipolar output. The op-amp transfer curve states that the output voltage, V_{out} is given by

$$V_{out} = - A (V_{-} - V_{+}) = -A (\Delta V) \qquad (1\text{-}1)$$

where A is the open loop gain, V_{-} is the inverting input voltage and V_{+} is the non-inverting input voltage. The negative sign in front of the gain term A inverts the output. The gain A can be defined as the ratio of the magnitude of the output voltage V_{out} to the input difference voltage ΔV. In practical op-amps, the gain can be from 10,000 to 20,000,000. Only a very small input signal is required to generate a large output. For example, if the op-amp gain is one million, a 5 microvolt input would drive the op-amp output to 5 volts.

Most op-amps are bipolar. This means that the output can be a positive or negative signal. As a result, two power supply voltages are required to power the op-amp. In this text, we will assume that the supply voltages for all op-amp circuits are +15 and –15 volts. The output voltage can never exceed the power supply voltage. In fact the rated op-amp output voltage V_{max} is often a volt or so smaller than the power supply voltage. This limit is often referred to as the + or – rail voltage.

LabVIEW Demo 1.1: Op-Amp Gain

Launch the LabVIEW program entitled OpAmp1.vi from the chapter 1 library. Click on the **Run** button to power up your op-amp.

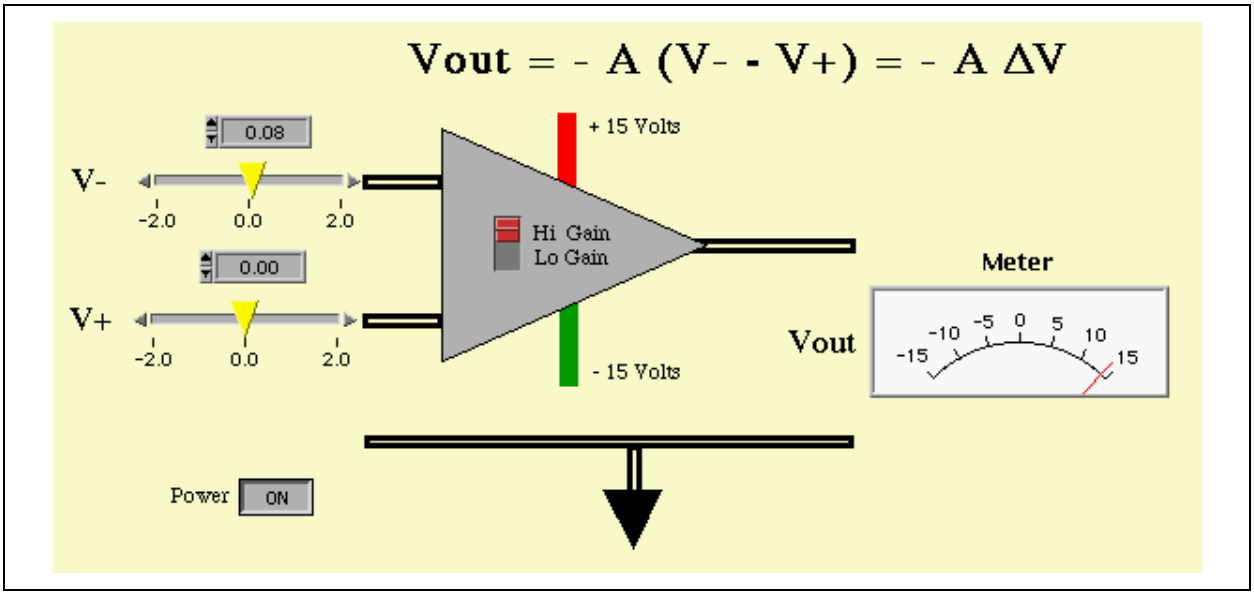


Figure 1-1. Open Loop Op-Amp Circuit

Investigate the sensitivity and sign of the output voltage as the input signal levels V_- and V_+ are varied. There are two choices for the op-amp gain. The *Lo Gain* position sets A = 10 and allows the viewer to see how the amplifier functions. The *Hi Gain* position sets A=100,000 and is more representative of a real op-amp. Note that the rail voltages are about 1 volt less than the power supply. When the output is at the rail voltage, the op-amp is said to be saturated. For *Hi Gain*, it seems that the op-amp is almost always saturated in this open loop configuration.

A better view of the transfer curve is to plot the output voltage as a function of the input differential voltage, ΔV.

LabVIEW Demo 1.2: Op-Amp Transfer Curve

Launch the LabVIEW program called OpAmp2.vi from the chapter 1 library. This program is similar to the previous program, except that the ground and power supply lines have been removed. These lines must always be connected in a real circuit but often are not shown in schematic diagrams. A X-Y graph has been added to dynamically display the transfer curve. Run the program as in the previous demo.

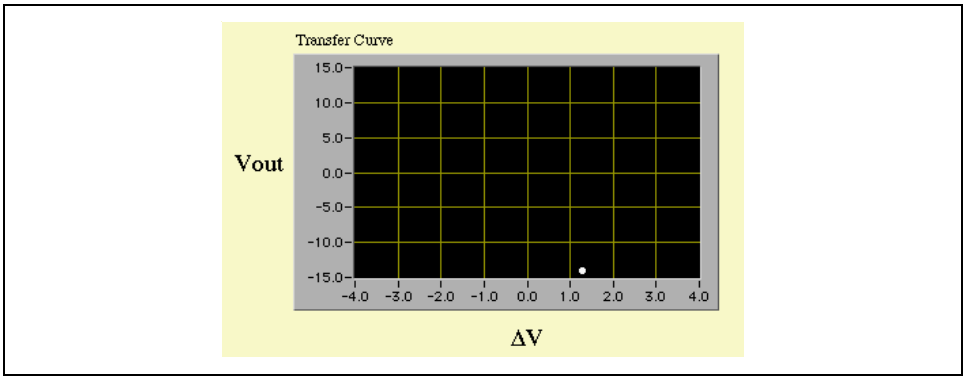


Figure 1-2. Transfer Curve Display for Open Loop Op-Amp

Again the **Lo Gain** button is used to observe the amplifier operation. Use the **Hi Gain** setting to simulate a real Op-Amp. By selecting various input voltage levels, the complete transfer curve can be traced out. Two colored LED displays straddle the meter to indicate when the amplifier saturates either at the + or – rail.

Closed Loop Op Amp Circuits

High gain amplifiers are difficult to control and keep from saturation. With some external components part of the output can be fed back into the input. For negative feedback, that is the feedback signal is out of phase with the input signal, the amplifier becomes stable. This is called the closed loop configuration. In practice, feedback trades off gain for stability, as much of the open loop gain A is used to stabilize the circuit. Typical op-amp circuits will have a closed loop gain from 10 to 1000 while the open loop gain ranges from 10^5 to 10^7. If the feedback is positive, the amplifier becomes an oscillator.

Inverting Amplifier

The following circuit (probably the most common op-amp circuit) demonstrates how a reduction in gain produces a very stable linear amplifier. A single feedback resistor labeled R_f is used to feed part of the output signal back into the input. The fact that it is connected to the negative input indicates that the feedback is negative. The input voltage V_1 produces an input current i_1 through the input resistor R_1. Note the differential voltage ΔV across the amplifier inputs (–) and (+). The plus amplifier input is tied to ground.

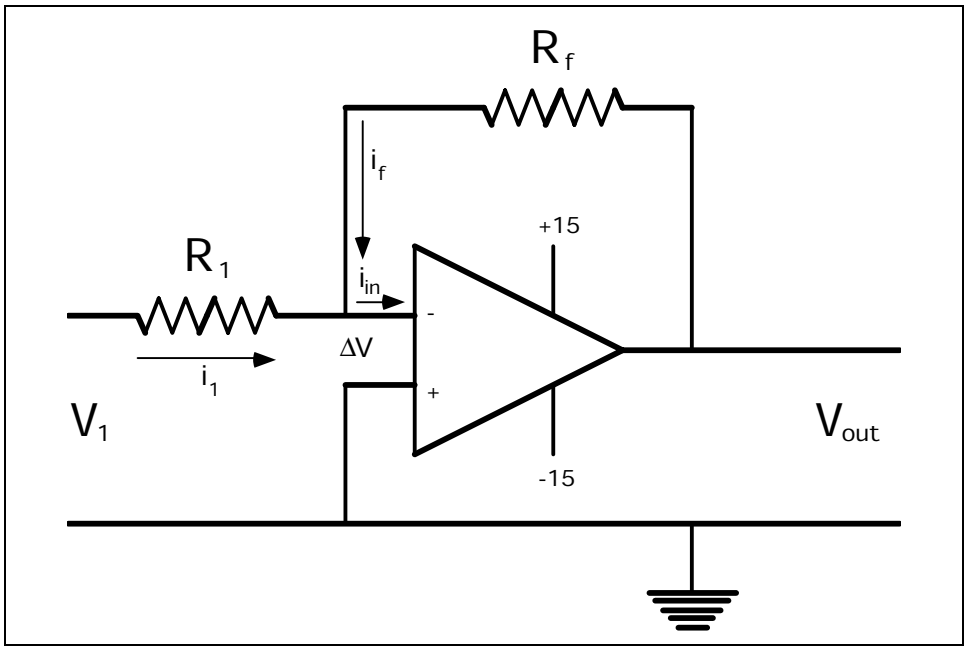


Figure 1-3. Schematic Diagram for an Inverting Op-Amp Circuit

Kirchoff's laws and the loop equations are used to develop the transfer characteristic.

$$\text{Input loop} \qquad V_1 = i_1 R_1 + \Delta V \tag{1-2}$$

$$\text{Feedback Loop} \qquad V_{out} = - i_f R_f + \Delta V \tag{1-3}$$

$$\text{Summing Point} \qquad i_1 = - i_f + i_{in} \tag{1-4}$$

$$\text{Gain Equation} \qquad V_{out} = - A \Delta V \tag{1-5}$$

Solving these four equations yields

$$V_{out} = i_{in}/Z - (V_1/ R_1)/Z \tag{1-6}$$

where the close loop impedance $Z = 1/R_f + 1/AR_1 + 1/AR_f$.

The feedback and input resistor are usually large (kΩ's) and A is very large (>100,000), hence $Z = 1/R_f$. Furthermore ΔV is always very small (a few microvolts) and if the input impedance, Z_{in} of the amplifier is large (usually about 10 MΩ) then the input current $i_{in} = \Delta V/ Z_{in}$ is exceedingly small and can be assumed to be zero. The transfer curve Equation 1-5 then becomes

$$V_{out} = - (R_f / R_1) V_1 = - (G) V_1 \tag{1-7}$$

The ratio (R_f / R_1) is called the closed loop gain G and the minus sign tells us that the output is inverted. Note that the closed loop gain can be set by the selection of two resisters R_1 and R_f.

LabVIEW Demo 1.3: Inverting Op-Amp

Launch the LabVIEW program called OpAmp3.vi from the chapter 1 program library. This program simulates in a very real way the operation of a simple op-amp configured as an inverting amplifier. Click on the **Run** button to observe the circuit operation. One can change the resistance by click-and-dragging on the slide above each resistor or by entering a new value in the digital display below each resistor. The input voltage can be changed by clicking on the thumb-wheel arrows or entering a new value into the input digital display. Vary the feedback resistor, the input resistor and the input voltage to verify that the output follows the transfer Equation 1-6.

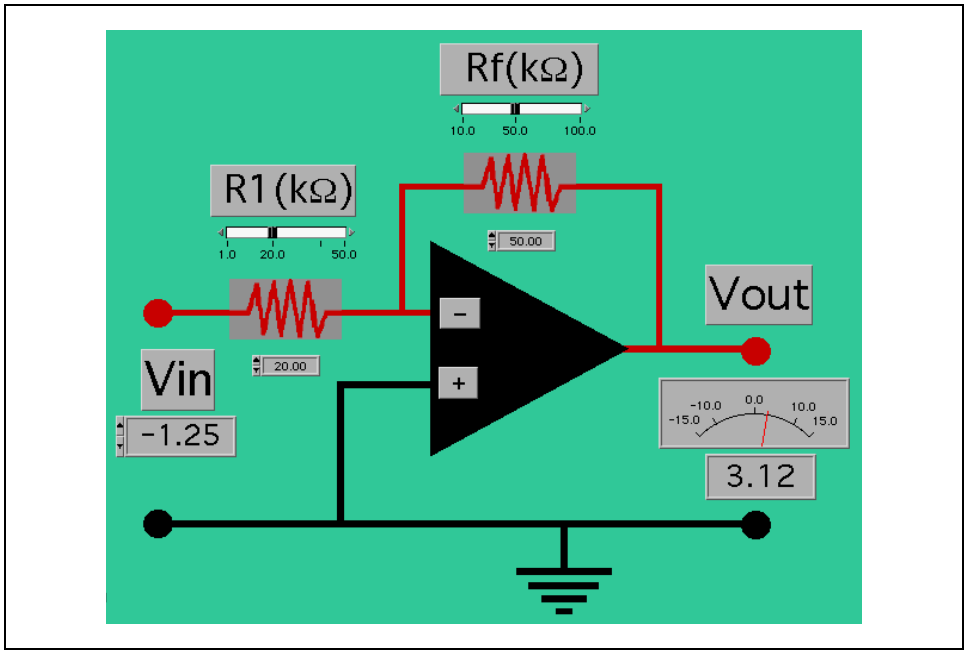


Figure 1-4. LabVIEW Simulation for an Inverting Op-Amp Circuit

Questions

What happens when the output voltage tries to exceed the power supply voltage of + or – 15 volts?

What happens when the input voltage reaches the power supply voltage?

What happens when the input voltage exceeds the power supply voltage by 1 or 2 volts?

Real Inverting Op-Amp Circuit

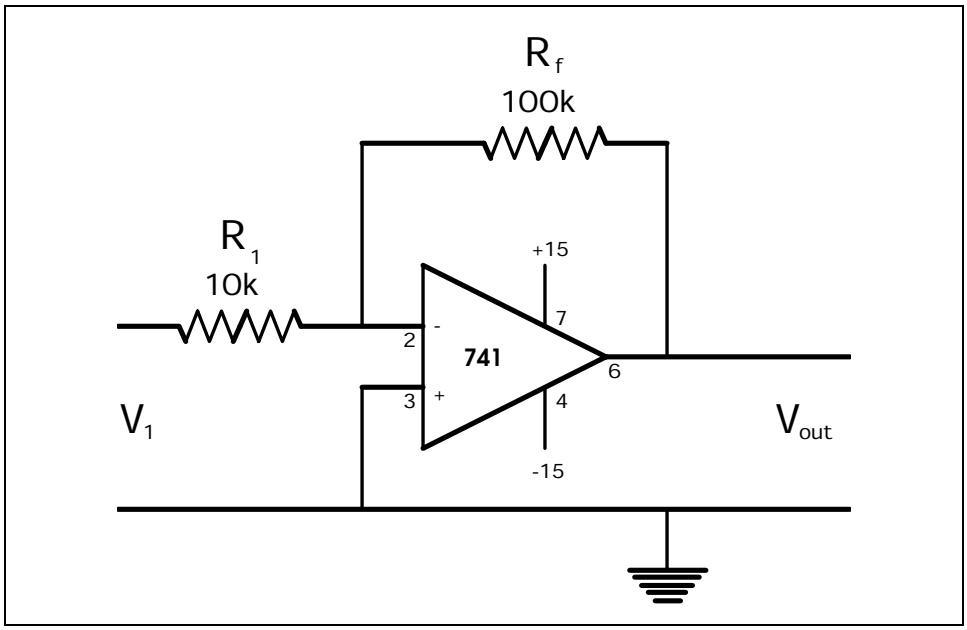


Figure 1-5. Schematic Diagram for Inverting Amplifier with Gain of 10

LabVIEW Challenge: LabVIEW Inverting Op-Amp Simulation (Version 2)

In the program OpAmp3.vi, replace the simple transfer curve Equation 1-7 with the more correct expression Equation 1-6. You will need a new control on the front panel so the open loop gain A can be varied from 10,000 to 1,000,000. Investigate for what values of R_1 and R_f is the simple transfer curve not a good approximation. Save your program as OpAmp3_2.vi

eLab Project 1

Objective

The objective of this electronic lab is to demonstrate the easy of building an amplifier with a precise gain and determine the amplifier accuracy.

Procedure

Build the inverting amplifier circuit of Figure 1-5 and shown pictorially below. The circuit requires a popular 741 op-amp, a few resistors and two power supplies. These can be found at a local electronics supply store. Set the input voltage to be in the range –1 to +1 volts.

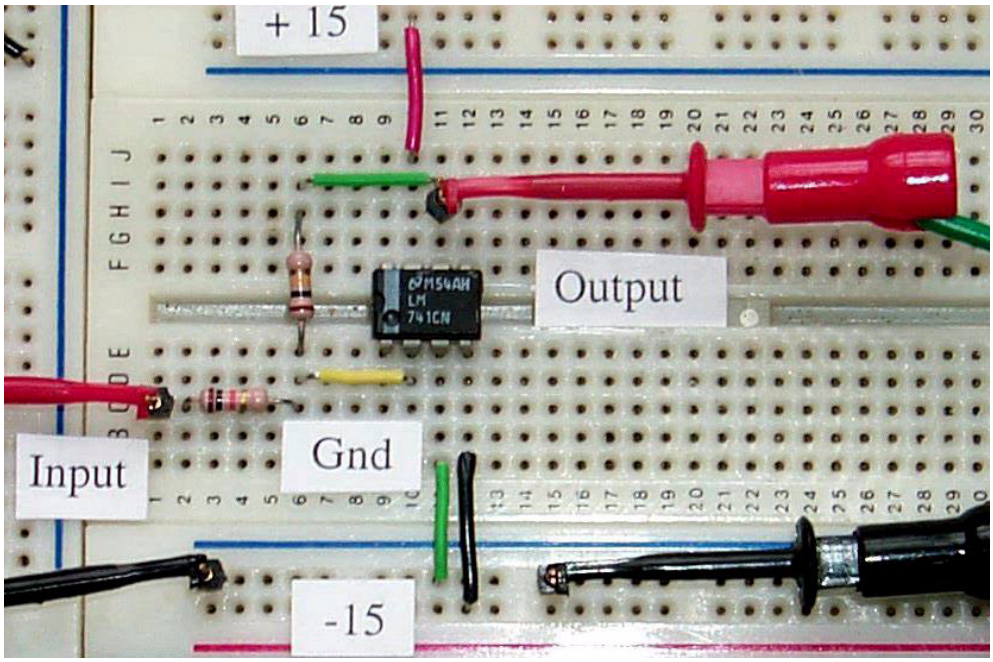


Figure 1-6. Component Layout for an Inverting Op-Amp Circuit

Before powering up the circuit, measure the feedback resistor, the input resistor and the input voltage (not connected to the circuit). Calculate the expected output from the transfer Equation 1-7. Estimate the error for each measurement and calculate the expected error. Now connect all the components, power up the circuit and measure the output voltage.

Fill in the chart

R_f (kΩ)	R_1(kΩ)	Gain(R_f/R_1)	V_{in}	V_{out} (Calculated)	V_{out} (Measured)

How does the measured output voltage compare with V_{out} calculated. You should be impressed!

Computer Automation 1: The Basics

In measuring the characteristic properties of a device, it is often necessary to measure the output signal over a range of input conditions. For example, the inverting amplifier has a unique transfer curve as long as the output stays within the rail voltage limits. This restriction puts a limit on the range of input signal levels that a device functions as a linear amplifier. Computer automation allows a range of test voltages to be output and the response

measured, displayed and analyzed. In this lab, we look at computer generation of a test signal and measurement of the amplifier response.

Launch the LabVIEW program entitled TestAmp1.vi from the chapter 1 library.

This program uses the DAQ card to generate a DC test signal between –0.5 and +0.5 volts and present it as an output on one of the DAQ card lines. The program then measures the response on an input line of the DAQ card and displays it on a front panel indicator.



Note The DAQ card Analog Output and Analog Input functions need to be configured for bipolar operation (–5 to +5 V range). Run the op-amp from a (±) 5 volt power supply.

After wiring the DAQ lines to you test circuit, click on the **Run** button to power up the test circuit. Enter a variety of input signal levels and plot the transfer curve (Measured Signal versus Input Signal). The graph will be similar to that derived from the LabVIEW Simulation for an Inverting Op-Amp Circuit (OpAmp3.vi) only now you are looking at a real device.

Questions for Consideration

What is the measured value of the + rail voltage?

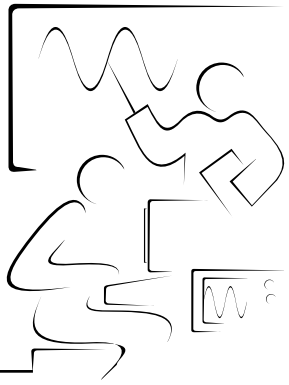
What is the measured value of the – rail voltage?

What is the output voltage when the input signal is zero? This is called the offset voltage.

Over what range of input signals is the amplifier linear?

What is the Gain of inverting amplifier circuit?

Lab 2
Operational Amplifier Circuits



Lab 1 demonstrated that the simple transfer curve Equation 1-7 was an excellent representation of a real op-amp circuit. The primary assumption was that the input differential voltage ΔV was so small it could be ignored. This assumption can be stated in several different ways. In most circuits ΔV can be replaced by a virtual short between the (–) and (+) input so that the voltage at the (–) input is essentially the same as at the (+) input. Another way is that the current flowing into the op-amp i_{in} is so small it can be neglected. Yet a third way states that the input impedance of the op-amp Z_{in} is exceedingly large. An ideal op-amp embodies all these properties and most op-amp circuit equations for gain, input and output impedance can be derived using this op-amp model.

An ideal op-amp has the following properties:

- The open loop gain is infinite and $\Delta V = 0$.
- No current flows into or out of the input leads.
- There is no offset voltage or current.
- Input impedance of the op-amp Z_{in} is infinite.
- The output impedance Z_{out} is zero.

In most common operating regions, the ideal op-amp approximation is sufficient to derive useful mathematical expressions to model the operation of real op-amps. Let’s take a second look at the inverting op-amp circuit.

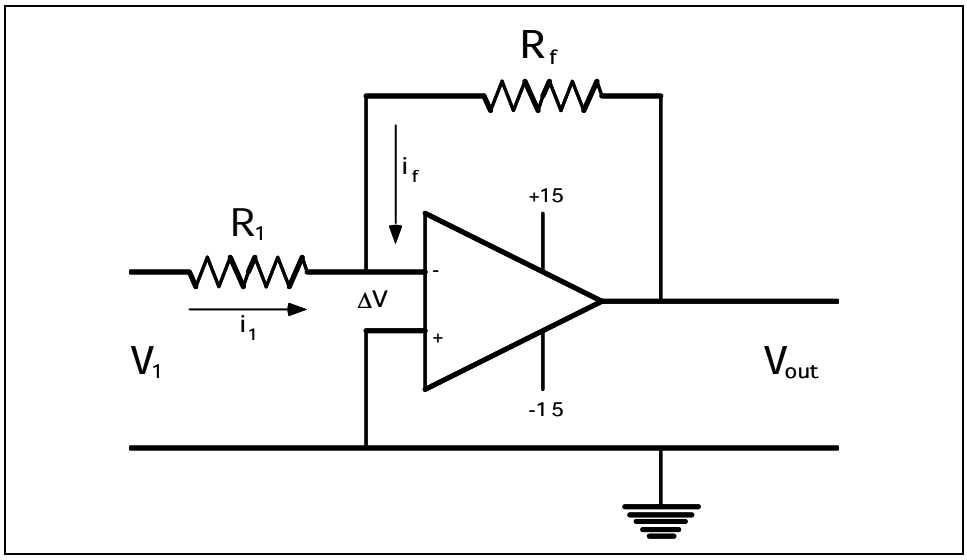


Figure 2-1. The Inverting Op-Amp Circuit

Inverting Op-Amp Revisited

The inverting op-amp circuit basically multiplies the input signal by a negative constant. The magnitude of the constant is just the closed loop gain (R_f / R_1) and the sign inverts the output signal polarity. The (–) input is in effect shorted to ground and the input current i_1 is calculated from Ohm's law for the input loop as (V_1/R_1). In this configuration the (–) input is often called a virtual ground as the (–) input is effectively at ground. Kirchoff's second law states that the sum of all the currents at any node must be zero, i.e $i_1 + i_f + i_{in} = 0$. Property 2 states that the current i_{in} into the op-amp is zero, hence $i_1 + i_f = 0$. For the output loop, $V_{out} = i_f R_f$.

These results lead directly to the transfer equation

$$Vout = - (R_f / R_1) Vin . \quad (2\text{-}1)$$

It is straight-forward to show that while the input impedance of the op-amp is infinite (property 4), the input impedance of the inverter circuit is in fact R_1.

LabVIEW Demo 2.1: The Inverting Op-Amp

Launch the LabVIEW program entitled Inverting.vi from the chapter 2 program library. Click on the **Run** button to power up the inverting circuit. Click and drag on the input slider to show the inverting feature of this circuit. Try other values for R_1 and R_f.

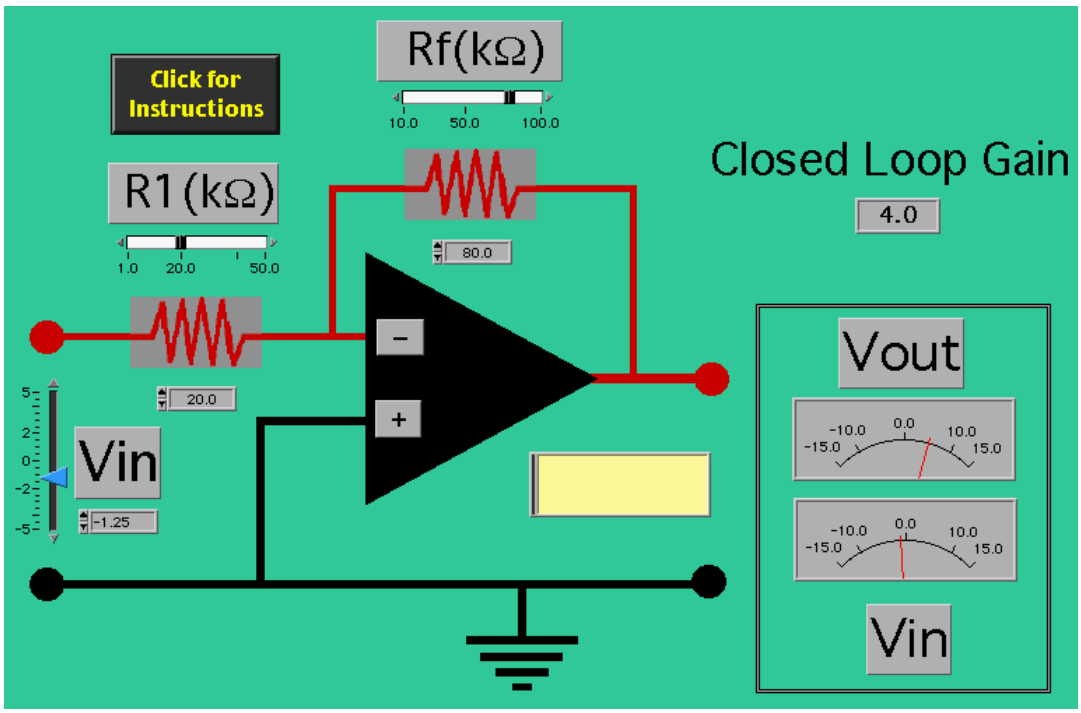


Figure 2-2. LabVIEW Simulation of an Inverting Op-Amp Circuit

When $R_f = R_1$ the closed loop gain equals one, G = 1. The op-amp circuit executes the mathematical function, negate. If V_{in} is positive, then V_{out} is negative or if V_{in} is negative, then V_{out} is positive.

Noninverting Op-Amp Circuit

A noninverting op-amp circuit can be configured from the previous circuit by tying the input resistor, R_1 to ground and placing the input signal on the (+) input.

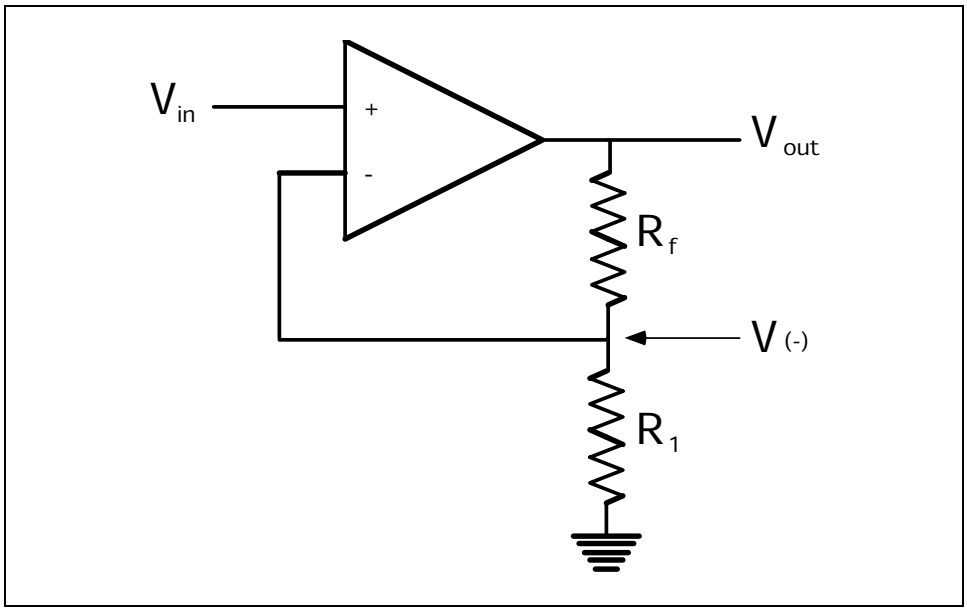


Figure 2-3. Schematic Diagram for a Noninverting Op-Amp Circuit

The output voltage is dropped across a voltage divider made up of the feedback resistor R_f and input resistors R_1. The voltage at the center tap V(–) is just

$$V(-) = [R_1/(R_1+ R_f)]V_{out} \quad (2\text{-}2)$$

According to the ideal op-amp properties (1), the input op-amp voltage ΔV is zero, hence $V_{in} = V(–)$. Rearranging the equation yields

$$V_{out} = (1+ R_f / R_1)\ V_{in} \quad (2\text{-}3)$$

This is a general purpose amplifier with a closed loop gain $G = (1+ R_f / R_1)$ that does not change the sign of the input signal. It can be shown that the input impedance for this circuit Z_i is very large and given by

$$Z_i \sim Z_{in}\ [R_1/(R_1+ R_f)]\ A \quad (2\text{-}4)$$

where Z_{in} is the input impedance of a real op-amp (about 20 MΩ). You can also show that the output impedance, Z_o of the circuit goes to zero as the open loop gain A becomes large. Thus the op-amp in the noninverting configuration effectively buffers the input circuitry from the output circuitry but with a finite gain.

LabVIEW Demo 2.2: The Noninverting Op-Amp

Launch the LabVIEW program entitled NonInverting.vi from the chapter 2 program library. Click on the **Run** button to power up the circuit. Click and drag on the input slider to show the noninverting feature of this circuit. Try other values for R_1 and R_f.

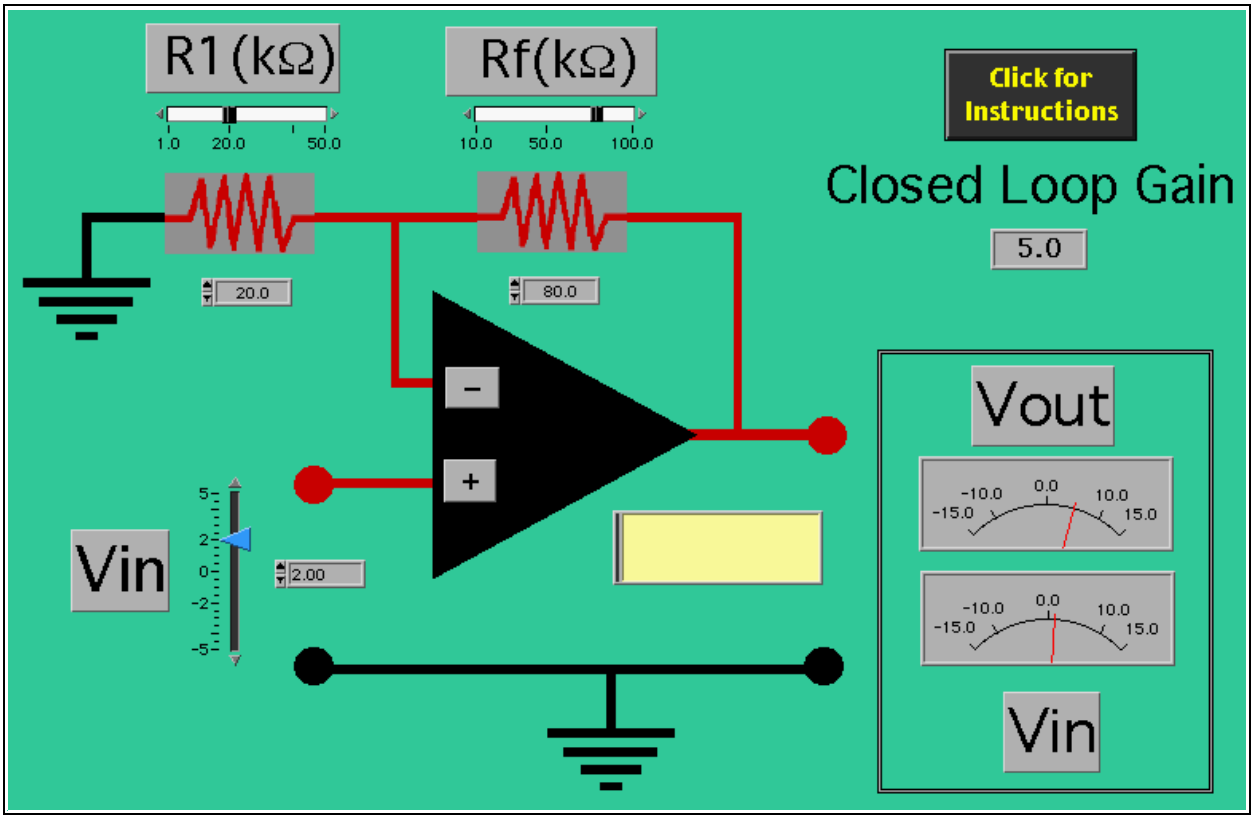


Figure 2-4. LabVIEW Simulation of an Noninverting Op-Amp Circuit

A special case of this circuit is when $R_f = 0$ and there is no input resistor R_1. In this case, $V_{out} = V_{in}$, $Z_i = Z_{in}A$ and $Z_o = Z_{out}/A$. This configuration is called a buffer or a unity gain circuit. It is somewhat like an impedance transformer which has no voltage gain but can have large power gains.

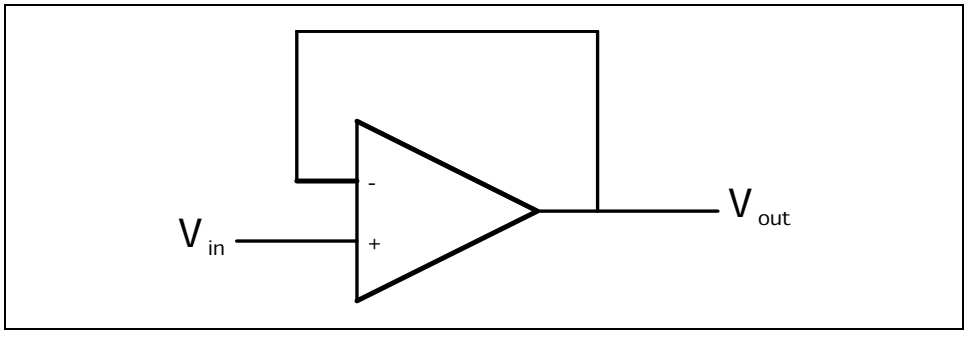


Figure 2-5. Unity Gain Op-Amp Circuit

Difference Amplifier

The difference op-amp circuit applies the same gain (R_f/R_1) to each of the differential inputs. The result is that the output voltage is the difference between the two input signals multiplied by a constant.

$$V_{out} = (R_f / R_1) (V_2 - V_1) \quad (2\text{-}5)$$

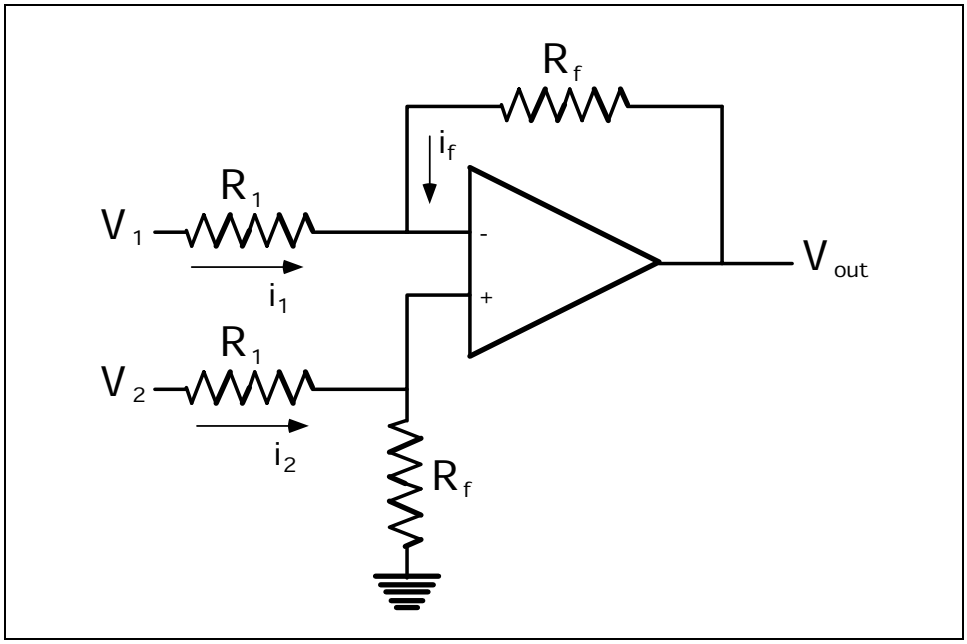


Figure 2-6. Schematic Diagram for a Op-Amp Difference Circuit

Using the ideal op-amp assumptions, one can write the voltage at the noninverting input (+) as

$$V(+) = [R_f /(R_1 + R_f)] V_2 \quad (2\text{-}6)$$

From the input loop 1 $$i_1 = [V_1 - V(+)] / R_1 \quad (2\text{-}7)$$

From the output loop $$i_f = - [V_{out} - V(+)] / R_f \quad (2\text{-}8)$$

and at the summing point $$i_1 = - i_f \quad (2\text{-}9)$$

Substituting for the currents, eliminating V(+) and rearranging yields the difference Equation 2-5.

LabVIEW Demo 2.3: Difference Op-Amp Circuit

Launch the LabVIEW program entitled Difference.vi from the chapter 2 program library. Click on the **Run** button to power up the difference circuit. Investigate the input-output relationship.

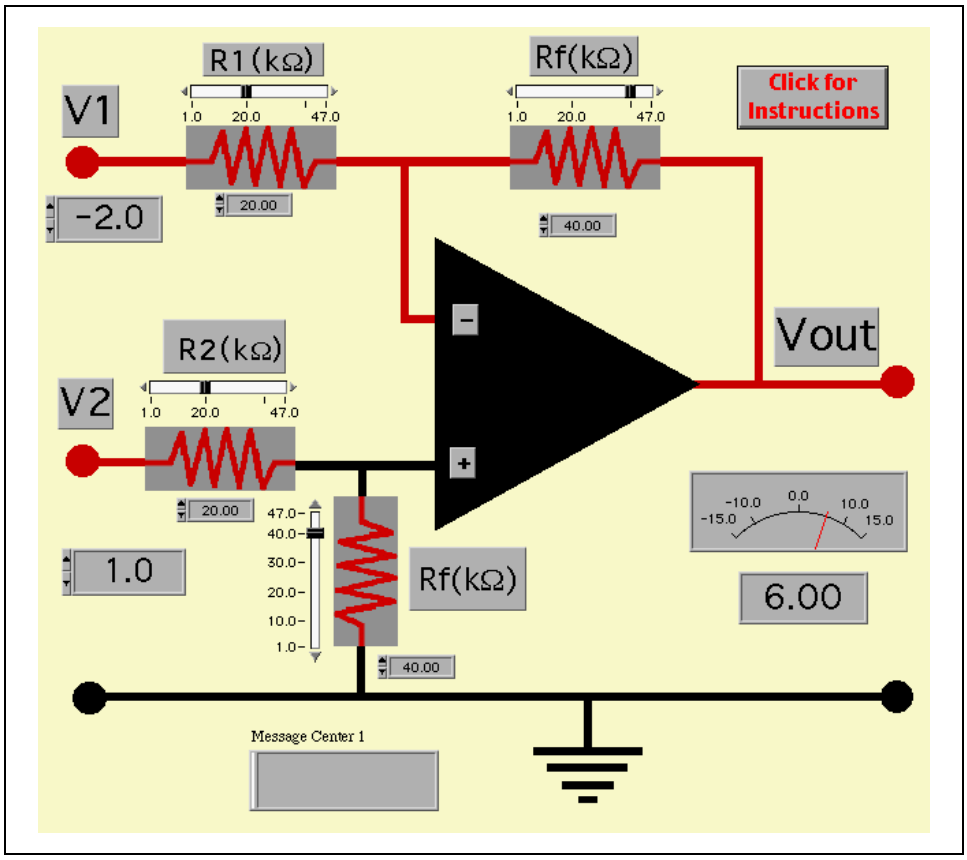


Figure 2-7. LabVIEW Simulation of a Difference Op-Amp Circuit

Note that the difference equation is only valid when the input resistors are equal and the feedback resistors are equal. For a real op-amp difference circuit to work well, great care is required to select matched pairs of resistors. When the feedback and input resistors are equal, the difference circuit executes the mathematical function, subtract.

Op-Amp Integrator Circuit

In the op-amp integrator circuit, the feedback resistor of the inverting circuit is replaced with a capacitor. A capacitor stores charge Q and an ideal capacitor having no leakage can be used to accumulate charge over time. The input current passing through the summing point is accumulated on the feedback capacitor C_f. The voltage across this capacitor is just equal to V_{out} and is given by the relationship $Q = CV$ as $Q = C_f V_{out}$. Recall that the current $i = dQ/dt$. Combining these two identities yields

$$i_f = C_f (dV_{out}/dt) . \quad (2\text{-}10)$$

From the ideal op-amp approximations, $i_1 = V_{in} / R_1$ and $i_1 = - i_f$

$$V_{in} /R_1 = - C_f (dV_{out} /dt) \quad (2\text{-}11)$$

or in the integral form

$$V_{out} = - (1/R_1C_f) \int V_{in}\, dt \qquad (2\text{-}12)$$

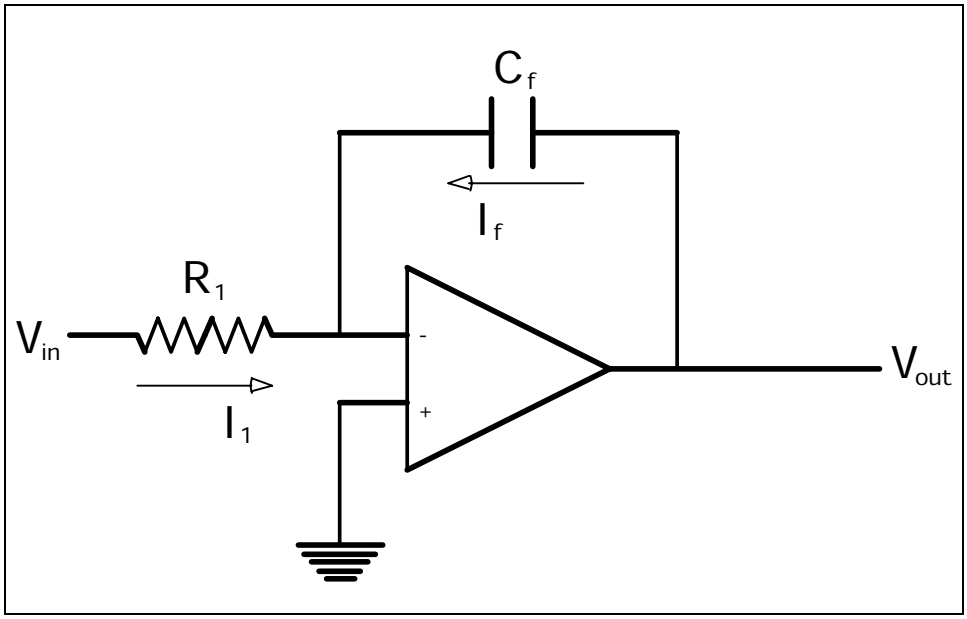


Figure 2-8. Schematic Diagram for an Op-Amp Integrator

The output voltage is the integral of the input voltage multiplied by a scaling constant ($1/R_1C_f$). The unit of R is ohms and C is farads. Together the units of (RC) are seconds. For example, a 1 µf capacitor with a 1MΩ resistor gives a scaling factor of 1/second.

Consider the case where the input voltage is a constant. The input voltage term can be removed from the integral and the integral equation becomes

$$V_{out} = - (V_{in} / R_1C_f)\, t + \text{constant} \qquad (2\text{-}13)$$

where the constant of integration is set by an initial condition such as $V_{out} = V_o$ at $t = 0$.

This equation is a linear ramp whose slope is $-(V_{in}/RC)$. For example, with $V_{in} = -1$ volt, C = 1 µf and R= 1 MΩ, the slope would be 1 volt/sec. The voltage output would ramp up linearly at this rate until the op-amp saturated at the + rail voltage. The constant of integration can be set by placing an initial voltage across the feedback capacitor. This is equivalent to defining the initial condition $V_{out}(0) = V_{constant.}$ At the start of integration or t = 0, the initial voltage is removed and the output ramps up or down from that point. The usual case is when the initial voltage is set to zero. Here a wire is shorted across the feedback capacitor and removed at the start of integration.

LabVIEW Demo 2.4: Integrator Circuit

Launch the LabVIEW program entitled Ramp.vi from the chapter 2 program library. A switch is used to short (set the initial condition) or open (let circuit integrate). Click on the **Run** button to power up the integrator circuit. Initially the output capacitor is shorted, hence the output is zero. Click on the thumb-wheel markers of the *Switch Control* to open and close the switch. Open the switch and watch the output voltage increase linearly. Investigate the output voltage as you change the slope parameters (V_{in}, R_1 and C_f). If the output saturates, restore the circuit to its initial state by shorting the capacitor.

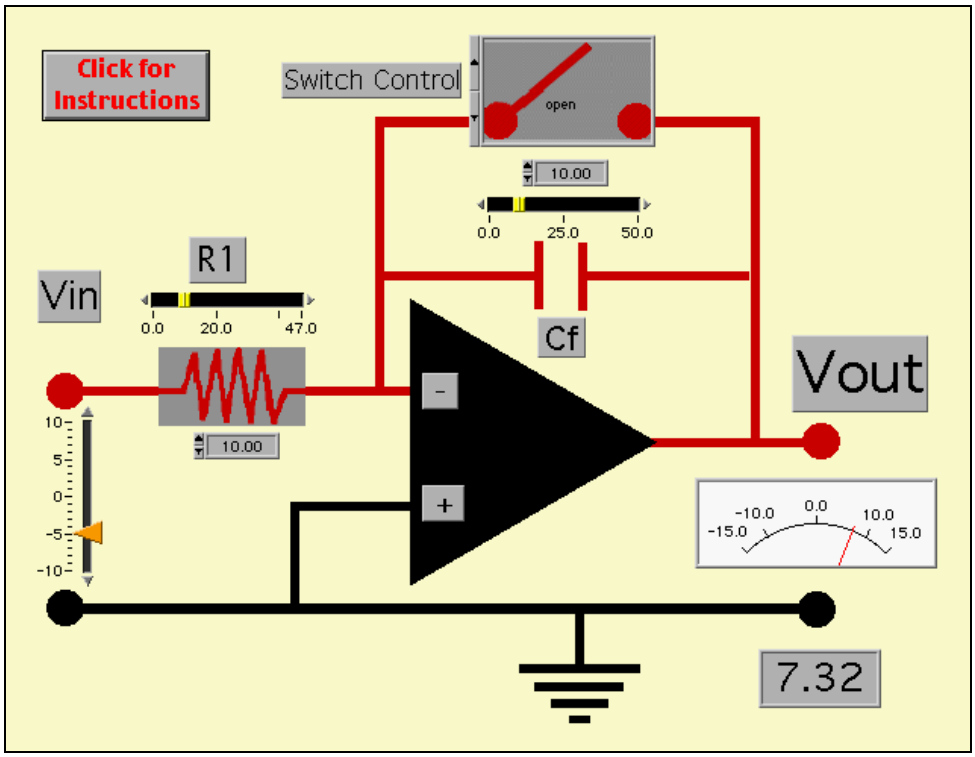


Figure 2-9. LabVIEW Simulation of an Op-Amp Integrator

For a constant input, this circuit is a ramp generator. If one was to momentarily short the capacitor every time the voltage reached say 10 volts, the resulting output would be a sawtooth waveform. In another program called Sawtooth.vi, a chart output has been added and a pushbutton placed across the capacitor to initialize the integrator. By clicking on the push button at regular intervals, a sawtooth waveform can be produced. Try it! Does this demonstration suggest a way to build a sawtooth waveform generator?

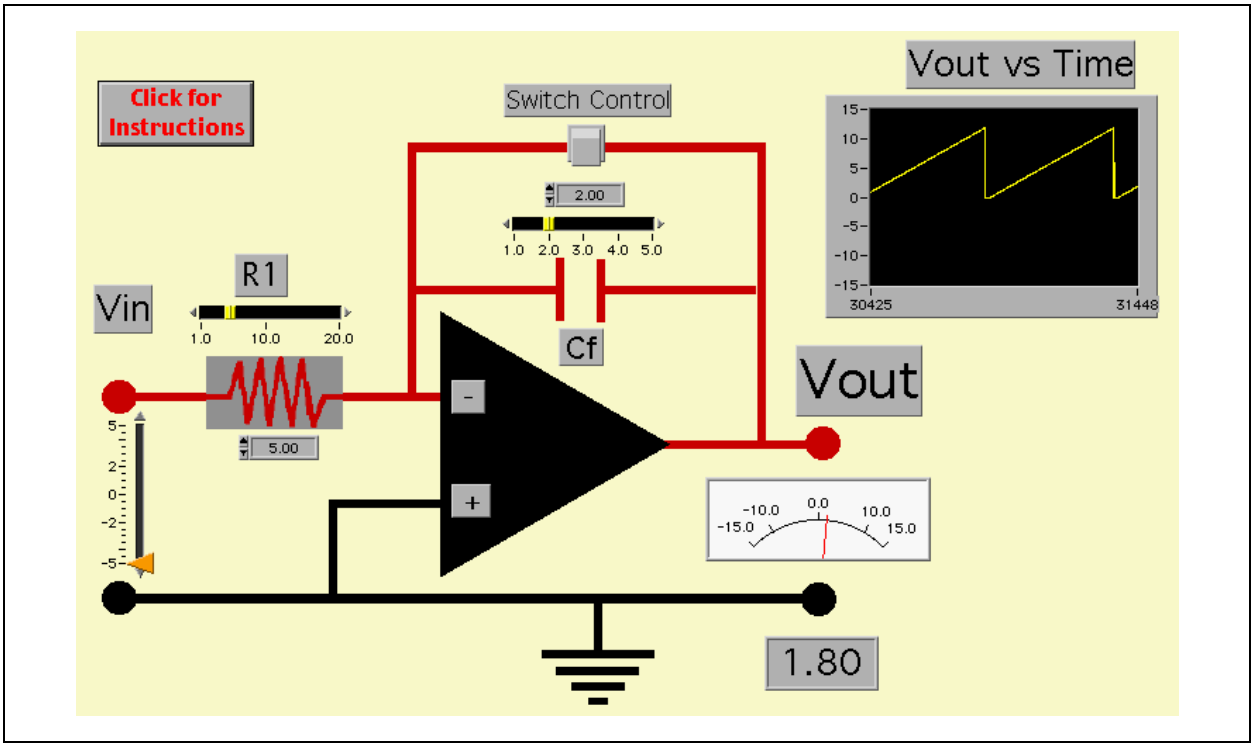


Figure 2-10. LabVIEW Op-Amp Integrator used to Generate a Sawtooth Waveform

LabVIEW Challenge

How would you modify the integrator simulation to generate a triangular waveform?

Op Amp Summing Circuit

The op-amp summing circuit is a variation of the inverting circuit but with two or more input signals. Each input V_i is connected to the (–) input pin through its own input resistor R_i. The op-amp summer circuit exploits Kirchoff's 2nd law which states that the sum of all currents at a circuit node is zero. At the point V(–), $i_1 + i_2 + i_f = 0$. Recall that the ideal op-amp has no input current (property 2) and no offset current (property 3). In this configuration, the (–) input is often called the summing point, V_s. Another way of expressing this point, is that at the summing point, all currents sum to zero.

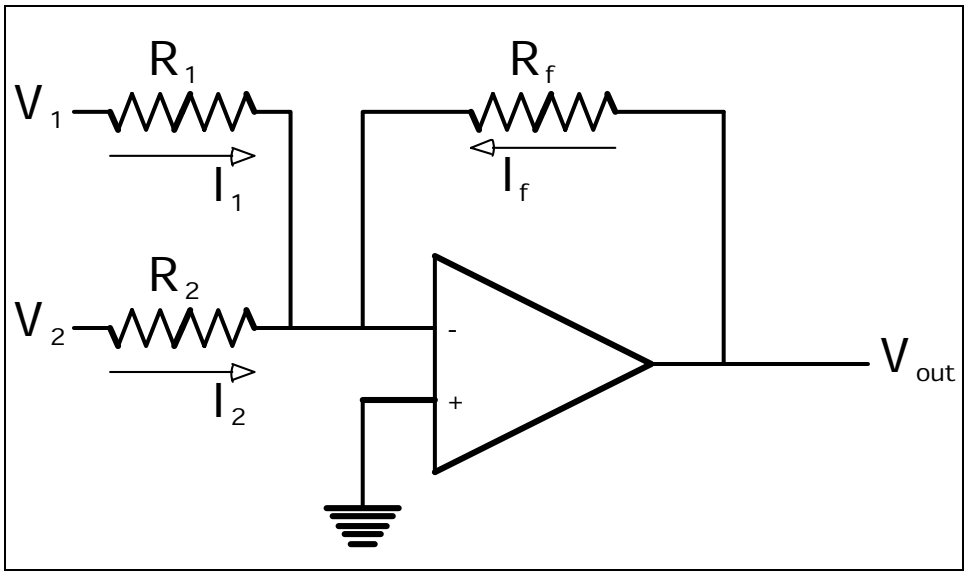


Figure 2-11. Schematic Diagram for an Op-Amp summing Circuit

For the input loop 1 $i_1 = V_1 / R_1$ (2-14)

For the input loop 2 $i_2 = V_2 / R_2$ (2-15)

For the feedback loop $i_f = - (V_{out} / R_f)$ (2-16)

Combining these equations at the summing point yields

$$V_{out} = - R_f (V_1 / R_1) - R_f (V_2 / R_2) \quad (2\text{-}17)$$

If $R_1 = R2 = R$, then the circuit emulates a true summer circuit.

$$V_{out} = - (R_f / R) (V_1 + V_2) \quad (2\text{-}18)$$

In the special case where $(R_f / R) = 1/2$, the output voltage is the average of the two input signals.

LabVIEW Demo 2.5: Summing Circuit

Launch the LabVIEW program entitled Summer.vi from the chapter 2 program library. Two inputs V_1 and V_2 can be added together directly when $R_1=R_2=R_f$ or added together each with its own scaling factor R_f / R_1 or R_f / R_2 respectively. Click on the **Run** button to power up the summing circuit. This is a very powerful circuit which finds its place as a solution in many instrumentation circuits.

eLab Project 2

Objective

The objective of this electronic lab is to build an op-amp circuit which sums two independent and separate input signals.

Procedure

Build the summer op-amp circuit of Figure 2-12 and shown pictorially below. The circuit requires a 741 op-amp, a few resistors and two power supplies. Set the input voltage levels to be in the range –1 to +1 volts.

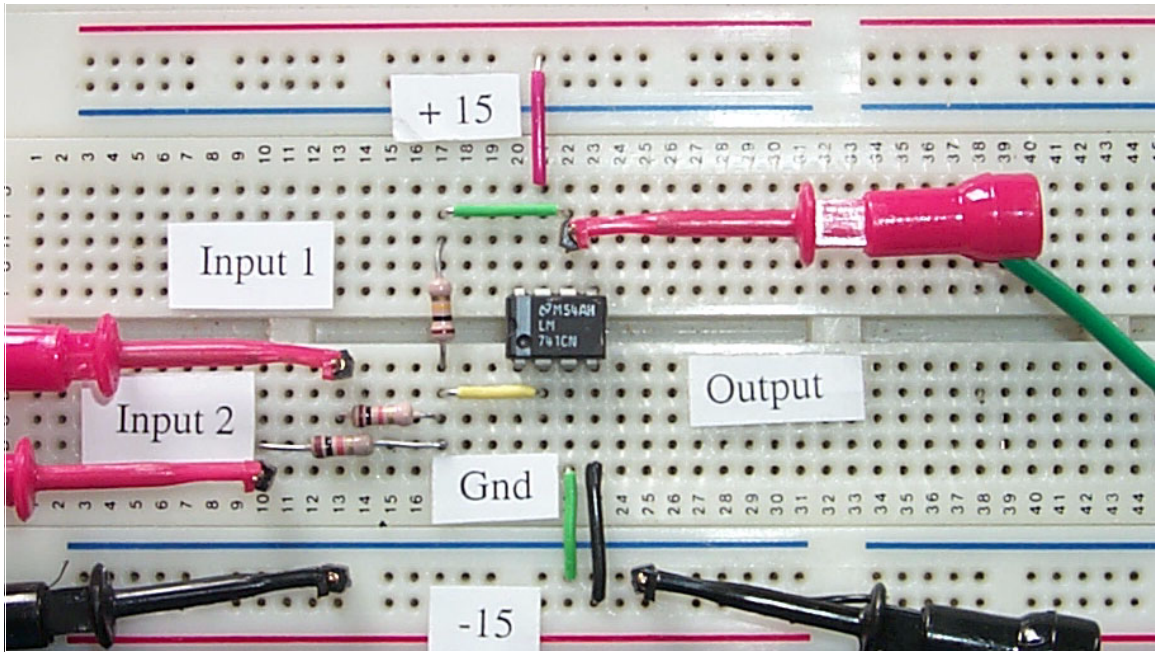


Figure 2-12. Component Layout for an Op-Amp Summing Circuit

For a simple summer, choose $R_1 = R_2 = R_f = 10\ k\Omega$.

For a summing amplifier with a gain of 10, choose $R_1 = R_2 = 10\ k\Omega$ and $R_f = 100\ k\Omega$.

For an averaging circuit, choose $R_1 = R_2 = 10\ k\Omega$ and $R_f = 5\ k\Omega$.

Measure the inputs and output with a digital voltmeter or a DAQ card configured as a voltmeter.

Computer Automation 2: Op-amp Transfer Curve

In assessing the characteristic properties of a device, a graphical representation of the transfer curve provides a unique visualization tool that summarizes all the measurements. Computer automation allows a range of test voltages to be output and the response measured, displayed and analyzed. In this lab, we look at computer generation of test signals and a measurement of the amplifier response displayed in a graphical format.

Launch the LabVIEW program entitled OpAmpTester2.vi from the chapter 2 library. This program uses an analog-output channel on a DAQ card to generate DC test signals and a single analog-input channel to measure the circuit response. The LabVIEW program displays the op-amp response for each input signal and records the transfer curve on a front panel graph. The scan range, scan rate and number of test points can be selected from front panel controls. To save a test set in a spreadsheet format, click on the **Save Data** button.



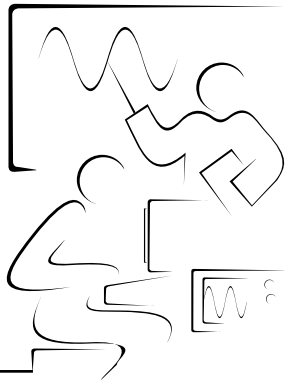
Note Without conditioning, the DAQ card reads signals in the bipolar range –5 to +5 volts. If using the DAQ card without conditioning, set the op-amp power supplies to –5 and +5 volts.

If using the summer circuit of eLab Project 2, then set Input 2 of the op-amp circuit to a constant (usually 0 volts), while the other channel Input 1 steps through a range of input signal levels. After wiring the DAQ lines to you test circuit, set the **Start Measurements** button to (On) and enter a range of test voltages. Click on **Run** to observe the op-amp transfer curve. Observe the ± rails voltage levels and determine the gain of the circuit.

LabVIEW enhancements to the user Interface

- Add a second output channel to the DAQ card so that op-amp summing characteristics can be displayed.
- Create an alarm indicator which lights whenever the output signal level saturates.
- Design a LabVIEW VI to automatically measure the op-amp gain and the rail voltage levels.

A solution can be found on the WEB site
`sensor.phys.dal.ca/LabVIEW`



Lab 3
Semiconductor Diodes

A pn junction is formed by fusing together semiconductor material doped with an excess of electrons called n-type, with semiconductor material doped with a deficiency of electrons (holes) called p-type. The letter ‘n’ stands for the negative, the sign of the electron charge and the letter ‘p’ stands for positive, the average charge in a region deficient in electrons. When the two types of material are butted together, a rearrangement of charge in the neighborhood of the junction causes a potential barrier to be formed between the ‘n’ and ‘p’ side. In order to conduct, majority charge carriers must overcome this potential. The magnitude of the potential wall V_b is a property of the undoped semiconductor material and for silicon V_b is about 0.6 volts.

In a real circuit, an external battery is used to modify the potential wall. In the reverse bias direction, the space charge increases, the width of the depletion increases and the effective potential as seen by the majority carriers becomes higher making it even more difficult for conduction to occur.

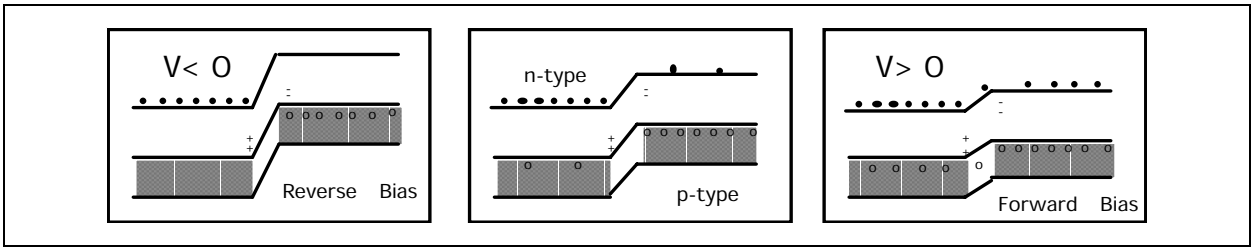


Figure 3-1. Energy Level Diagrams for Reverse, Zero and Forward Biased Diode

In the forward biased direction, the opposite occurs. The effective potential wall reduces in height and conduction can occur. The magnitude of the conduction depends on the probability that the majority carriers can surmount the barrier height. This probability follows a Maxwell-Boltzman distribution, hence the conduction is exponential with the applied voltage.

The current I flowing through a pn junction can be approximated by the expression

$$I = I_o \{\exp(eV/\eta kT) - 1\} \quad (3\text{-}1)$$

where I_o is the reverse bias saturation current, e is the electron charge, V is the applied voltage, k is the Boltzman constant, T is the absolute temperature and η is a property of the junction material.

Let's look at the diode Equation 3-1 in three different limits

1. Reverse Bias (V large and negative)

$$I = - I_o \quad (3\text{-}2)$$

   [In practice, I_o is a few microamps]

2. Forward Bias (V > 0.1 and positive)

$$I = I_o \exp(eV/\eta kT) \quad (3\text{-}3)$$

   [At room temperature, e/kT is about 40 Volts^{-1} and $I = I_o \exp(40\ V)$]

3. Zero Bias (V~0 volts)

$$I = I_o (e/\eta kT)\ V \quad (3\text{-}4)$$

   [In this limit, the exponential term can be expanded in a power series]

Comparing Equation 3-4 with Ohm's Law (V=IR), shows that the term $(\eta kT/eI_o)$ has units of resistance and its magnitude is a property of the diode. At other points, $\Delta V/\Delta R$ or the $(\text{slope})^{-1}$ on the [I-V] characteristic is called the dynamic resistance.

LabVIEW Demo 3.1: Current-Voltage Characteristic of a Silicon Diode

Load the LabVIEW program Diode IV.vi. Ensure the power switch is on and then click on the **Run** button. Investigate the I-V characteristic of a silicon signal diode.

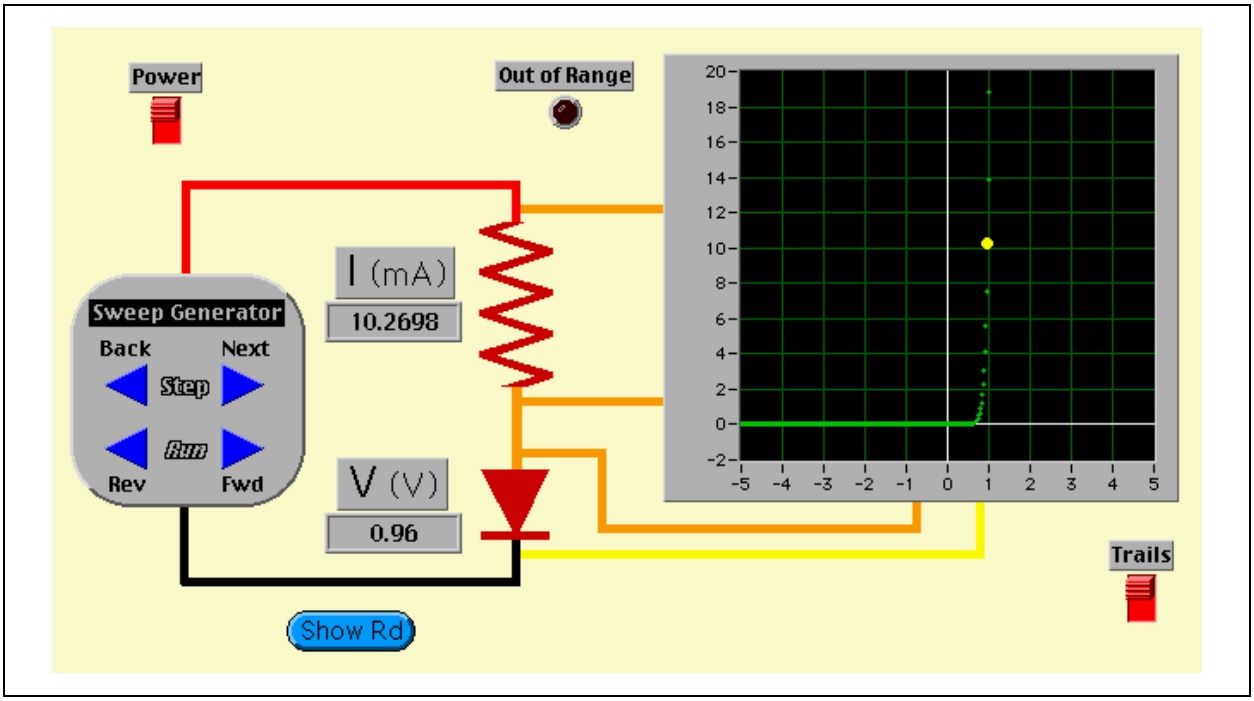


Figure 3-2. LabVIEW Simulation Circuit to Measure the [I-V] Characteristic Diode Curve

The voltage is applied to the diode by clicking on the controls of a Sweep Generator (variable power supply). Clicking on the **Fwd** or **Rev** buttons sweeps the voltage. In the *Step* mode, the buttons **Next** and **Back**, increment or decrement the applied voltage one step (0.02 volts) at a time. By clicking on the *Trails* switch, the individual current and voltage measurements will be marked on the graph.

The dynamic resistance R_d ($\Delta V/\Delta I$) is defined as the inverse of the tangent to the I-V curve at the operating voltage. In the forward biased region, the resistance is small and conduction occurs easily. In the reverse biased region, the resistance is very large and conduction is difficult. Switching the applied voltage polarity from + (forward bias) to – (reverse bias) is like switching a resistor from a low state to a high state. Investigate the dynamic resistance of the silicon diode by clicking on the **Show Rd** button and changing the operating point. The diode's ability to switch resistance from a high to low state was exploited in the early digital logic circuits employing combinations of diodes and resistors to build DRL (Diode-Resistor logic) devices.

What is the dynamic resistance at plus and minus 0.6 volts?

Semiconductor Diodes

When a junction is formed, some of the carriers in each material diffuse across the junction into the other side. That is, some of the electrons go to the p-type material and an equal number the holes go to the n-type material. This continues until the separation of charge forms a dipole layer near the junction, which in turn creates an electric field across the junction. At equilibrium no more current flows and a potential difference or barrier exists at the junction boundary. The magnitude of the potential barrier is a property of the host semiconductor material. For conduction to occur in the forward biased region of the I-V characteristic curve, the applied voltage must be greater than this barrier. Extrapolation of the I-V curve back to the voltage axis yields a threshold voltage which is close (within 10%) to the energy gap of the host semiconductor.

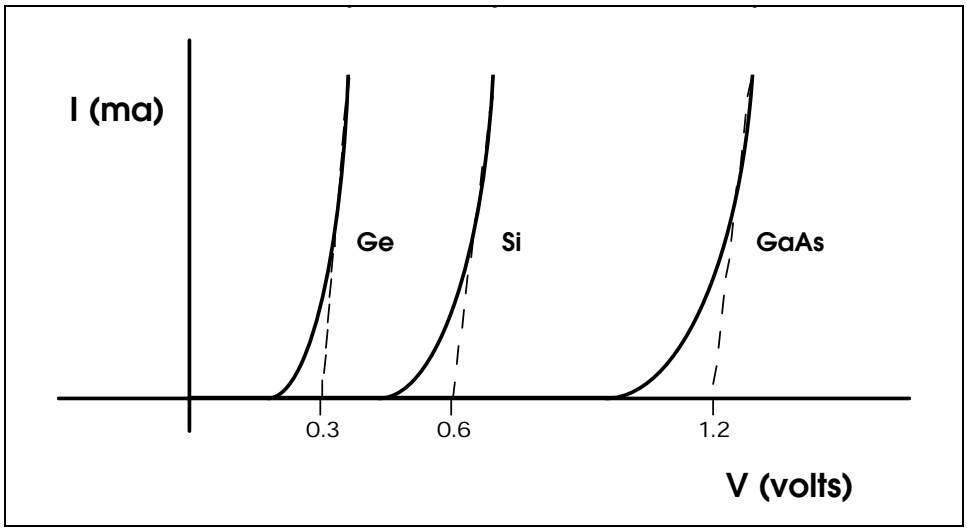


Figure 3-3. The [I-V] Characteristic Curves for Ge, Si and GaAs Diodes

For germanium the threshold voltage is 0.3 volts, for silicon the threshold voltage is 0.6 volts and for gallium arsenic the threshold voltage is 1.2 volts.

LabVIEW Demo 3.2: Forward Bias Properties

Load the LabVIEW program Diode2.vi from the chapter 3 program library. Ensure the power switch is on and then click on the **Run** button. This simulation plots the forward bias characteristics of diodes manufactured from three of the most popular semiconductor materials: silicon, germanium and gallium arsenic. Click on the thumb-wheel selector to change the material type. From the diagram make an estimate of the threshold voltage for each type.

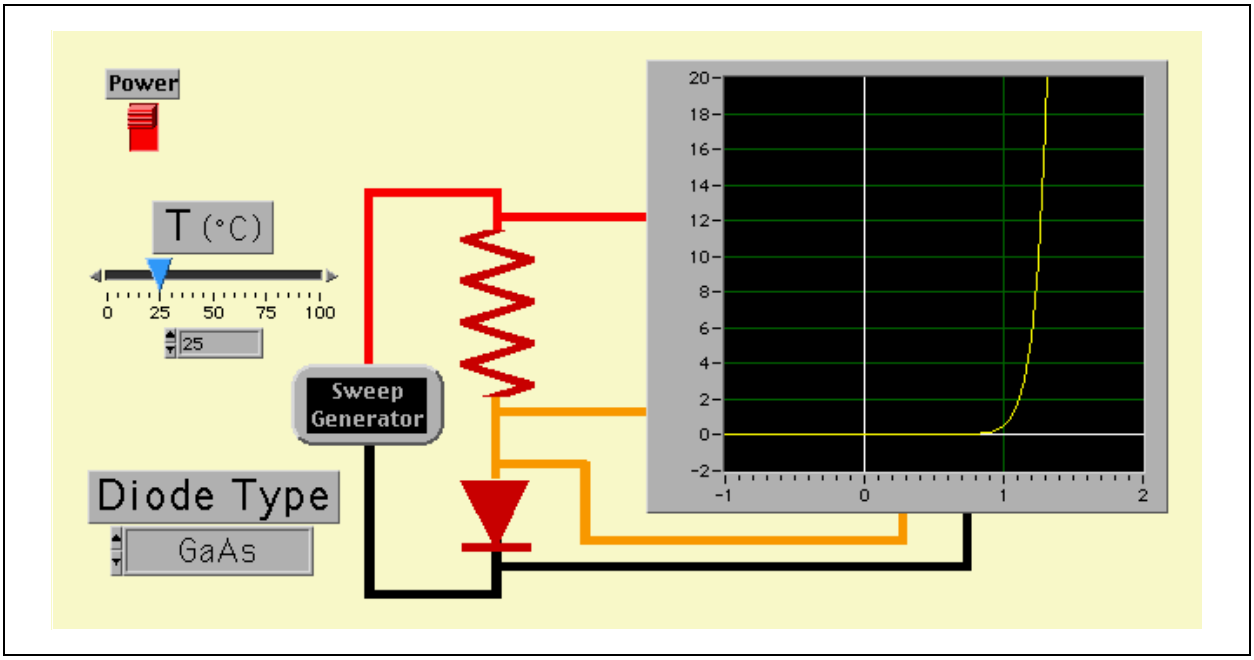


Figure 3-4. LabVIEW Simulation of the [I-V] Characteristic Curve of Ge, Si and GaAs Diodes

It is clear from the diode equation that the current flowing through a diode depends critically on the ambient temperature. In the above simulation, the ambient temperature can be varied by dragging the temperature slider. Investigate the temperature dependence of the diode I-V characteristic curve.

LabVIEW Exercise

Using the Diode2.vi program, make a plot of the voltage across the silicon diode versus temperature at a constant current of 10 ma. This voltage level is strongly dependent on temperature. Do diodes make good thermometers?

LabVIEW Demo 3.3: Reverse Bias Properties

Load the LabVIEW program Diode3.vi from the chapter 3 program library. Ensure the power switch is on and then click on the **Run** button. This simulation plots the reverse bias characteristics for Zener and Avalanche diodes. Click on the thumbwheel selector switch to change the diode type.

A Zener diode is heavily doped so that at a particular reverse voltage, the diode will switch from a normally high resistance state to a low resistance state. In Diode3.vi, the Zener voltage is at –12 volts. Zener diodes are used in all types of circuits to limit voltage to a particular designer maximum value.

All diodes if pushed far enough into the reverse bias region will eventually breakdown in an avalanche mode. Free electrons are accelerated by the

applied negative voltage to such a high velocity that on collision with an atom more electrons are freed which in turn are accelerated and collide with more atoms. As the process continues, the current rises exponentially and the diode will destroy itself unless the current is limited. A special type of diode called an avalanche photodiode exploits the avalanche charge multiplication to become a very sensitive light sensor.

The Photodiode

All diodes are light sensitive. The reverse biased saturation current depends on the density of free electrons and holes and for photodiodes I_o is called the dark current. Light shining onto a diode junction creates additional free electron-hole pairs. In reverse bias, large voltages can be applied to a diode. The free carriers are swept across the junction by the reverse voltage and result in a photocurrent. The magnitude of the current depends on the intensity of the light striking the junction region. Photodiodes are manufactured to optimize this effect.

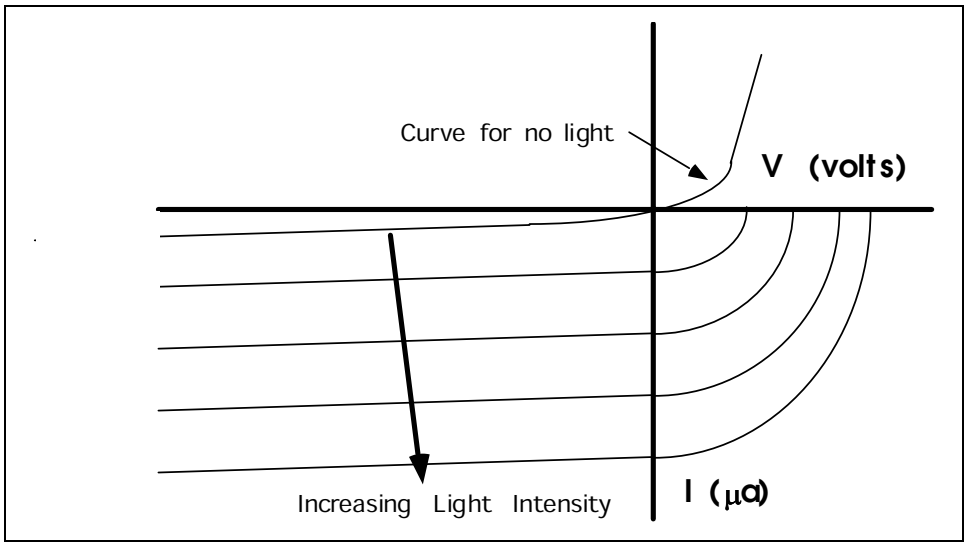


Figure 3-5. The [I-V] Characteristic Curve for a Photodiode

The I-V characteristic of a photodiode displays how light shinning on the diode junction shifts the characteristic curve away from the dark current curve. Photocurrents are in the microamp region, a factor of 1000 times smaller than currents flowing in the forward biased region. Precise measurements of light intensity require that the dark current to be subtracted from measured photocurrents.

LabVIEW Demo 3.4: The Photodiode [I-V] Characteristic Curve

Load the LabVIEW program PhotoDiode.vi from the chapter 3 program library. Ensure the power switch is on and then click on the **Run** button. This simulation plots the diode I-V characteristic curve as the intensity of a light source is varied. Click and drag on the *Light Intensity* slider. Note that the characteristic curve is also sensitive to the temperature. Precision measurements require that photodiodes be held at a constant temperature.

Simple Light Meter Using a Photodiode/Op-amp

The photocurrent i_p is directly proportional to the applied light intensity I_L. The proportionality constant R is called the responsivity and its value depends on the wavelength of the applied light and the host semiconductor material.

$$i_p\ (\mu amp) = R\ I_L(\mu watts) \tag{3-5}$$

For silicon photodiodes, R = 0.5 μamp/μwatt at 680 nanometers.

Recall the transfer curve for the inverting op-amp, Equation 3-5

$$V_{out} = -\ (R_f / R_1)\ V_1 \tag{3-6}$$

It can be written as

$$V_{out} = -\ (V_1/R_1)\ R_f = -\ i_1\ R_f \tag{3-7}$$

where i_1 is the current flowing in the input loop.

An op-amp configured in this manner is called a *current-to-voltage converter.* The output voltage is the product of the current flowing into the summing point times the feedback resistance. A photodiode is a current generator, hence the photocurrent i_p is the input current i_1 and the photodiode/op-amp transfer equation is just

$$V_{out} = -\ i_p\ R_f = -\ R\ I_L\ R_f \tag{3-8}$$

LabVIEW Demo 3.5: Photodiode/Op-amp Photometer Properties

Load the LabVIEW program Photometer.vi from the chapter 3 program library. Ensure the power switch is on and then click on the **Run** button. This simulation plots the photometer response curve *Vout* versus *Light Intensity* as the intensity of a light source is varied. Click and drag on the **Light Intensity** rotary knob.

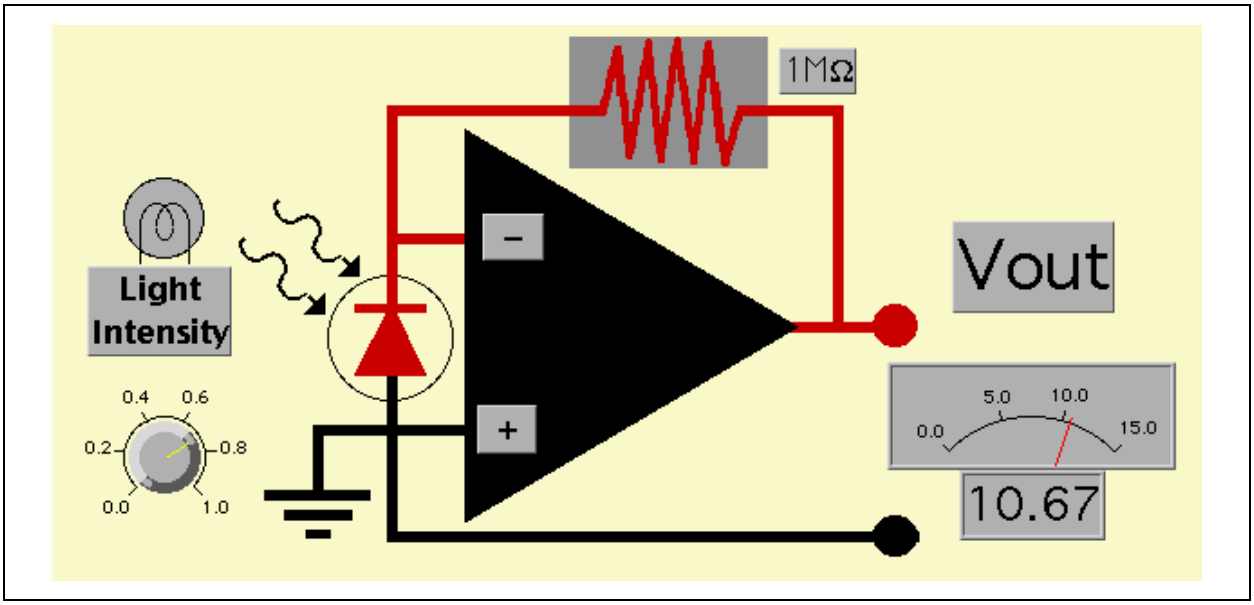


Figure 3-6. LabVIEW Simulation of a Simple Light Meter Using and Photodiode and Op-Amp

In general the photodiode characteristic curve is also sensitive to the temperature. Precision measurements require that photodiodes be held at a constant temperature.

LabVIEW Challenge

Design a LabVIEW program which includes the wavelength dependence of the Responsivity R into the simulation Demo 3.5. Over the visible region, R is approximately linear with values of 0.5 µA/µW in the deep red (680 nm) and 0.14 µA/µW in the deep violet (400 nm).

eLab Project 3

Objective

The objective of this electronic lab is to build an sensor circuit to measure light intensity.

Procedure

Build an op-amp current-to-voltage circuit shown in Figure 3-6 or displayed pictorially below. The circuit requires a 356 FET input op-amp, a resistor, a photodiode and two power supplies. If a photodiode is not available, it can be replaced with a Light Emitting Diode. LEDs are efficient light sources when forward biased and can be used in reverse or zero bias as a photodiode.

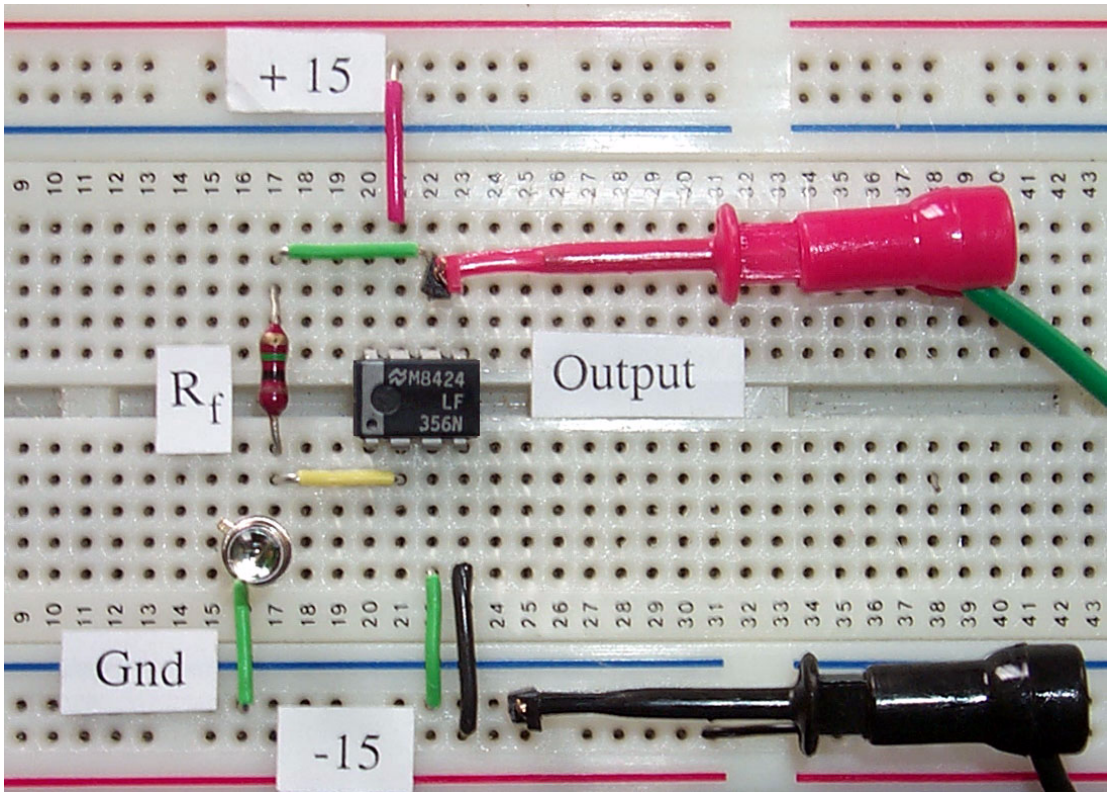


Figure 3-7. Component layout for Op-Amp Light Meter

Most photodiodes generate a photocurrent of a few microamps in a moderately bright light field. If $R_f = 1M\Omega$, then the light meter output voltage will be a few volts.

Investigate the voltage output during sunrise, sunset, or the passage of clouds.

LabVIEW Challenge: Night-time Speed Detector

Place two light meters 100 feet apart along a busy road. As a car passes a detector, the voltage level will rise dramatically. Log the detector signals and measure the time between each rising signal. Dividing the elapsed time between detector rising signals into the distance between the detectors gives the speed of a passing vehicle.

Computer Automation 3: I-V Characteristic Curve of a Diode

In assessing the characteristic properties of a device such a diode, a graphical representation of the current-voltage [I-V] curve under various input conditions completely defines the operation of the device. Computer automation allows a range of test signals under a variety of conditions to be output to the device under test. The measured response together with the

input conditions can be displayed and analyzed. In this lab, we look at the I-V characteristic curve for a diode under test as one of the environmental conditions (temperature or light intensity) is varied.

Launch the LabVIEW program entitled TestDiode.vi from the chapter 3 library. This program uses an output channel on the DAQ card to generate DC test signals for the automated testing a diode circuit similar to Figure 3-4. The scan range, rate and number of test points can be selected from front panel controls. Two input channels on the DAQ card measure the current and voltage of the photodiode at the operating point. The program displays the family of transfer curves on a front panel graph. To save a test set in a spreadsheet format, click on the **Save Data** button.

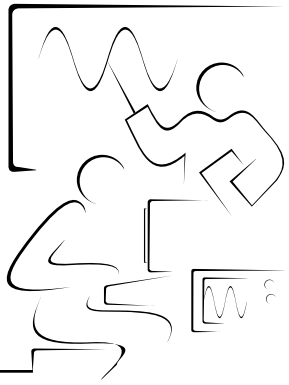
Connect the diode and current limiting resistor to the DAQ output. In most cases, the DAQ output will have to be buffered to provide the required current at the maximum forward biased limit. Chose a resistor value of (<1 kΩ) so as to produce a voltage signal in the 1-5 volt range when the diode is forward biased. Click on **Run** to observe the transfer characteristic curve.

LabVIEW Enhancements

Change the operating temperature and collect a family of [I-V] curves.

Use a LED to illuminate a photodiode and collect a family of curves in the reverse bias region.

Lab 4
Op-Amp AC Characteristics



In the earlier labs, the input signal level was assumed to be constant or at least slowly varying. Most analog circuits are AC (alternating current) and as such the small signal AC response of an op-amp is one of the most importance properties. The AC frequency characteristic is best described in term of a Bode plot where the gain is plotted on a log scale on the vertical axis and the frequency is plotted on a log scale on the horizontal axis. Log plots allow the gain and frequency to be plotted over a wide dynamic range. Special regions on the Bode plot show up as a straight line where the response curve follows a simple power law.

The open loop gain A was described earlier as the ratio of the change in the output voltage to the change in the input voltage (V_{out}/V_{in}). In the limit of zero Hertz, the open loop gain is independent of frequency and written as A(0). Gain can also be expressed in decibels as

$$N(dB) = 20 \log_{10}(V_{out}/V_{in}) = 20 \log_{10}(A) \qquad (4\text{-}1)$$

For example: a typical op-amp with an open loop gain A(0) = 100,000 has N(0) = 100 dB. An ideal Bode plot for such an op-amp might have the following response curve.

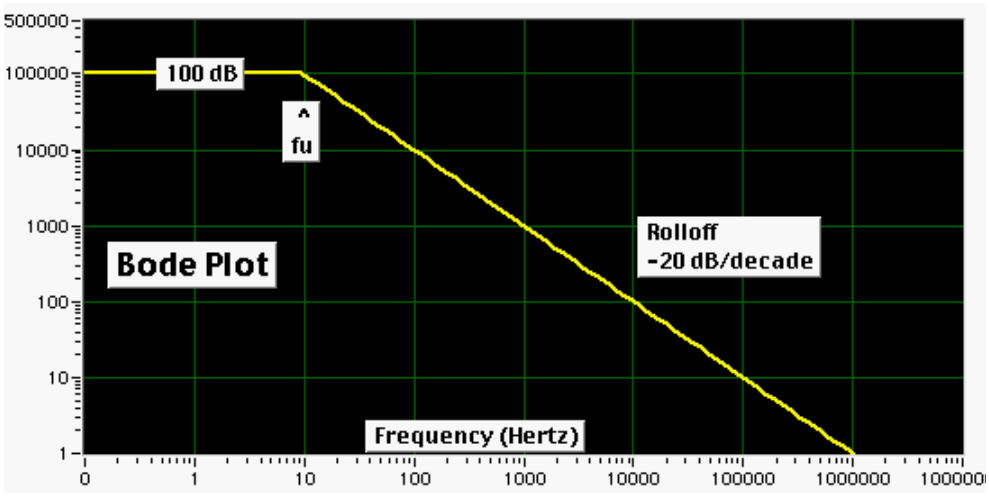


Figure 4-1. Bode Plot for an Open loop Op-Amp Circuit

The open loop gain (100 dB) is a constant for all frequencies up to about 10 Hertz. Above this frequency, called the upper frequency cutoff point f_u, internal components (mainly capacitors) have a dramatic effect on the frequency response. The response curve falls off or rolls off with a slope of –20 dB/decade. This is indicated on the Bode plot as the straight line for all frequencies greater than the cutoff point. Below f_u, the op-amp response is independent of frequency and can be represented by A(0) or N(0), while for frequencies greater then f_u, the response is strongly frequency dependent.

An amplifier's bandwidth BW is defined as the difference between the upper and lower frequency cutoff points ($BW = f_u - f_l$). Recall that op-amps are DC coupled so the low frequency cutoff is at 0 Hertz. Hence the bandwidth of the op-amp is just f_u.

A second special frequency $f_u(0dB)$ occurs where the response curve cuts the horizontal axis at a gain of 1 or 0 dB. This point is called the unity gain bandwidth BW(0dB). In the above example this point occurs at 1,000,000 Hertz. Here the unity gain bandwidth $BW(0dB) = f_u(0dB) = 1$ Mhz. It is interesting to note that at these two frequencies f_u and $f_u(0dB)$, the gain-bandwidth product (GBW) is a constant.

$$\text{At } f_u \qquad GBW = 100{,}000 \times 10 \text{ Hz} = 10^6 \tag{4-2}$$

$$\text{At } f_u \text{ (0dB)} \qquad GBW = 1 \times 1{,}000{,}000 = 10^6 \tag{4-3}$$

In fact, the frequency at the intersection of all constant gain lines with the response curve displays this property. The gain-bandwidth product is a constant and its value is a property of each op-amp. When negative feedback applies, this relationship provides a quick way to calculate the upper frequency cutoff point for different gains.

LabVIEW Demo 4.1: Ideal Frequency Response Curve (Open Loop)

Load the program called Bode1.vi from the chapter 4 program library. The open loop gain has been set to 100 dB with an upper frequency cutoff at 10 Hertz. Click on the **Run** button to display the Bode plot. Investigate the ideal Bode plot by varying the open loop gain and the cutoff point.

–3 dB Cutoff Point

A more precise definition of the cutoff point is the frequency at which the gain has fallen to one half of A(0), that is when $(V_{out} / V_{in}) = 1/2$. In decibels this is

$$N(dB) = 20 \log_{10}(1/2) = -3 \text{ dB} \quad (4\text{-}4)$$

In an op-amp with N = 100 dB, the upper frequency cutoff point is the frequency where the gain has fallen to (100 - 3) = 97 dB. On the Bode plot this limit is shown as a horizontal line at N = 97 dB. In the previous section, f_u sometimes called the corner frequency was found from the intersection of the two straight line regions, A(0) and the roll off line.

A more exact definition of the gain curve is

$$A(f) = A(0) / \sqrt{[1 + (f^2 / f_u^2)]} \quad (4\text{-}5)$$

Note that the gain curve (see Figure 4-2) is smooth near the upper frequency cutoff point.

In decibels, the above equation is

$$N(f) = 20 \log_{10}(A(0)) - 20 \log_{10} \sqrt{[1+f^2 / f_u^2]} \quad (4\text{-}6)$$

At the frequency where $f = f_u$, $N(f_u) = 20 \log_{10}(A_0) - 20 \log_{10}(\sqrt{2})$ or

$$N(f_u) = N(0) - 3 \text{ dB7} \quad (4\text{-}7)$$

Thus the upper frequency cutoff point is given by the intersection of the –3 dB line with the open loop op-amp frequency curve N(f).

LabVIEW Demo 4.2: Frequency Response Curve (Open Loop)

Load the program called Bode2.vi from the chapter 4 program library. The open loop gain has been set to 100 dB with an upper frequency cutoff at 10 Hertz. Click on the **Run** button to display the Bode plot. The heavy line which surrounds the smooth response curve is the ideal approximation used in LabVIEW Demo 4.1. Near the sharp corner f_u, the more exact frequency

response curve is shown as the smooth curve. You can vary the open loop gain and the upper frequency cutoff point. A comparison of the –3 dB cutoff frequency and the corner frequency can be seen by zooming in near the upper cutoff frequency.

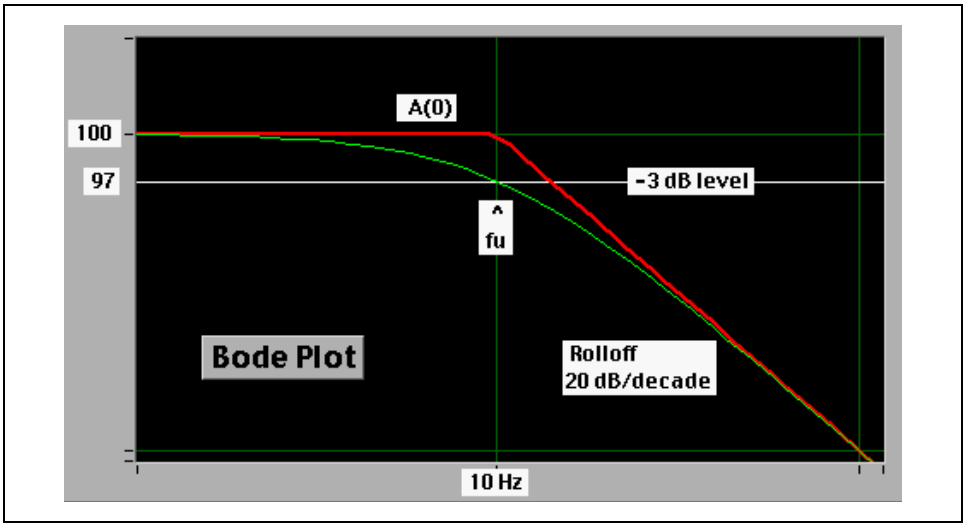


Figure 4-2. Bode Plot for an Ideal and Normal Op-Amp Circuit

The ideal Bode plot with A(0) = 100,000 and f_u = 10 Hz is shown as the heavy line. A white line below A(0) shows the –3db level. The more precise gain is shown as the curved line. The intersection of the –3 dB line with the exact gain curve yields the upper frequency f_u(–3 dB) point. Note the closeness of this frequency to the corner frequency of the ideal op-amp curve. This is the reason why the gain-bandwidth approximation can be used to estimate the upper frequency cutoff point in real circuits.

Frequency Response of Closed Loop Gain Circuits

Circuits with negative feedback (closed loop) have a much smaller gain than the open loop value. Circuit stability is traded off against gain. The closed loop bode plot can be found by replacing A(0) in Equation 4-5 with G(0) then

$$G(f) = G(0) / \sqrt{[1 + (f^2 / f_{u'}^2)]} \qquad (4\text{-}8)$$

where $f_{u'}$ is defined as the –3 dB point for the G(f) curve. Take for example our typical op-amp with A(0) =100,000 and f_u =10 Hertz. In a closed loop circuit with a gain G(0) = 1000, the upper frequency point calculated from the GBW=10^6 would be 1000 Hertz.

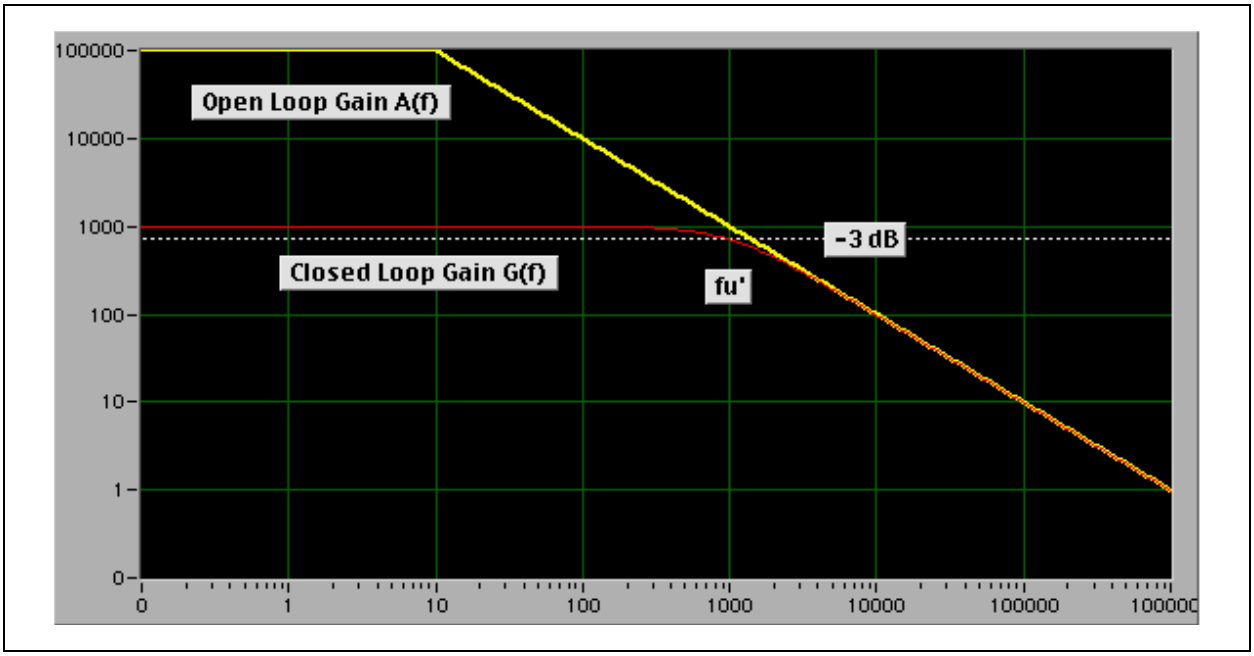


Figure 4-3. Closed Loop Bode Plot for Op-Amp Circuit with G = 1000

The heavy line is the open loop frequency response curve (ideal) and the curved line is the closed loop frequency response curve. The region between the two curves is where negative feedback trades off gain for stability. As long as A(f) is much greater than G(f), the op-amp circuit is stable. As the operating frequency approaches the closed loop cutoff frequency $f_{u'}$, G(f) becomes close to A(f) and the curves merge. At frequencies higher than the cutoff point, the closed loop gain curve becomes the open loop curve and the response curve is strongly frequency dependent at –20 dB/decade.

LabVIEW Demo 4.3: Dynamic Frequency Response Curve (Closed Loop)

Load the program called Bode3.vi from the chapter 4 program library. Click on the **Run** button to activate the circuit. The closed loop gain can be set by clicking and dragging on the **Gain** slider. Investigate how the closed loop gain is always contained inside the open loop ideal gain curve. Note the shape of the closed loop gain curve at unity gain.

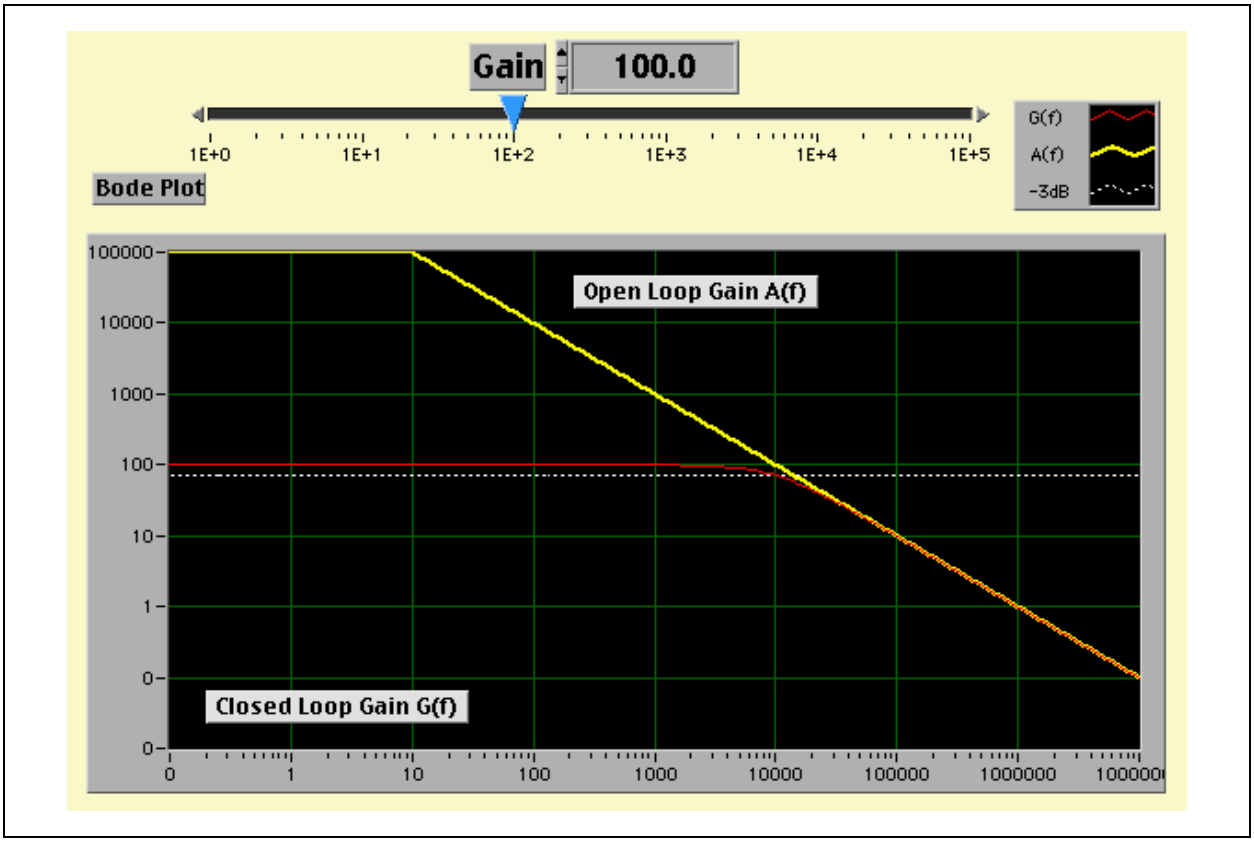


Figure 4-4. Closed Loop Bode Plot for Op-Amp Circuit with G = 100

How does the upper frequency cutoff point $f_{u'}$ vary with gain?

What can you say about the closed loop Gain-Bandwidth product?

LabVIEW Challenge

Design a LabVIEW calculator to calculate the upper frequency cutoff point using the gain-bandwidth product and the closed loop gain. Design a LabVIEW calculator (Version 2) to calculate the upper frequency cutoff given the input resistor, feedback resistor, open loop and unity gain values.

eLab Project 4

Objective

To investigate the frequency response of an inverting op-amp circuit with a gain of 10 to 1000.

Procedure

Build an inverting op-amp circuit of Figure 4-5. The circuit requires a 741 op-amp, three resistors and two power supplies. If R_f = 100 kΩ and R_1 = 1 kΩ, then the closed loop gain G(0) = (R_f /R_1) at 0 Hertz is 100 or

N(0) = 40 dB. For the 741 op-amp, the unity gain-bandwidth is about 1.5 MHz and the open loop gain is about 200,000. The GBW equation predicts f_u = 7.5 Hz. For a closed loop gain of 100, then the upper frequency cutoff f_u, should be about 15 kHz. Repeat the calculation when R_1= 10 kΩ in the circuit below.

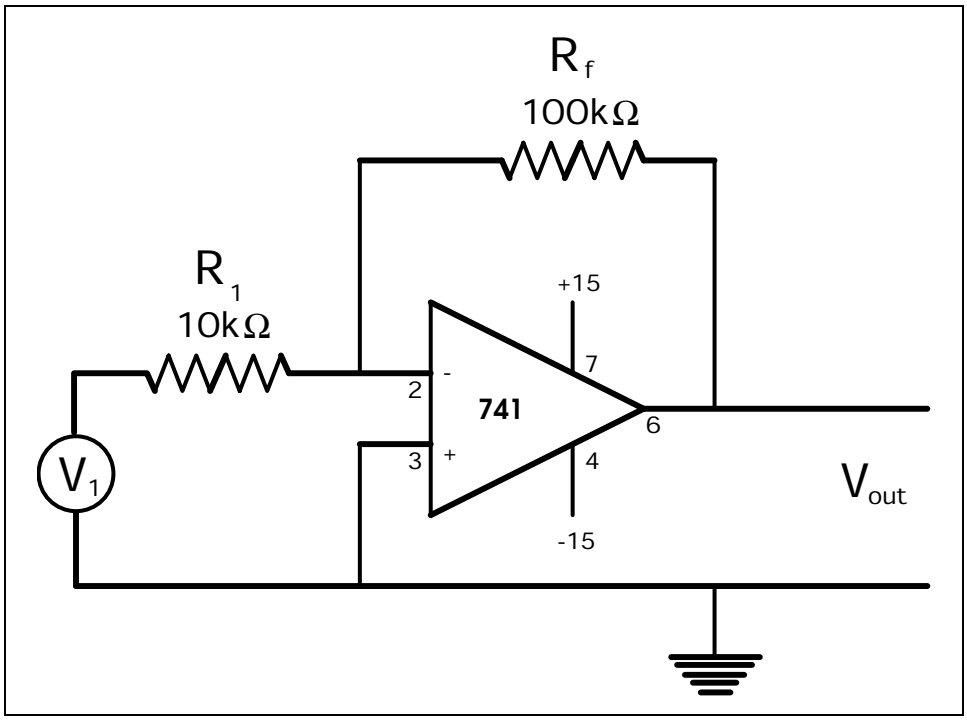


Figure 4-5. Schematic Diagram of Inverting Op-Amp Circuit

Use a function generator set to sine wave with an output signal level of 5mV (peak-peak). Use a good oscilloscope or a high speed DAQ card to measure the output signal level. In all cases, it is wise to measure the input signal level and compute the gain from the expression V_{out}/V_{in}. In choosing the test frequencies, select the decade range then multiply by 1, 2, 4, and 8. This gives an approximately uniform set of points on a log f scale. Graph the Bode plot, that is the gain in decibels as a function of $\log_{10}$ of the frequency. Compare the measured upper cutoff frequency with the predict value.

Computer Automation 4: Stimulus Signals

Computer automation allows a range of periodic stimulus signals to be applied to a device or circuit under test. The response to this stimulus can be used to characterize the device or ensure that it falls within specifications. The most general form of a periodic stimulus is

$$V = V_0 + A[\ Fcn(f, \theta, t)]$$

where V_0 is a DC voltage level often called the offset voltage, f is the frequency of the periodic signal, θ is the phase of the signal, and t is time. While the functional shape, Fcn of the waveform can be varied, the most common waveforms are sinusodial, square wave, sawtooth and triangle. In this lab, we look at stimulus signals generated by a LabVIEW program and observed on an oscilloscope connected to the DAQ card, an analog out channel.

Launch the LabVIEW program entitled FunctionGenerator4.vi from the chapter 4 library. This program uses an output channel on the DAQ card to generate AC and DC test signals for the automated testing applications. The scan range, rate and number of test points can be selected from front panel controls. The default parameters are set for a sinusodial waveform

$$V = V_0 + A \sin(2 \pi f t + \theta)$$

where $V_0 = 0$ volts, A = 2.0 volts, f = 20 Hz and $\theta = 0$.

Connect the oscilloscope to DAQ pins for device(1)/channel(0). Click on **Run** to start the signal generation. Observe the signal on the oscilloscope as the offset voltage, amplitude, frequency and phase are varied. Try the other waveforms Triangle, Square and Sawtooth.

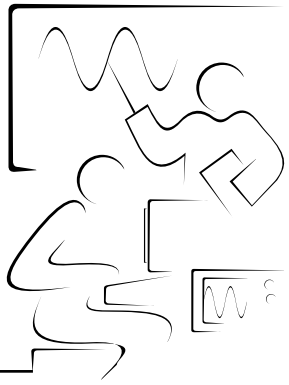


Note The maximum frequency that the DAQ can output depend on the type and specifications of the DAQ card available.

LabVIEW Techniques

On the diagram panel of the main program, open up the sub-VI called Compute waveform.vi to see how the different waveforms have been created. This program called Function Generator4.vi is an adaptation of a program called Function Generator.vi found in the `LabVIEW/Examples/daq/anlogout/anlogout.llb` library file.

Lab 5
Op-Amp Filters



In the last lab, we discovered that the frequency response curve of op-amp circuits with resistive elements was dominated by the intrinsic frequency dependence of the op-amp. In this lab, capacitive and inductive elements are introduced into the input and feedback loops. These elements have their own frequency dependence and they will dominate the frequency response of the gain curve. In many cases, the frequency response curve can be tailored to execute specialized functions such as filters, integrators and differentiators. Filters are designed to pass only specific frequency bands, integrators are used in proportional control circuits and differentiators are used in noise suppression and waveform generator circuits.

Impedance

A network of resistors, capacitors and/or inductors can be represented by the generalized impedance expression

$$\mathbf{Z} = \mathbf{R} + j\mathbf{X} \tag{5-1}$$

where **R** is the resistive component and **X** is the capacitive/inductive component called the reactance. The complex symbol j indicates that the reactive component is shifted in phase by 90° from the resistive component. Complex notation will be used in the analysis of op-amp circuits in this lab. The voltage **V** and current **I** are in general a vector or a phasor with both real and imaginary terms.

Ohm's law tells us that there is a direct relationship between the voltage across a resistor and the current flowing through that resistor. Assuming that the AC current $i = i_o \sin(\omega t)$, then the voltage across a resistor is

$$V_R = iR = i_o \sin(\omega t)\, R \tag{5-2}$$

where $\omega = 2\pi f$ and f is the frequency measured in cycles per second or Hertz. The amplitude of V_R is just ($i_o R$). Resistance is real and always positive. In complex notation, the voltage across a resistor is

$$V_R = i_o R \exp(j\omega t) \tag{5-3}$$

For an inductor, the magnitude of the reactance or equivalent resistance X_L is (ωL). Lenz's law tells us that the voltage across an inductor is proportional to the derivative of the current. Assuming that the current is given by $i = i_o \sin(\omega t)$, then the voltage across the inductor is

$$V_L = L\,(di/dt) = L\,\omega\,i_o \cos(\omega t) \tag{5-4}$$

Recalling that $\cos(x) = \sin(x+90°)$, then Equation 5-3 becomes

$$V_L = i_o \sin(\omega t + 90°)\,(\omega L) \tag{5-5}$$

This expression look like Ohm's law, Equation 5-2 where (ωL) is the equivalent of "resistance" but with a phase shift of 90°. The equivalent complex "resistance" is called the reactance $\mathbf{X_L} = \mathbf{j}\omega L$ and the 90° phase shift is represented by the complex operator **j**. In complex notation

$$\mathbf{V}_L = (j\omega L)\, i_O \exp(j\omega t) \tag{5-6}$$

For a capacitor, the magnitude of the reactance or equivalent resistance X_C is ($1/\omega C$). The charge Q on a capacitor is directly proportional to the voltage across the capacitor ($Q = CV$). Recalling the definition of current $i = dQ/dt$, one can write this relationship as

$$i = C\,(dV/dt) \tag{5-7}$$

Solving for V in Equation 5-7 and integrating yields

$$V_C = (1/C) \int i_o \sin(\omega t)\, dt = (1/\omega C)\, i_o(-\cos\omega t) \tag{5-8}$$

With the identity $-\cos(x) = \sin(\omega t - 90°)$, then

$$V_C = (1/\omega C)\, i_o \sin(\omega t - 90° \tag{5-9}$$

This expression look like Ohm's law, Equation 5-2 where ($1/\omega C$) is the "resistance" but with a phase shift of - 90°. The equivalent complex "resistance" is called the reactance $\mathbf{X_C} = 1/j\omega\mathbf{C}$ and the 90° phase shift is represented by the complex operator j. In complex notation

$$\mathbf{V}_C = (1/j\omega C)\, i_o \exp(j\omega t) \tag{5-10}$$

In summary

- Resistance (**R**) is real and its magnitude is R.
- Reactance for an inductor ($\mathbf{X}_L = j\omega L$) is imaginary and its magnitude is ωL.
- Reactance for a capacitor ($\mathbf{X}_C = 1/j\omega C$) is imaginary and its magnitude is $1/\omega C$.

Low Pass Filter

A simple low pass filter can be formed by adding a capacitor C_f in parallel with the feedback resistor R_f of an inverting op-amp circuit.

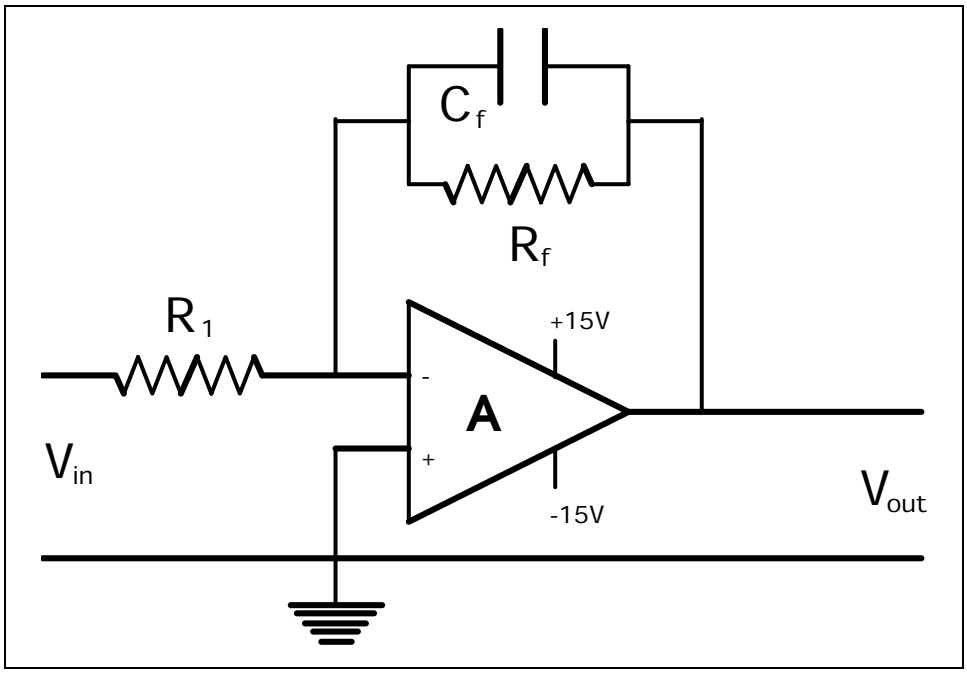


Figure 5-1. Low Pass Op-Amp Circuit

Recall that "resistors" in parallel add as reciprocals. Hence the feedback network of these components can be represented by a single feedback impedance Z_f where

$$1/\mathbf{Z}_f = 1/R_f + 1/\mathbf{X}_c \qquad (5\text{-}11)$$

Inverting and rationalizing leads to the expression

$$\mathbf{Z}_f = (R_f - j\omega C_f R_f^2)/(1+\omega^2 C_f{}^2 R_f^2) \qquad (5\text{-}12)$$

The feedback impedance has both a real and an imaginary term, both of which are frequency dependant. The voltage transfer equation can be written as

$$\mathbf{V}_{out} = (\mathbf{Z}_f / R_1)\, \mathbf{V}_{in} \tag{5-13}$$

Solving for the gain (V_{out}/V_{in}) leads to a simple equation

$$G(f) = G(0)/\sqrt{(1+f^2/f_u^2)} \tag{5-14}$$

where $G(0) = (R_f/R_1)$ is just the closed loop gain with no capacitor. This equation looks suspiciously like the intrinsic frequency dependence of the op-amp, Equation 4-5. And it is, except that now upper frequency cutoff point f_u is related to the feedback network and given by

$$2\pi f_u = 1/ R_f C_f \tag{5-15}$$

The closed loop cutoff point is always less than the open loop frequency cutoff. Note as before, the gain falls to 1/2 or –3 dB at f_u and the filter bandwidth is just f_u.

LabVIEW Demo 5.1: Simple Low Pass Filter

Load the program called LowPass.vi from the chapter 5 program library. Click on the **Run** button to see the Bode plot. Investigate the position of the upper frequency cutoff point as the feedback capacitor or feedback resistor is varied. Note the response curve when the gain G(0) is changed by varying R_1or R_f. For convenience the open loop curve with A(0) = 100 dB and an open loop cutoff frequency at 10 Hertz is also shown.

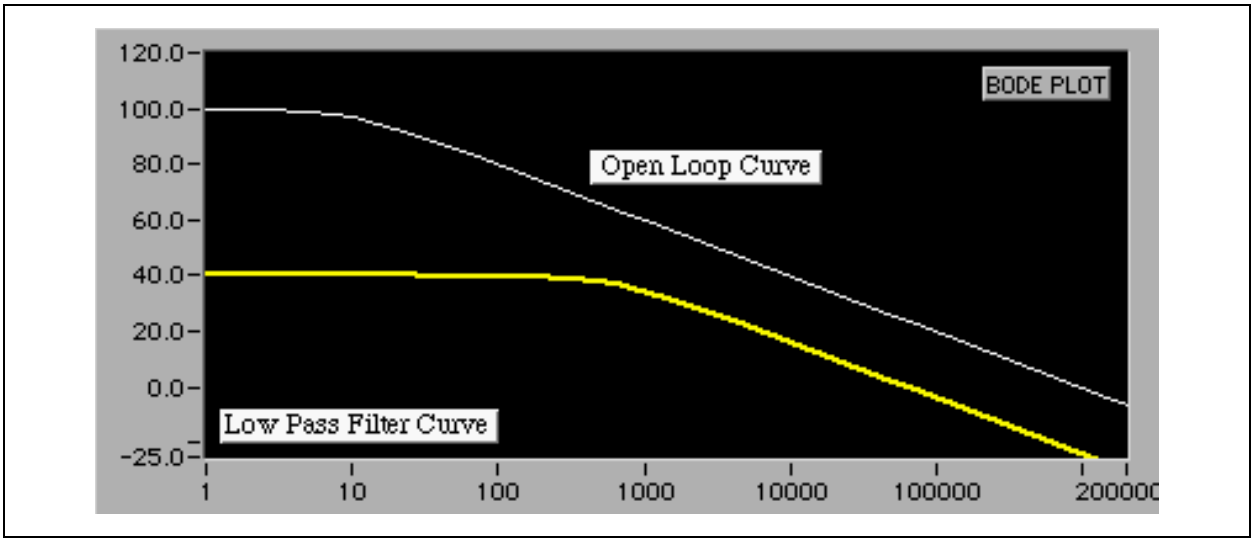


Figure 5-2. Bode Plot of an Op-Amp Low Pass Filter

All frequencies with f is less than f_u have a constant gain while all frequencies with f greater then f_u are attenuated. A filter which displays this property is called a low pass filter. For high frequencies, one notes that the response curve rolls off with the same slope of –20 dB/decade as the open loop response curve. What is happening here?

Look at the feedback network impedance in the limits where $f<f_u$ and $f>f_u$. Calculating $\mathbf{Z}_f$ or using the LabVIEW vector calculator shows in the limit of

$$\text{low frequencies} \quad (f< f_u),\ \mathbf{Z}_f \rightarrow R_f \tag{5-16}$$

$$\text{high frequencies} \quad (f> f_u),\ \mathbf{Z}_f \rightarrow 1/j2\pi fC_f \tag{5-17}$$

At low frequencies, the reactance of the capacitor is so large, that all the current flows through R_f and the gain is just (R_f/R_1). At high frequencies, the capacitor reactance is low and the current readily flows through the capacitor not the resistor. Now the gain is $(1/j2\pi f\, R_1C_f)$ and falls off inversely with frequency. On the Bode plot, this region is a straight line with a negative slope of 20 dB/decade.

When a square wave is integrated, what waveform do you find? That is right, a triangular wave. Just like in Lab 2 for the DC integrator, the capacitor C_f allows charge to accumulate on the feedback capacitor in the region where $f> f_u$. A low pass filter in this frequency range integrates the waveform so that a square wave input becomes a triangular wave output. AC integrators find extensive use in analog computation circuits.

High Pass Filter

A simple high pass filter can be formed by adding a capacitor C_1 in series with the input resistor R_1 of an inverting op-amp circuit.

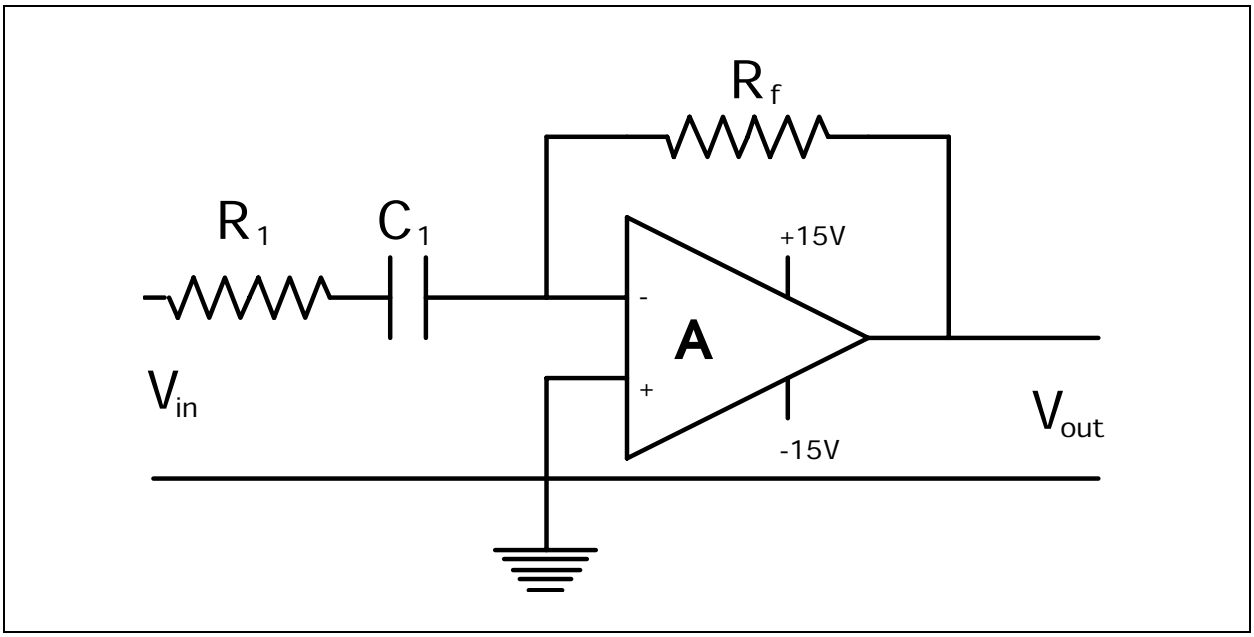


Figure 5-3. High Pass Op-Amp Circuit

Recall that "resistors" in series add serially. The input network of components can be represented by a single feedback impedance $\mathbf{Z}_1$ where

$$\mathbf{Z}_1 = R_1 + \mathbf{X}_c \tag{5-18}$$

Substituting the definition of reactance for a capacitor leads to

$$\mathbf{Z}_1 = (R_1 - 1/j\omega C_1) \tag{5-19}$$

The complex transfer equation for gain can be written as

$$\mathbf{V}_{out} = (R_f / \mathbf{Z}_1)\ \mathbf{V}_{in} \tag{5-20}$$

Solving for the gain ($\mathbf{V}_{out} / \mathbf{V}_{in}$) leads to

$$G(f) = G(0)/\sqrt{(1+f_l^2/f^2)} \tag{5-21}$$

where $G(0) = R_f / R_1$. This is similar in form to the previous Equation 5-13 except that the frequency ratio is inverted. Here f_l is a low frequency cutoff point and is governed by the input components R_1,C_1 and the equation

$$2\pi f_l = 1/ R_1 C_1 \tag{5-22}$$

In this configuration, the op-amp circuit is AC coupled and no DC signal can pass. Only AC signals with a frequency greater than the low frequency cutoff point will be amplified fully. Note at $f = f_l$, the gain has fallen to 1/2 or –3 dB. The filter bandwidth is now ($f_u - f_l$) where f_u is the closed loop gain upper cutoff frequency.

LabVIEW Demo 5.2: Simple High Pass Filter

Load the program called HighPass.vi from the chapter 5 program library. Click on the **Run** button to see the Bode plot. Investigate the position of the low frequency cutoff point as the input capacitor or resistor is varied. Note also the response curve when the gain G(0) is changed by varying R_1or R_f. For convenience the open loop curve with A(0) = 100 dB and an open loop cutoff frequency at 10 Hertz is also shown.

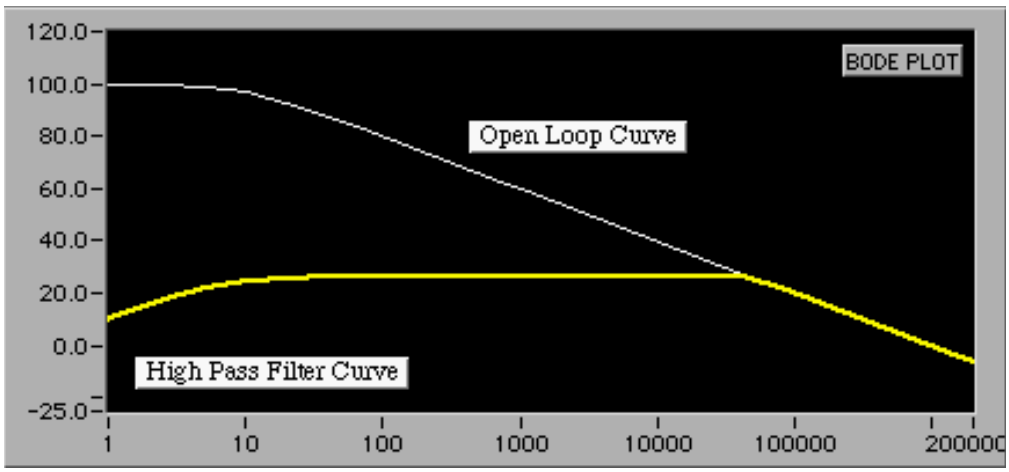


Figure 5-4. Bode Plot of an Op-Amp High Filter

All frequencies greater than f_l have a constant gain (up to the open loop cutoff) while all frequencies less than f_l are attenuated. A filter which displays this property is called a high pass filter. For low frequencies, the response curve rolls off with a slope of 20 dB/decade. What is happening here?

Look at the input network impedance in the limits where $f < f_l$ and $f > f_l$. Calculating $\mathbf{Z}_1$ or using the LabVIEW vector calculator show that in the limit of

$$\text{low frequencies} \quad (f < f_l),\ \mathbf{Z}_1 \rightarrow 1/j2\pi f C_1 \tag{5-23}$$

$$\text{high frequencies} \quad (f > f_l),\ \mathbf{Z}_1 \rightarrow R_1 \tag{5-24}$$

At low frequencies, the reactance of the capacitor is so large that current is strongly attenuated and the gain ($j2\pi f R_f C_f$) increases linearly with frequency up to f_l. On the Bode plot, this region is a straight line with a positive slope of 20 dB/decade. At high frequencies, the capacitor reactance is low and the current readily flows through the input capacitor. The gain acts as if there were no capacitor in the input loop and the gain is constant (R_f / R_1) up to the open loop frequency response curve.

What happens when a triangular waveform is applied to a high pass filter in the region where the gain is frequency dependent? That's right, the output is a square wave. The harmonic components of the triangular wave are strongly modified so that the input signal is differentiated. AC differentiators find extensive use in analog computation circuits and noise suppression circuits.

Bandpass Filter

A bandpass filter passes all frequencies between two cutoff points at a low and a high frequency. An ideal bandpass filter would be infinity sharp at the cutoff points and flat between the two points. Real bandpass filters with names like Chebyshev, Butterworth and Elliptic come close to the ideal but never quite make it. A simple bandpass filter can be made by combining the simple high pass and low pass circuit of the previous sections.

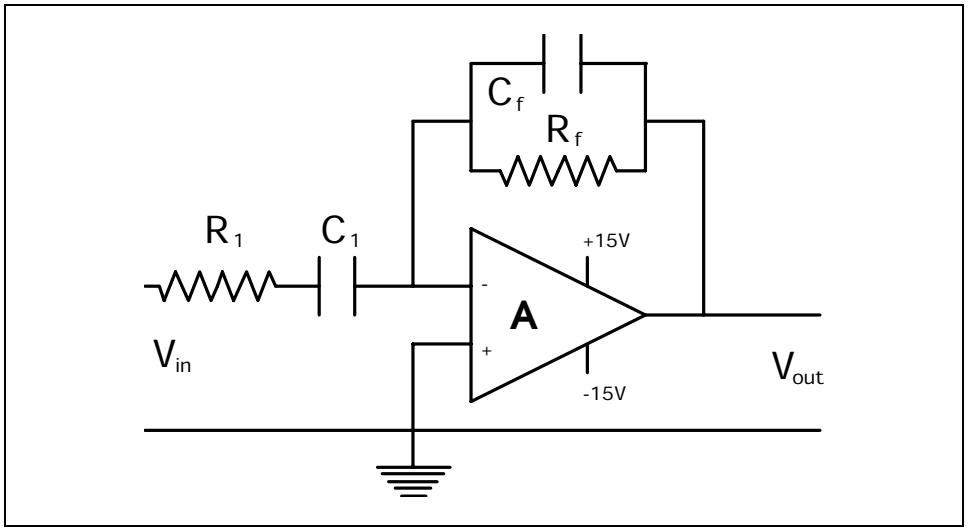


Figure 5-5. Schematic Diagram of a Op-Amp Bandpass Filter

Both the input and feedback loop impedances are now complex and the gain is

$$G(f) = |\mathbf{Z}_f / \mathbf{Z}_1| \tag{5-25}$$

Solving this gives the frequency dependent gain

$$G(f) = G(0) / [\sqrt{(1+f_l^2/f^2)}][\sqrt{(1+f^2/f_u^2)}] \tag{5-26}$$

with a low frequency cutoff point f_l (Equation 5-22) and a high frequency cutoff point f_u (Equation 5-15). The bandwidth of the band pass filter is given from the intersection points of the –3 dB line with G(f) or simply $BW = (f_u - f_l)$.

LabVIEW Demo 5.3: Simple Band Pass Filter

Load the program called BandPass.vi from the chapter 5 program library. Click on the **Run** button to see the Bode plot. Investigate the shape of the band pass filter curve when the key components R_1, C_1, R_f or C_f are varied. For convenience the open loop curve with A(0) = 100 dB and an open loop cutoff frequency at 10 Hertz is also shown.

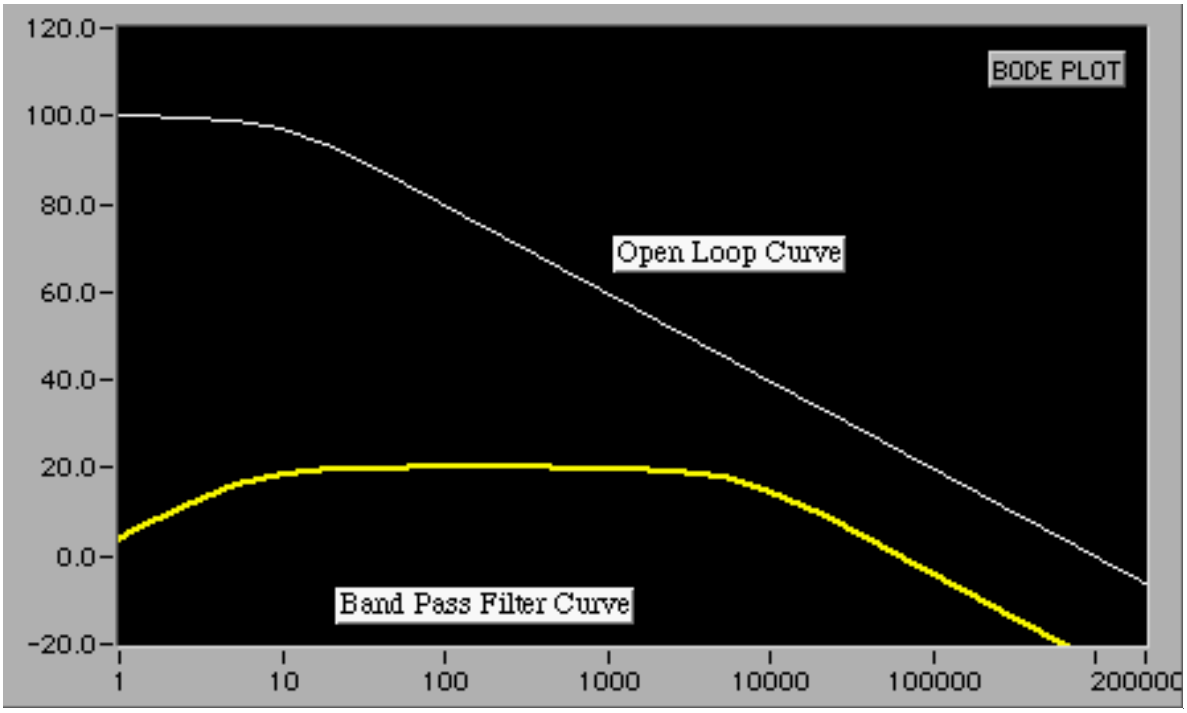


Figure 5-6. Bode Plot of an Op-Amp Bandpass Filter

What shape does the bandpass filter response curve take when $f_u = f_l$?

Such a curve selects one frequency above all the others.

LabVIEW Challenge

What happens when a square wave is used as the source waveform V_{in} for a low pass filter?

A square wave is made up of a fundamental sine wave at frequency f and higher odd harmonics at 3f, 5f ,7f etc. The amplitudes of each frequency component are 1, 1/3, 1/5, 1/7 etc. When a square wave is applied to the filter in the region where the gain is frequency dependent, the harmonics are rapidly attenuated, so much so that the output voltage is modified or filtered into a triangular waveform.

Design a LabVIEW program which adds the fundamental and three harmonics of a square wave and is displays the resultant waveform for one complete cycle. Apply this waveform to an op-amp with a gain of 1000 and an upper cutoff frequency at the waveform fundamental frequency. What is the amplitude for each component? Add these components to see an approximation of a triangular wave

eLab Project 5

Objective

To study the frequency response of a bandpass filter and its dependence on a series capacitor in the input loop and a parallel capacitor across the feedback resistor.

Procedure

Build a real bandpass filter using the circuit shown below. With a function generator as a source of sine waves measure the frequency characteristics and determine the Bode plot.

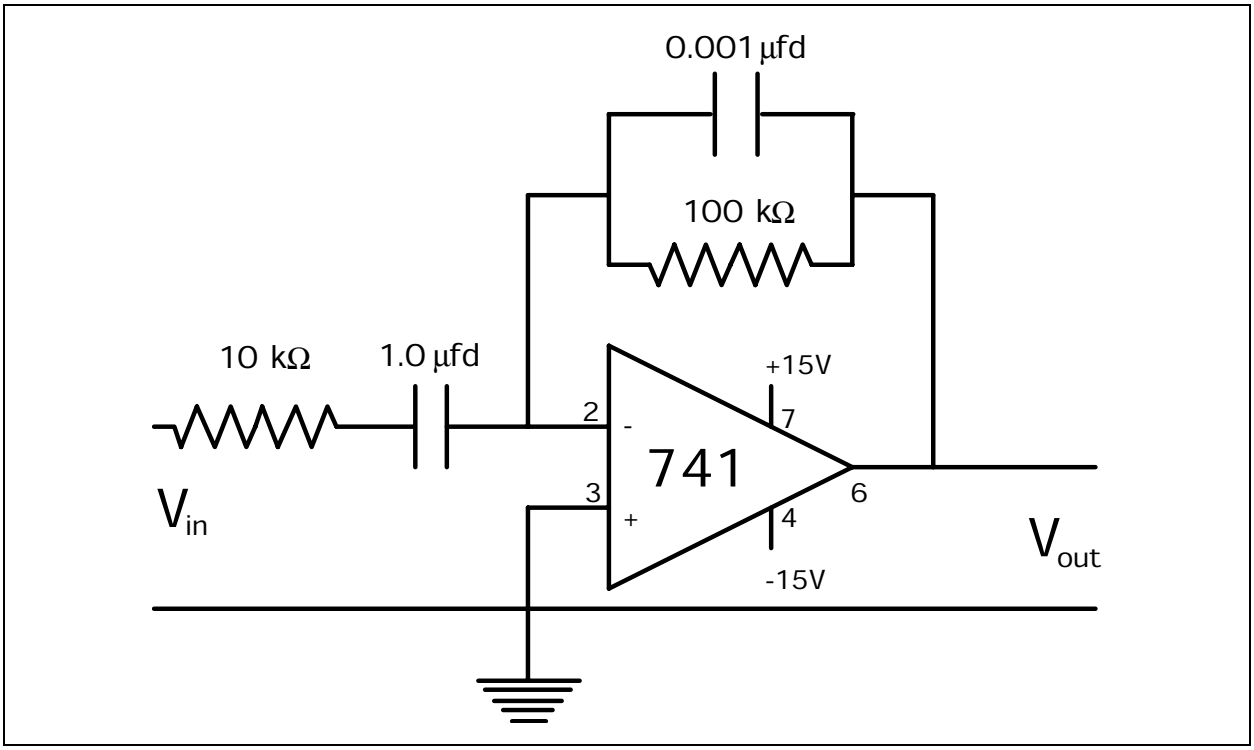


Figure 5-7. Schematic Diagram of a Bandpass Filter

The circuit requires a 741 op-amp, two resistors, two capacitors and two power supplies. Choosing $R_f = 100\ k\Omega$ and $R_1 = 10\ k\Omega$ gives the closed loop gain of 10 or 20 dB in the bandpass frequency region. Chose $C_1 = 1\mu f$ and $C_f = 0.001\mu f$. Chose a function generator set to sine wave with an

amplitude of 50 mV as the input voltage V_{in}. Component layout is shown below.

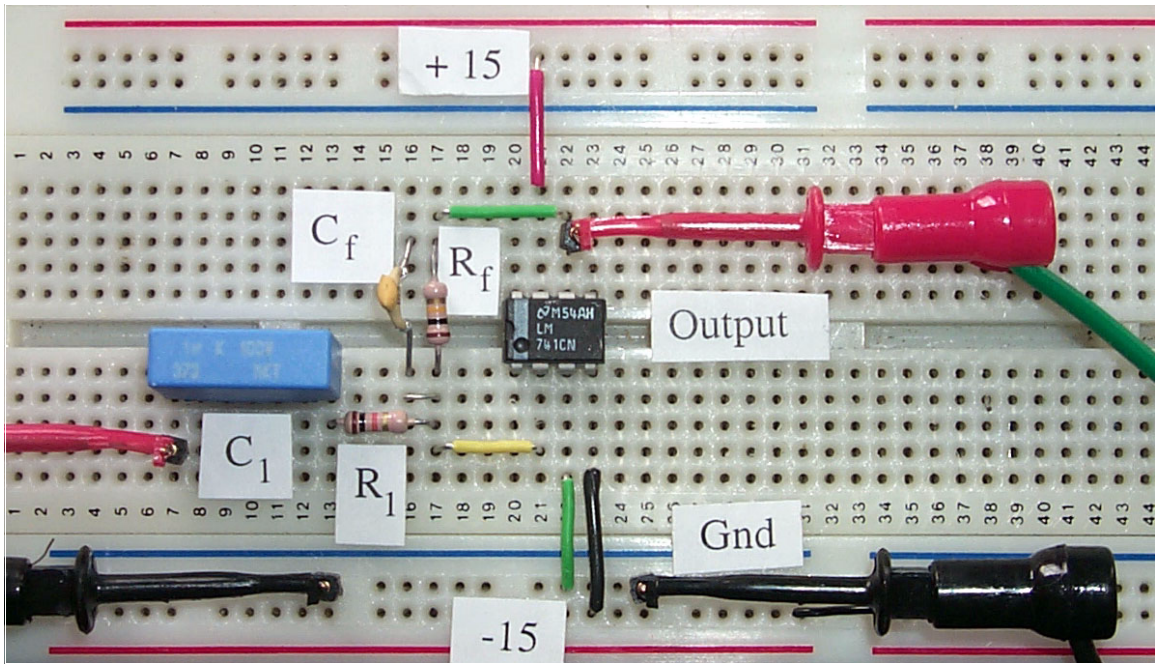


Figure 5-8. Component Layout of an Op-Amp Bandpass Filter

Use an oscilloscope or a high speed DAQ card to measure the output signal level. In all cases, it is wise to measure the input signal level and compute the gain from the expression V_{out}/V_{in}. In choosing the test frequencies, select the decade range then measure at multiples of 1, 2, 4, and 8. This gives an approximately uniform set of points on a log f scale. Graph the Bode plot, that is the gain in decibels as a function of $\log_{10}$ of the frequency.

From the key variables R_1, C_1, R_f or C_f calculate the lower and upper frequency cutoff points. How do these points compare with the actual measured –3dB points on the Bode plot?

Computer Automation 5: Response to Stimulus Signals

Computer automation is all about the automated measurement, analysis and reporting of the response of devices or systems under test. For AC stimulus, the response of interest could be the amplitude, the frequency or the phase content. In all cases, a representative sample of the signal in the form of an array is the most convenient to analyse. LabVIEW has many array VIs that enable the amplitude to be measured in units of peak, peak-peak, average or rms signal level. The frequency of sinusodial signals can be measured eloquently with frequency, period or counter VIs. The harmonic content of more complex stimulus signals can be analysed with FFT or Power

Spectrum VIs. Phase measurements require a reference signal and it is best to store the reference and response signals as an array. In this lab, we look on at sinusodial stimulus signals applied to a bandpass filter and observe the response on a LabVIEW graph.

Launch the LabVIEW program entitled Response 5.vi from the chapter 5 library. This program uses an input channel on the DAQ card to measure the circuit response signals. Connect a waveform generator sinusodial output (1volt peak signal level) to the input (pin 3) of the bandpass filter, eLab 4. Choose components so that the low frequency cutoff is about 50 Hertz. Click on **Run** to start the data collection and observe the waveform as the stimulus is varied from 1 to 100 Hertz. Adjust the stimulus frequency until the measured response is –3dB below the input level. This frequency is the low frequency cutoff point. How does it compare with the value predicted from Equation 5-21?

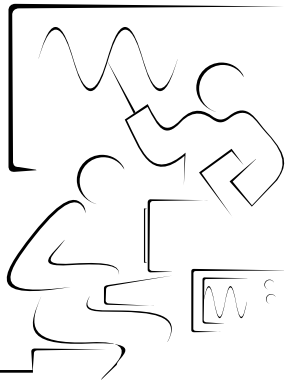
LabVIEW Enhancements

Design a LabVIEW VI to determine the peak, peak-peak or rms signal amplitude.

Replace the waveform generator with a LabVIEW generator.

Design a LabVIEW VI to automatically sweep the input frequency and determine the low frequency cutoff point.

Lab 6
The 555 Timer Chip
Astable Circuit



Introduction

The 555 IC is unique in that it simply, cheaply, and accurately serves as a free-running astable multivibrator, square-wave generator, or signal source, as well as being useful as a pulse generator and serving as a solution to many special problems. It can be used with any power supply in the range 5-18 volts, thus it is useful in many analog circuits. When connected to a 5-volt supply, the circuit is directly compatible with TTL or CMOS digital devices. The 555 timer can be used as a monostable multivibrator (one-shot), as an astable multivibrator (oscillator), as a linear voltage ramp generator, as a missing pulse detector, as a pulse width modulator and in many other applications.

Clocked digital logic devices are synchronous with an internal clock of some form. Computer and real time clocks use crystal controlled oscillators as the internal standard. Slower devices such as digital multimeters and consumer electronics often use oscillators whose timing is dependent on the charging and discharging of a simple RC network. In this lab, we look at one such device, the 555 timer chip, as a free-running (astable) oscillator.

555 Timer Chip

The astable configuration of the 555 circuit, shown below uses two resistors and a capacitor to define the oscillator frequency. The voltage across the external capacitor is measured at the trigger and threshold inputs (pins 2 and 6 respectively). Depending on the magnitude of this voltage, an internal RS flip-flop may be set or reset. This output places the circuit into a charge or discharge cycle. On charging, the capacitor voltage rises to 2/3 V_{cc} and on discharge the capacitor voltage falls to 1/3 $V_{cc.}$ At the upper limit, the threshold input turns off the internal flip-flop, and at the lower limit, the trigger input turns it on. The output voltage (pin 3) is a buffered copy of the flip-flop output and hence is a digital signal. The resulting pulse waveform defines the 555 oscillator signal.

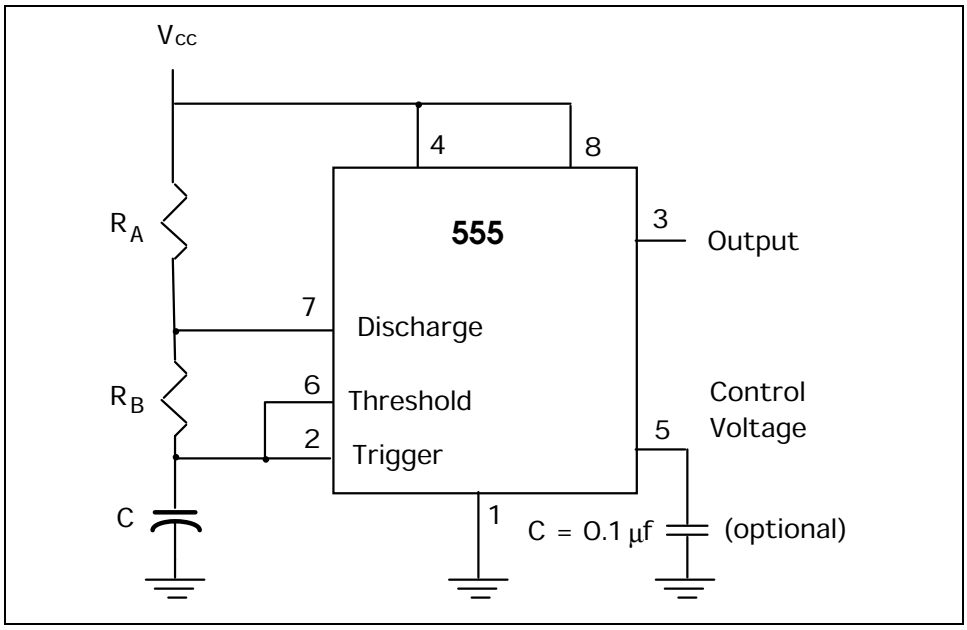


Figure 6-1. The Basic 555 Astable Circuit

The frequency of oscillation depends only on the resistor-capacitor chain (**R_A,R_B,C**) and is independent of the power supply voltage V_{cc}.

On charging, the external capacitor C charges through resistors R_A and R_B. The charging time t_1 is given by

$$t_1 = 0.693\ (R_A + R_B)\ C \qquad (6\text{-}1)$$

and this part of the cycle is signaled by a high level on the output (pin3).

On discharge, the external capacitor C discharges through the resistor R_B into pin 7 which is now connected internally to ground. The discharge time is given by

$$t_2 = 0.695\ R_B\ C \qquad (6\text{-}2)$$

and this part of the cycle is signaled by a low level on the output.

The total time for one oscillation (the period T) is given by the sum of these two times

$$T = t_1 + t_2 = 0.695(R_A + 2R_B)\ C \qquad (6\text{-}3)$$

The frequency F is given by the reciprocal of the period, or

$$F = 1.44/(R_A + 2R_B)C \qquad (6\text{-}4)$$

With the appropriate choices of external timing components, the period of the oscillation can range from microseconds to hours.

The duty cycle DC is the ratio of the time the output is low as compared to the period

$$DC = R_B/(R_A + 2R_B) \quad (6\text{-}5)$$

The duty cycle is always less than 50% or saying it another way, the off time $\mathbf{t_2}$ is always less than the on time $\mathbf{t_1}$. Thus the output of the 555 astable circuit is asymmetric. By making $\mathbf{R_B}$ large compared to $\mathbf{R_A}$, the waveform becomes more symmetric and the 555 output approaches a square wave.

LabVIEW Demo 6.1: The 555 Astable Oscillator Circuit

Load the program called 555Astable1.vi from the chapter 6 program library. Click on the **Run** button to activate the astable circuit. The output on pin 3 is a digital signal, it is either a high or low level.

Investigate how the output waveform changes with different values of $\mathbf{R_A}$, $\mathbf{R_B}$ or **C**.

Observe the output waveform and the duty cycle in the following cases:

- $\mathbf{R_A} > \mathbf{R_B}$,
- $\mathbf{R_A} < \mathbf{R_B}$,
- $\mathbf{R_A} = \mathbf{R_B}$.

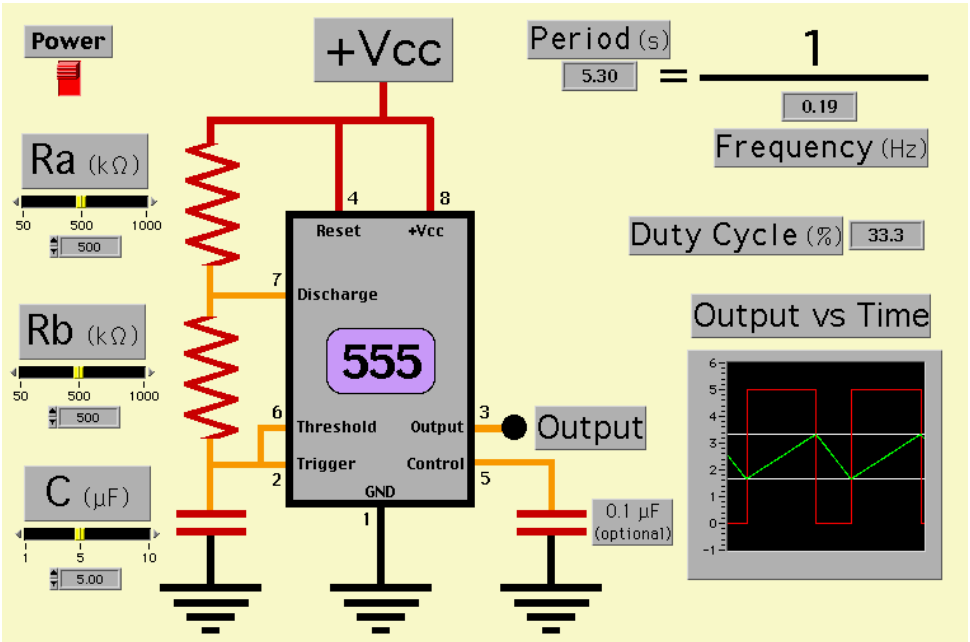


Figure 6-2. LabVIEW Simulation for a 555 Astable Circuit

A variable frequency source can be made by selecting capacitors whose values are decades (factors of ten) different from each other and a variable resistor for fine frequency tuning. In practice, $\mathbf{R_A}$ and $\mathbf{R_B}$ can have a resistance from 1 kΩ to 10 MΩ and the capacitor can range from 0.001 to 100 µf. These combinations give the 555 astable circuit truly a very wide frequency range.

How Does it Work?

The 555 timer is based on the sequential charging and discharging of the external capacitor. Two internal op-amps configured as comparators set the lower and upper voltage limits to 1/3 V_{cc} and 2/3 V_{cc}. The voltage across a capacitor at any time t is given by the expression

$$V(t) = V(0) \exp(-t/RC) \qquad (6\text{-}6)$$

where V(0) is the initial voltage and RC is a charging/discharge time constant.

LabVIEW Demo 6.2: 555 Astable Oscillator Timing Diagram

Load the program called 555Astable2.vi from the chapter 6 program library. Click on the **Run** button to activate the astable circuit. The timing diagrams for the output voltage (pin 3) and the capacitor voltage (pins 2 & 6) have been added to the front panel display.

While the output (pin 3) is high, the power supply (taken here as +5 volts) charges the capacitor through the resistors R_A and R_B and the capacitor voltage rises exponentially. When the voltage across the capacitor reaches a reference voltage of 2/3 V_{cc} (3.33 volts), the threshold comparator (at pin 6) triggers an internal flip-flop which resets the output (pin 3) low and starts the discharge cycle. The voltage at the upper limit is

$$3.33 = 1.67 \exp(-t_1/[R_A + R_B]C) \qquad (6\text{-}7)$$

Solving for t_1 in Equation 6-1 yields the time interval that the capacitor is charging. The timing diagram shows the charging cycle (green trace - capacitor voltage) as a positive ramp when the astable output (red trace - output pin 3) is at the high level. The two comparator limits 1/3 V_{cc} and 2/3 V_{cc} are shown as horizontal lines (white traces).

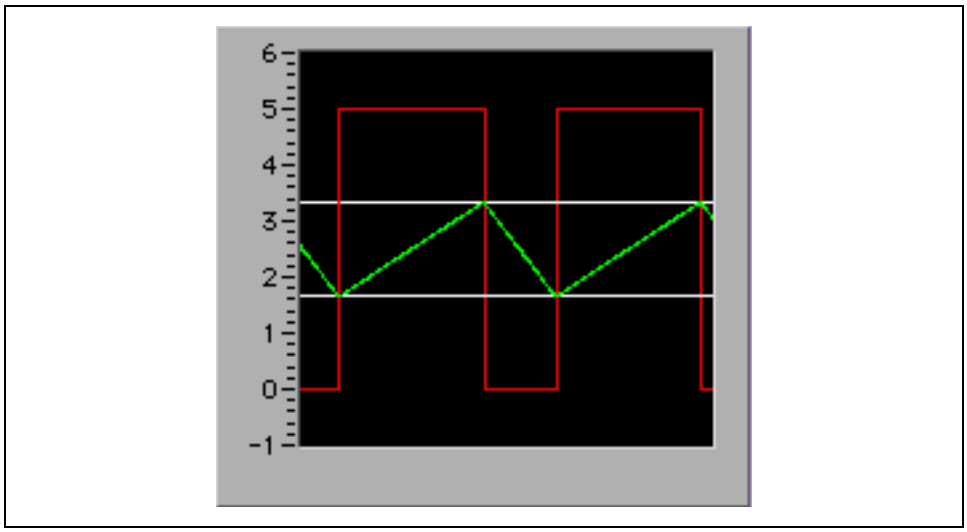


Figure 6-3. LabVIEW Display of the Charge and Discharge Cycles for a 555 Astable Circuit

When the capacitor voltage reaches the upper reference limit, the power supply is effectively removed from the capacitor circuit and pin 7 becomes internally connected to ground. The capacitor is allowed to discharge through the single resistor R_B. The discharge voltage at the lower limit is

$$1.67 = 3.33 \exp(-t_2/R_BC) \quad (6\text{-}8)$$

where t_2 is the discharge time constant. In the discharge cycle, the capacitor voltage ramps down (green trace) to the lower limit (1/3 V_{cc}). At this point the trigger comparator (pin 2) sets the flip-flop back to its high state and the cycle repeats.

LED Flasher

A flashing alert signal can be generated by driving a light emitting LED diode with a 555 astable circuit. The output (pin 3) is capable of sourcing a few milliamps or sinking up to 200 milliamps, more than enough current to brightly illuminate any light emitting diode.

LabVIEW Demo 5: The 555 LED Flasher Circuit

Load the program called 555Flasher.vi from the chapter 6 program library. A LED has been added to pin 3 and pulled up to V_{cc} through a series resistor. Click on **Run** to observe the LED flashing. A logic probe has also been added to pin 3. Whenever the output is high, it is red and whenever the output is low, it is black. The LED has the opposite state. Whenever the output is high, it is gray (off) and whenever the output is low, it is yellow (on). The output timing diagram and a frequency counter have also been added to the circuit.

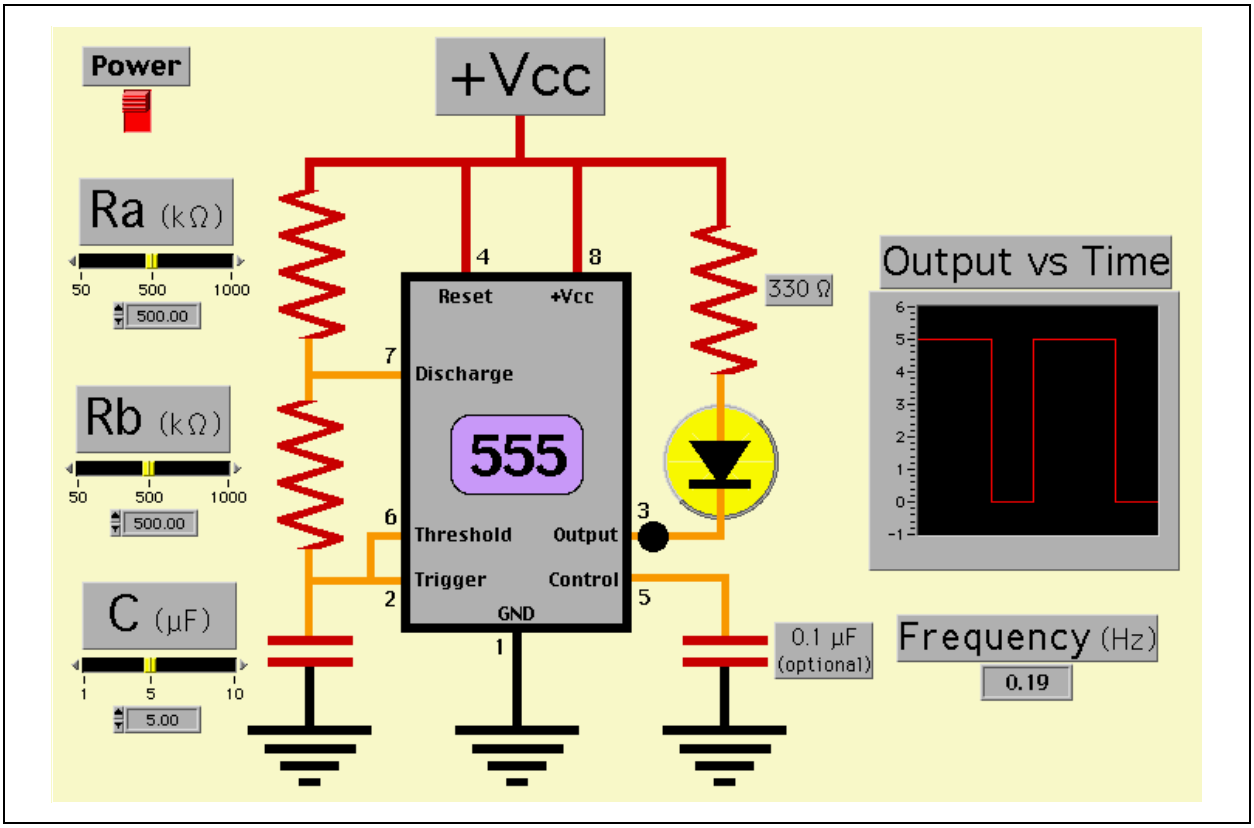


Figure 6-4. LabVIEW Simulation for a 555 LED Flasher Circuit

When the output (pin 3) is high, there is not enough voltage drop across the resistor and LED to turn the LED on. However when the output is low, current can flow through the LED (which is now forward biased) and into the output (pin 3) and out the ground lead (pin 1). The purpose of the resistor is to limit or to set the current when the LED is on. This resistor determines the brightness of the LED. Since the forward voltage across a silicon diode is 0.6 volts, and if the power supply is 5 volts, then (5 - 0.6) = 4.4 volts will be across the resistor. For a forward bias current of 13.3 ma (red LED brightly lit), the resistor should be about 330Ω.

Temperature Transducer

A transducer is an electronic circuit which converts a physical parameter such as temperature into an electrical signal so that it can be measured by conventional techniques. In this virtual experiment, a thermistor is used to convert temperature into a waveform whose off-time is directly proportional to temperature.

A thermistor is a device whose resistance is dependent on the device temperature. Thermistors are manufactured from semiconducting materials which accounts for their unusual conductivity.

Thermistors have three unique properties;

- The sensitivity or the change in resistance per degree Centigrade is large.
- The resistance decreases with increasing temperature (a negative temperature coefficient).
- The resistance has a nonlinear exponential response curve (often over six decades).

LabVIEW Demo 5: Temperature Transducer

Load the program called Thermometer.vi from the chapter 6 program library. A thermistor labeled Rb has been placed into a beaker of water. A gas burner controlled by a rotary valve allows you to heat the water to a known themperature. A thermometer has been added to the beaker to measure this temperature and it can be used to calibrate the thermistor. The thermistor replaces the resistor R_B in the 555 astable circuit. When run, the waveform will be displayed on an *Output vs Time* chart. By clicking and dragging the cursors, you can place the cursors on the appropriate transition to measure a time interval $\Delta t = t_2 - t_1$. You can measure the on-time, the off-time or the period. Activate the experiment by clicking on the **Run** button. Watch the waveform change as the liquid is heated or cooled by changing the gas flow.

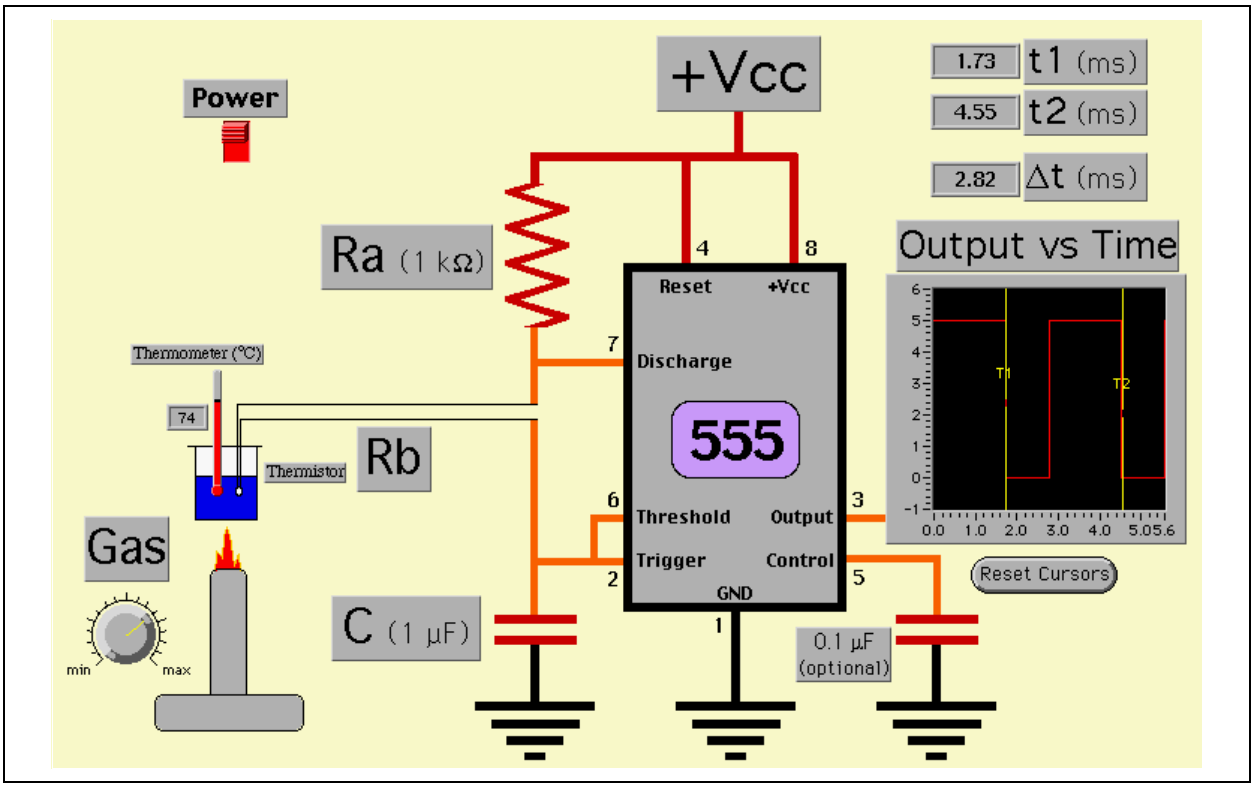


Figure 6-5. LabVIEW Simulation to Measure the Heating or Cooling Curve of Water

To measure the off-time, click and drag the cursor T1 to a falling edge and T2 to the adjacent rising edge such that T2>T1 and read the time from Δt indicator display.

Plot a graph of off-time of the thermistor circuit versus temperature as measured by the thermometer. Is this graph linear or nonlinear? Using Equation 6-2 and other component values (given in the above diagram), calculate the resistance of the thermistor for each temperature measurement.

LabVIEW Exercise

Plot a graph of the thermistor resistance versus temperature for this sensor to reveal the unique properties of a thermistor.

eLab Project 6

Objective

To study the waveforms from a 555 astable oscillator and its frequency, period and duty cycle dependence on a external chain of resistors and a capacitor.

Procedure

Build a LED flasher based on the circuit of Figure 6-1. Connect a 330 Ω resistor and red LED to the output (pin 3). Set R_A = 3.3 kΩ, R_B = 33 kΩ and C = 0.1 μF. The IC pinout and components can also be seen on the front panel of the program 555Flasher.vi, Figure 6-4. The component layout is shown below.

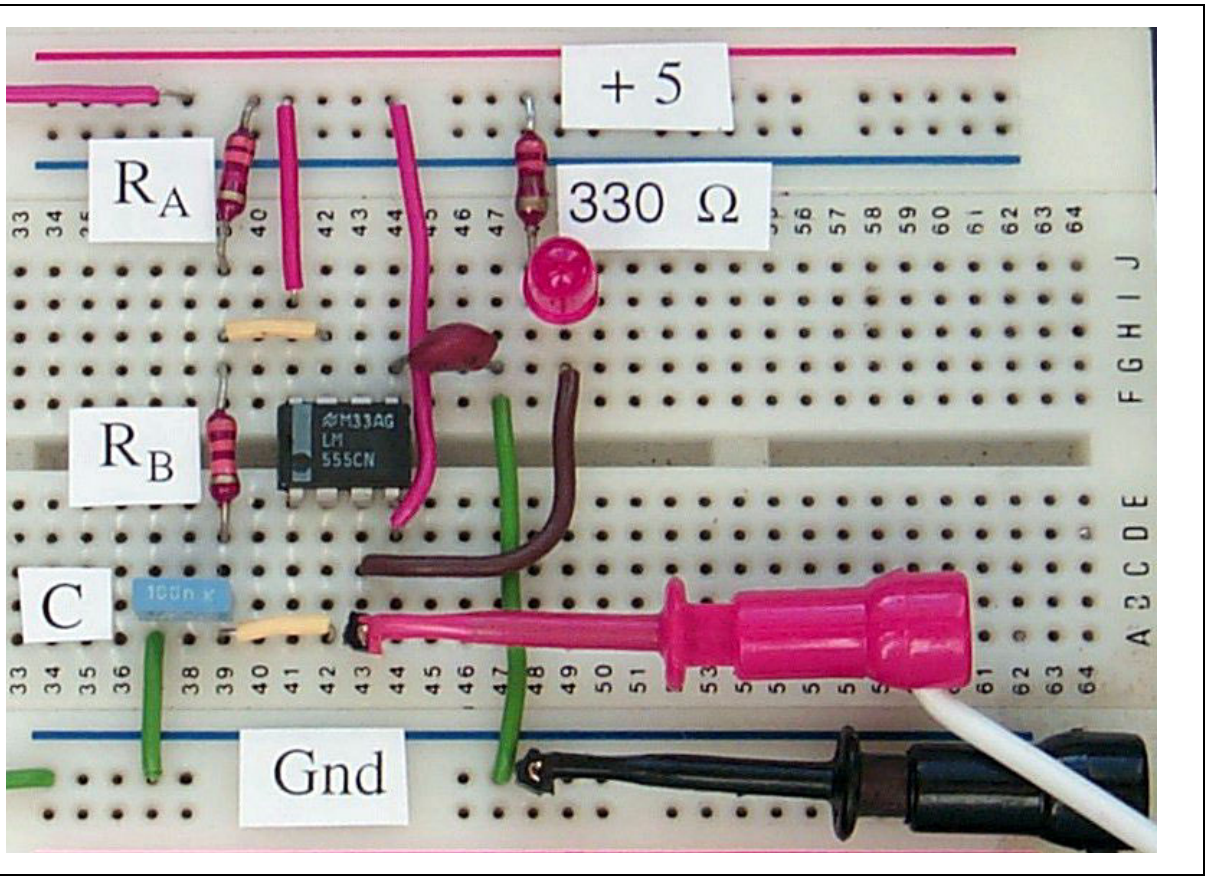


Figure 6-6. Component Layout of a LED Flasher Circuit Using the 555 Timer IC

Measure R_A, R_B and C separately before adding them into the circuit. Use Equations 6-3 though 6-5 to predict the oscillation period, the frequency and the duty cycle. Measure these same quantities on the output (pin 3) of the 555 IC. How close do the measured parameters agree with the calculated values?

Describe the appearance of the LED light.

Replace the 0.1 µF capacitor with a 1 µF capacitor and now describe the appearance of the LED light.

Computer Automation 6: Digital Signals

For digital signals, the amplitude is a constant and all information is carried in the time response be it frequency, period or duty cycle. In this lab, we will measure the digital frequency produced by a 555 timer chip driven from a +5 volt power supply. Use the eLab project 6 as the starting circuit. As in the eLab 6, choose R_A = 3.3 kΩ, R_B = 33 kΩ and C = 0.1 µF. Remove the LED from the circuit.

Launch the LabVIEW program entitled FrequencyLow.vi from the chapter 6 library. This program uses three internal counters on the DAQ card to measure TTL level digital signals in the frequency range f< 1 kHz. Ensure that the counters are connected externally as indicated on the front panel diagram.

Note That an external 7404 hex inverter chip is also required.

Connect the 555 output (pin 3) to the Counter2 input on the DAQ card.

Click on **Run** to make a frequency measurement. Verify that the measured frequency agrees with your frequency prediction based on the component values of R_A, R_B and C.

Circuit Enhancements

Replace the resistor with a variable resistor in the range 10–100 kΩ, and investigate the changes in frequency as the resistor is adjusted.

Replace the resistor with a thermistor or a photoresistor and investigate the changes in frequency with temperature or light intensity.

LabVIEW Enhancements

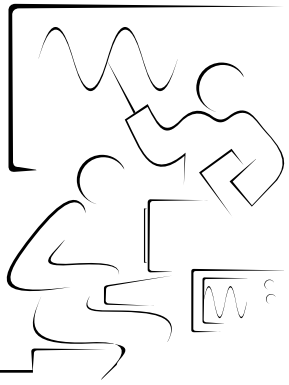
For frequencies greater the 1 kHz, a different VI is used.

Check your LabVIEW/examples/daq/counter library for a Vi called (Measure Frequency >1kHz.vi).

Note that different DAQ cards may use different timers.

Ensure you are using the correct library; 8253.llb or AMD9513.llb or DAQ-STC.llb.

Lab 7
The 555 Timer Chip Monostable Circuit



The 555 timer chip introduced in the last lab was configured as a free running astable multivibrator or oscillator. A different circuit allows the 555 timer chip to be configured as a monostable multivibrator or single pulse generator. In this configuration, the IC waits patiently for a trigger pulse which when received causes the output to change state for a fixed period of time related to an external capacitor and resistor, before returning to its initial state. The ability of the monostable to generate a single pulse of precise length is often referred to as a "one shot" circuit element. Many times in digital electronics, a precise delay is required to allow events to be measured, data be displayed for a specific period of time or allow a timing pulse to catch up in order to synchronize events with the clock signal. The 555 monostable is a good solution.

One-shots are circuits that generate a fixed-length output pulse after receiving an appropriate trigger signal. The length of the output pulse is generally determined by the charging of a capacitor through an external resistor. A trigger or start signal sets the output on and initiates the charging cycle. When the voltage on the capacitor reaches an upper threshold level of two thirds of the supply voltage, the output is turned off and the capacitor voltage returns immediately to the initial voltage, zero. The circuit is now ready for another trigger pulse.

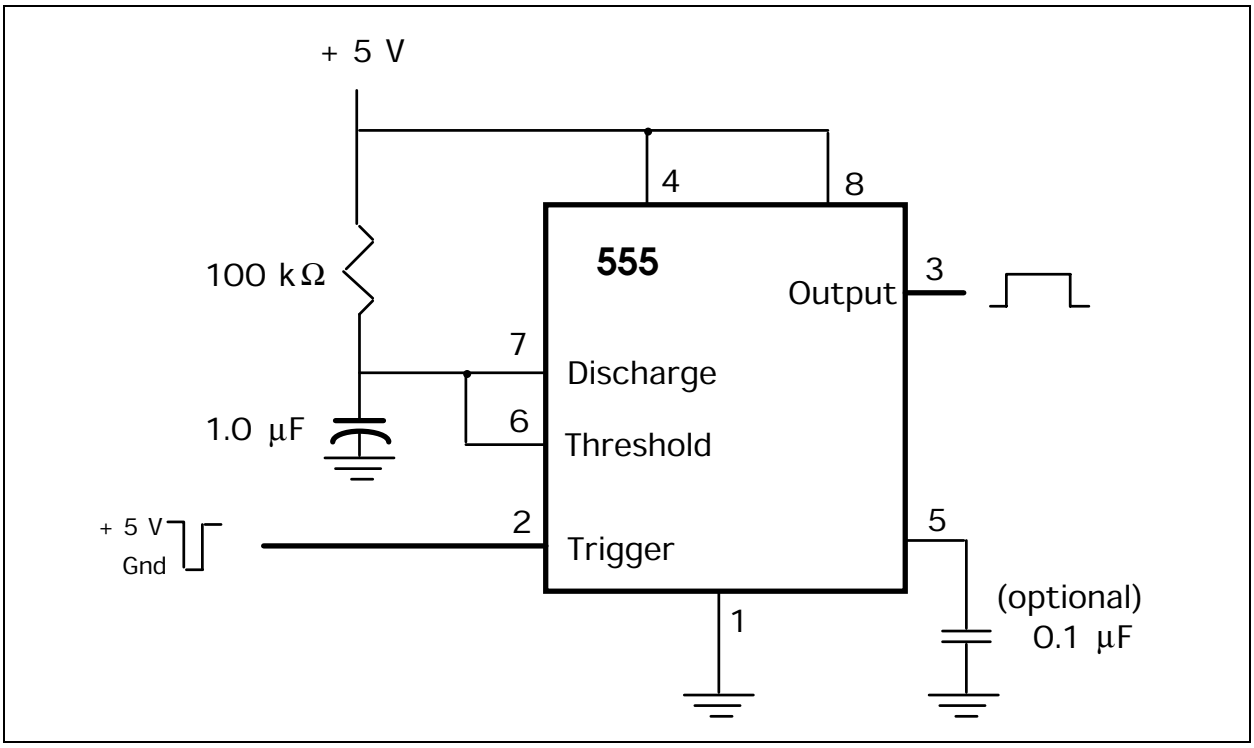


Figure 7-1. The 555 Timer IC Configured as a Monostable Circuit

The monostable arrangement of components requires only a single resistor and capacitor. The voltage across the capacitor is sampled on pins 6 and 7. A negative trigger pulse on pin 2 sends the output (pin 3) high for a time determined by the resistor and capacitor network. When the capacitor voltage reaches the threshold (2/3 V_{cc}), the output goes low. The on-time T_{on} is given by

$$T_{on} = 1.1\ R\ C. \qquad (7\text{-}1)$$

LabVIEW Simulation: Operation of the 555 Monostable Circuit

Load the program called Monostable1.vi from the chapter 7 program library. Activate the circuit by clicking on the **Run** button. Click on the trigger switch to fire the monostable. Investigate the on-time by changing the external resistor and capacitor values.

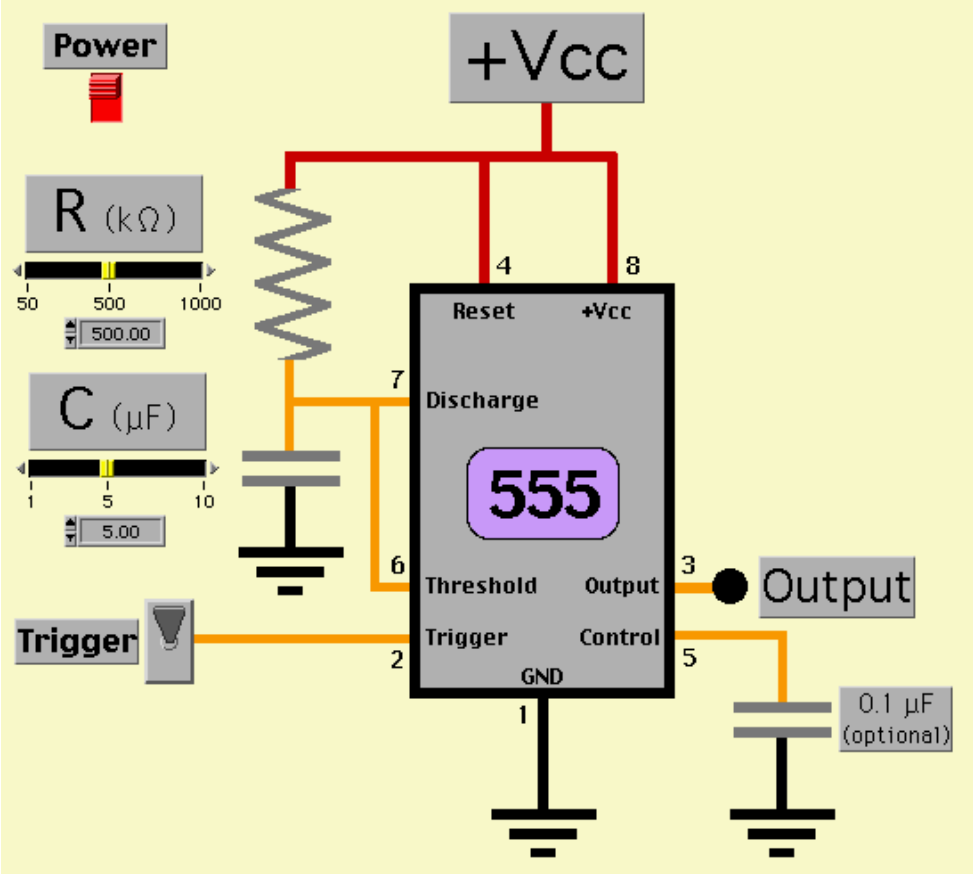


Figure 7-2. LabVIEW Simulation of a 555 Monostable Circuit

In general, the resistor can range from 1KΩ to 3.3MΩ and the capacitor from 500 pf to 10 µF. Thus the on-time can range from microseconds to hours.

The trigger input is normally high and momentarily bringing it low generates the trigger signal. It is important to remember that the trigger input must be brought high again after the triggered low state. For the 555 timer chip, the trigger pulse must be negative and narrower than T_{on}. Good design calls for a trigger pulse length about 1/4 T_{on} but shorter times often work well.

A graph of *Output vs Time* Figure 7-3 displays the operation of the monostable more clearly. On triggering, the output pulse (shown in red trace) jumps to the high (positive) state and an internal transistor switch (at pin 7) opens to allow the capacitor to charge. The power supply charges the capacitor through the external resistor. The capacitor voltage (green trace) increases “linearly” from 0 volts to 2/3 Vcc (yellow trace). At this point, the threshold comparator flips state and the internal transistor switch

is closed forcing the capacitor to discharge and the output to return immediately to zero volts.

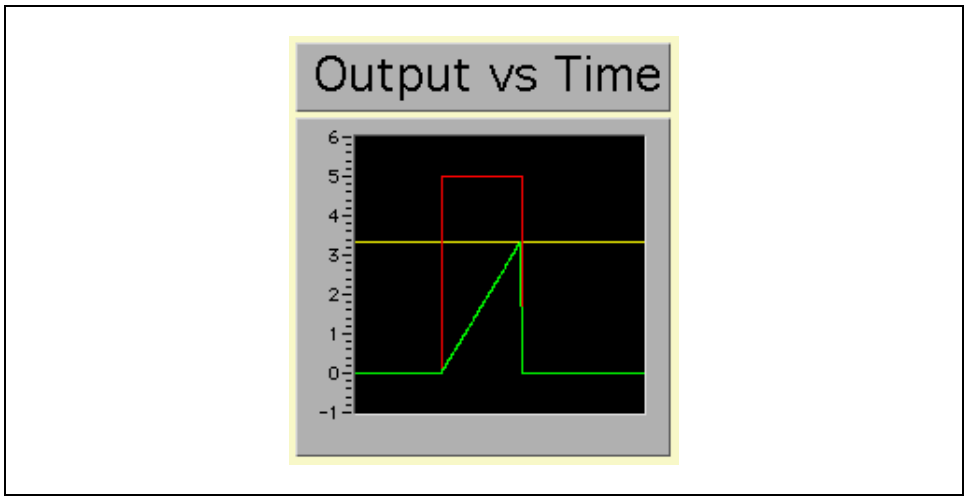


Figure 7-3. LabVIEW Display of the 555 Timing Voltages

In the simulation, 5 volts was chosen for the supply voltage so that the output is compatible with standard TTL digital chips. However the chip can be run at any voltage from 5 to 18 volts.

LabVIEW Simulation: Triggered LED Alarm

Load the program called Alarm.vi from the chapter 6 program library. A light emitting diode has been added to the output of the 555 monostable circuit. Watch the LED turn on and off, when triggered by clicking on the switch. See the output voltage change and measure the on-time. After activating the circuit with the **Run** button, click on the trigger switch to generate a single pulse.

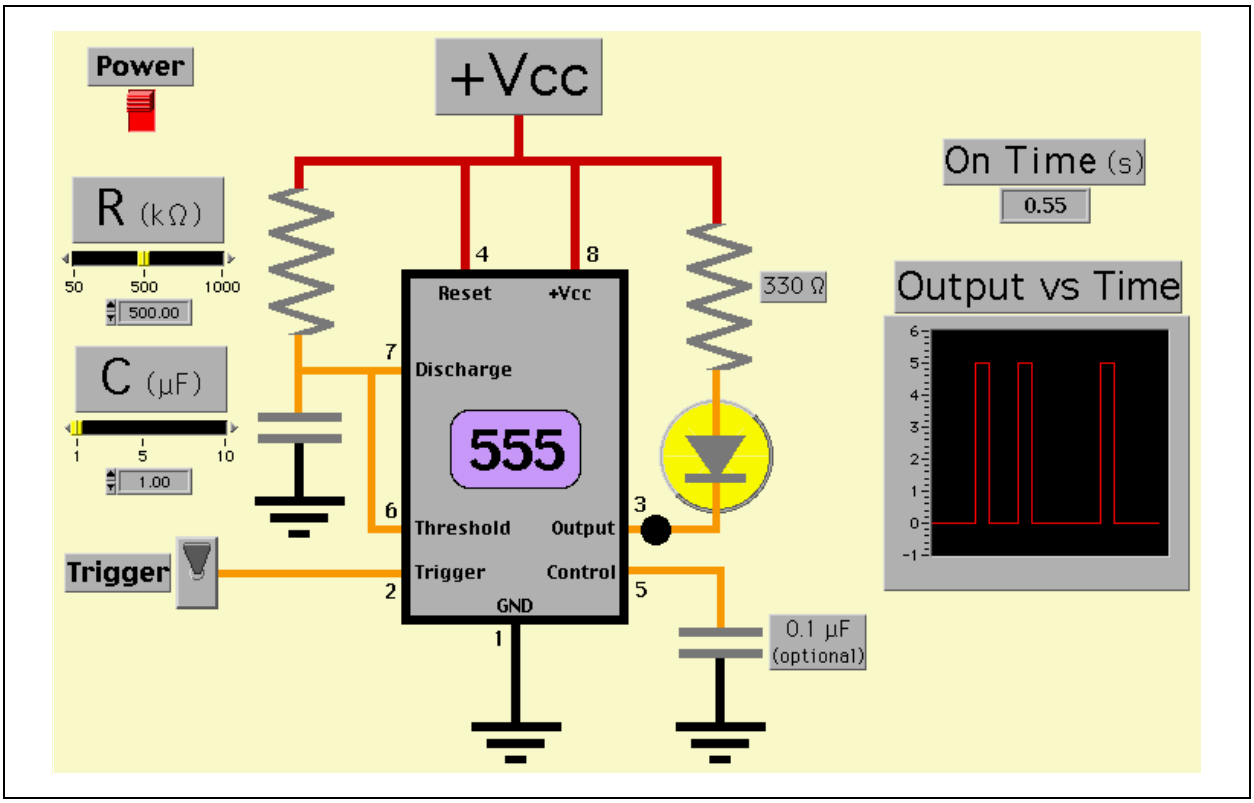


Figure 7-4. LabVIEW Simulation of a 555 Monostable with a LED Output

The LED is pulled high through a 330 Ω resistor whose magnitude was chosen to limit the current flowing through the LED. In the normal state, the output (pin 3) is low and current will pass through the LED and it will be on. When the output goes high, the LED turns off. A logic probe on pin 3 demonstrates the signal inversion of the LED pulled high.

Photoresistor Sensor

The resistance of a few semiconductors is strongly dependent on the amount of light impinging on the material. For these semiconductors, the energy gap is small enough so the photon energy can excite free carriers across the gap. The result is that current flowing through the sensor can be dramatically altered. The resistance of a typical photoresistor can change by six decades (1:1,ooo,ooo) in going from moonlight to sunlight. The resistance in absence of light, the so-called dark resistance is often in the megaohm region. As the light intensity increases, the resistance falls exponentially. In bright light the resistance is small, a few kilo-ohms or less. A plot of the device resistance versus light intensity displays an exponential variation. Plotting the device resistance as a function of the log of the light intensity displays a linear graph. On a logarithmic scale, the light intensity is measured in units of lux. Zero lux is no light while 10 lux corresponds to a bright flashlight beam. Cadmium selenide, a photoresistance material, has a

wavelength or colour response close to that of the human eye. The eye is most responsive to the yellow. These devices make good photometers in photography applications.

LabVIEW Simulation: Photometer

Load the program called Photometer.vi from the chapter 6 program library. In this simulation, one explores the 555 monostable as a light transducer (light is converted into a time interval). Recall that the on-time is directly proportional to the magnitude of the external resistor and capacitor. The charging resistor is replaced with a photoresistor. The on-time (1.1RC) is then a measure of the input light intensity . In this demonstration, the light intensity can be varied from 0 to 10 lux. Investigate the relationship between light intensity and T_{on}. Click and drag the **Light Intensity** vertical slider marker. To make a measurement click on the **Trigger** switch.

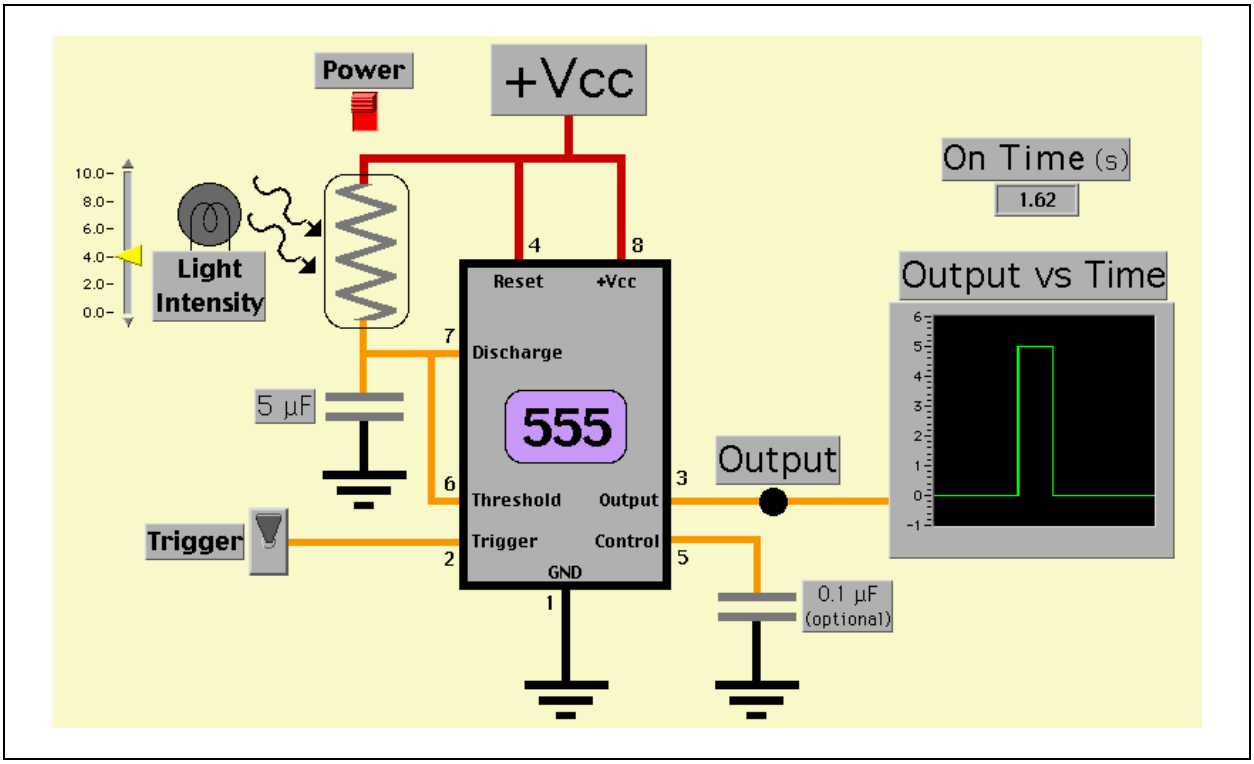


Figure 7-5. LabVIEW Simulation - Monostable Circuit to Measure Light Intensity

LabVIEW Exercise

Plot a graph of the photoresistance as measured from T_{on} versus the light intensity on a linear scale.

Hint: Convert the lux scale into a linear scale.

LabVIEW Simulation: Angular Displacement Transducer

In the early days of consumer electronics, the Apple II microcomputer used a 555 timer chip to read angular position of a game paddle. Whenever the software instruction INP(0) or INP(1) was executed, a 555 timer chip on input 0 or input 1 was triggered. An internal capacitor with an external potentiometer in the game paddle was used to read the angular position of the game paddle knob. When the paddle was rotated, a new resistance was set. The on-time of the 555 output was measured by counting the number of instruction cycles from the start of the trigger pulse until the monostable returned to the off state. The span was scaled from 0 at one end to 255 at the other end. The angular resolution was approximately one degree per count. It was used to play numerous computer games. With a game paddle on each input, two could play games or the paddles could be used together to plot points on an XY graph such as “Etch-a-Sketch”.

LabVIEW Simulation: X-Y Joystick

Load the program called XYJoystick.vi from the chapter 7 program library. The two game paddles are simulated using two LabVIEW virtual slide wires. Moving the slide causes a change in the resistance. For fixed capacitors, the on-time is then directly proportional to the resistance or angular rotation of the virtual knob. Two identical circuits have been provided so that both the X and Y motion of a cursor can be controlled. Note the variation in T_{on} for each channel as the slides are moved. The on-time is scaled to produce a number from 0 to 255.

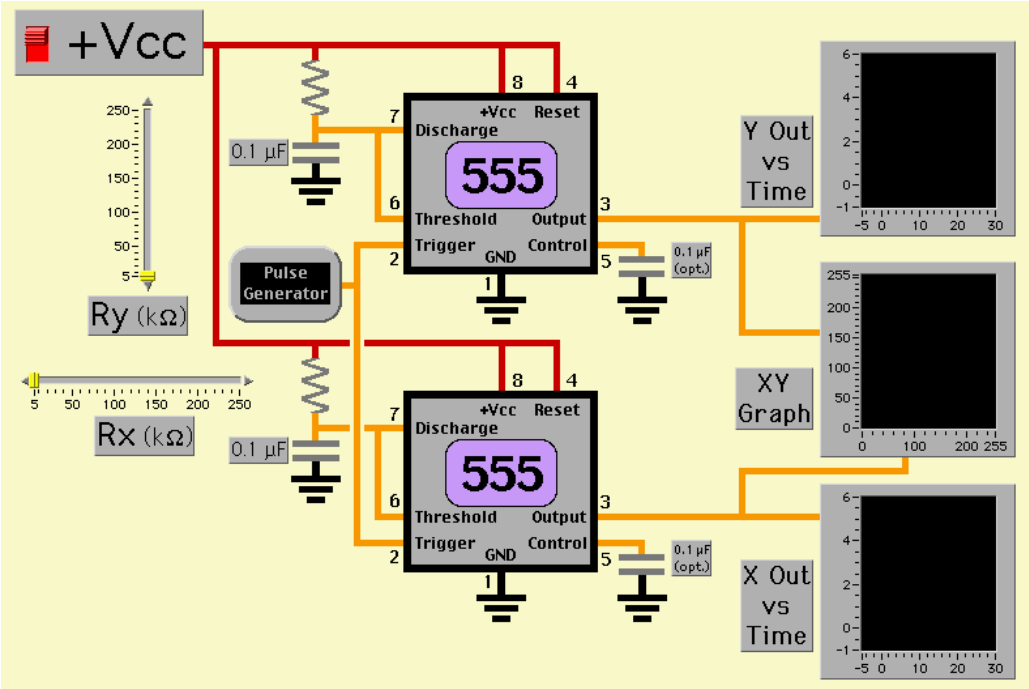


Figure 7-6. LabVIEW Simulation - Joy Stick Operation Using a Monostable Circuit

LabVIEW Challenge: Capacitance Meter

Design a LabVIEW simulation to demonstrate how a 555 Timer IC can be used to measure capacitance.

Hint: The capacitance is unknown and can vary over many decades. Choose a series of resistors of the same mantissa but different multipliers. For example: 1 kW, 10 kW, 100kW etc.

eLab Project 7

Objective

To study the application of a 555 Timer IC in a triggered alarm circuit.

Procedure

Build a monostable circuit based on the front panel Alarm.vi, Figure 7-4. Connect a 330 Ω resistor and red LED to the output (pin 3). Set R = 5.0 MΩ and C = 1.0 µF. A pushbutton is used as the triggering device. Each time the trigger is pushed, the output (pin3) goes low for a specific period of time. In order to invert the 555 output, a TTL buffer chip 7406 has been added. Its output (pin 4) now only goes high when the switch is triggered and stays high for the time set by the monostable circuit. The component layout is shown below.

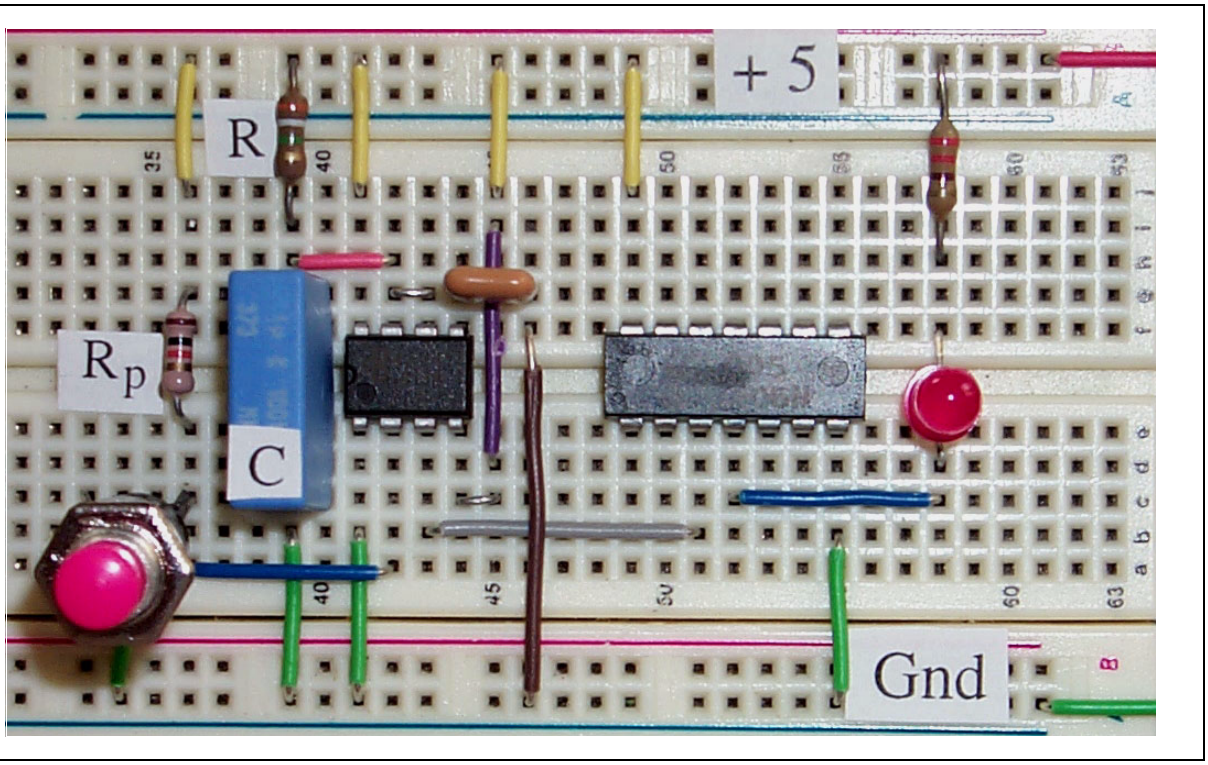


Figure 7-7. Component Layout for Triggered Alarm 555 Timer Circuit

Note 7406 is a TTL Hex Inverting Buffer, Input for Inverter No.2 is pin 3, Output for Inverter No.2 is pin 4, Power +5 volts is pin 14 and Gnd is pin 7.

Computer Automation 7: Measuring Time Interval

The monostable circuit of eLab 7 produces a pulse of fixed length each time the 555 timer IC is triggered. All information about the circuit is contained within the pulse length. In this lab on computer automation, a time interval counter is used to measure the pulse width.

Launch the program Pulse Width.vi from the program library of chapter 7. Note that some connections are require on the output of the DAQ card. Connect the output of counter0 to the clk or source input of counter1. Connect the output pin 3 of the 555 timer chip to the gate input of counter1.

Note If you are using the DAQ card with the AMD9513 or DAQ-STC counter/timer chip, then use Measure Long Pulse Width.vi from the AMD9513.llb or DAQ-STC.llb library.

Pulse Width.vi has a variable time limit. If during this time a pulse is detected then the pulse width is measured and the VI stops. If no pulse is detected, the VI stops after the time limit and a Boolean LED display is lite. Set the time limit to at least 10 seconds.

Run the program by clicking on the **Run** button. With the program running, now generate a trigger signal by momentarily pressing on the push button of eLab 7. Pulse Width.vi will report on a front panel, the width of the pulse generated by the 555 monostable circuit. Observe how the measurement accuracy depends on the timebase.

Circuit Enhancements

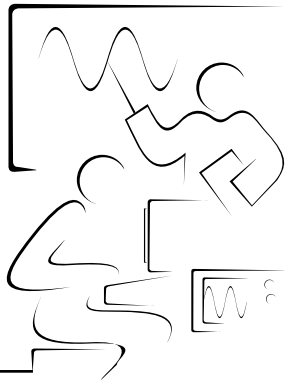
Replace the resistor with a variable resistor in the range 10–100 kΩ, and investigate the changes in pulse width as the resistance is changed.

Replace the capacitor with a variable capacitor in the range 0.05–1 μf, and investigate the changes in pulse width as the capacitance is changed.

LabVIEW Enhancements

Design a LabVIEW program which continuously monitor the 555 monostable circuit and reports the pulse width of each pulse generated by the trigger signal.

Lab 8
Voltage-to-Frequency Converters



Historically, voltage-to-frequency (V-F) converters were used as the input stage for digital recorders. A slowly varying input analog signal was converted into a frequency, then recorded on a conventional magnetic tape recorder. This combination provided a high precision analog recorder, whose output was a digital frequency. More recently, V-F converters are found in the front end of inexpensive digital voltmeters, and other low cost analog-to-digital circuits. The classic 555-timer chip studied in Lab 7 is a form of voltage-to-frequency converter.

The heart of a V-F converter is an integrating op-amp circuit. The input voltage is connected to the integrator, which ramps up to a preset voltage level. At this upper limit, the input is replaced with a reference input, but of the opposite polarity, and the output integrates down to a lower limit. At this limit, the reference input is removed and the input signal reconnected. The oscillator cycle begins again. The output is a logic low during signal integration, and a logic high during reference integration. The resulting waveform has constant on-time, and a variable off-time, proportional to the magnitude of the input signal. The output frequency is proportional to the input signal level.

A V-F converter consists of four fundamental op-amp building blocks: an electronic switch, an integrator, a comparator and a monostable.

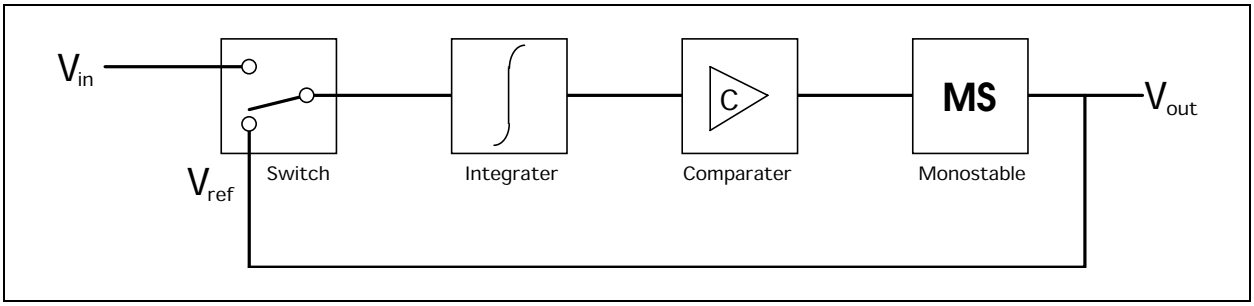


Figure 8-1. The Building Blocks of a Voltage-to-Frequency Converter

The input V_{in} (usually a negative voltage) is ramped up by the op-amp integrator. In the following example, the output rises from –10 volts towards 0 volts. An op-amp comparator is referenced at zero volts. At this upper limit, it switches from a high state to a low state creating a negative going pulse used to trigger the next stage, a 555 monostable. Once triggered, the output of the monostable (a positive voltage) replaces the input voltage. The integrator now ramps down until the monostable has timed out. The monstable voltage is replaced with the input signal and the cycle begins again.

In the following diagram, the input voltage is –5 volts. The upper reference level is 0 volts and the lower reference level is set by the monostable on-time. The capacitor voltage is shown as the heavy (red) trace. The monstable output goes from 0 to 5 volts (yellow trace). The comparator output is seem as the light line (green trace) which goes from +15 to –15 and back to +15 volts.

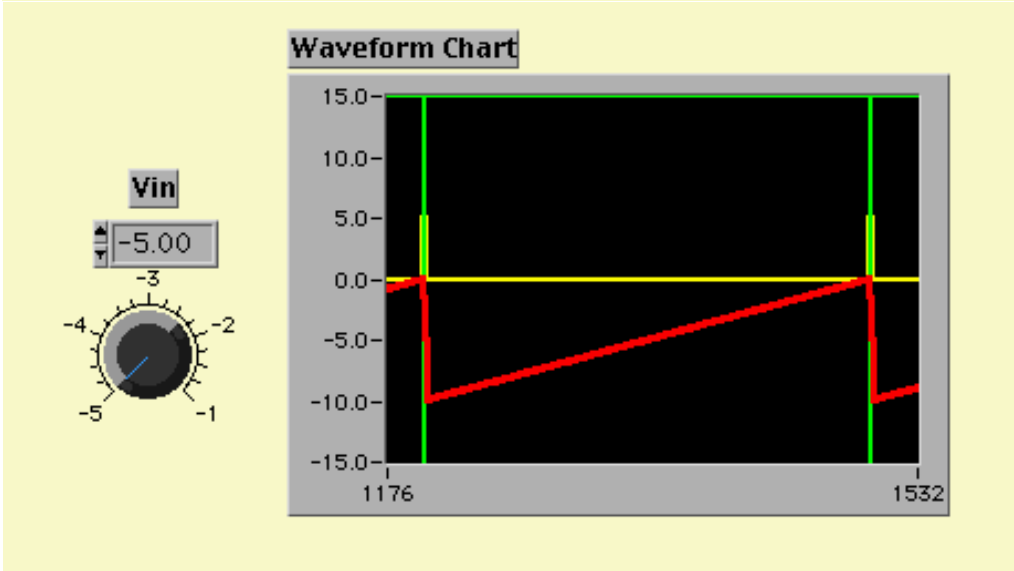


Figure 8-2. LabVIEW Display of the V-F Timing Diagrams

Reducing the input signal lowers the charging rate (slope of the heavy line), increasing the period and decreasing the frequency.

Block 1: The Op-Amp Integrator

When a capacitor is placed in the feedback loop of a conventional inverting op-amp circuit, the result is that the summing current is accumulated on the capacitor. The output voltage thus becomes the sum of all the input charges.

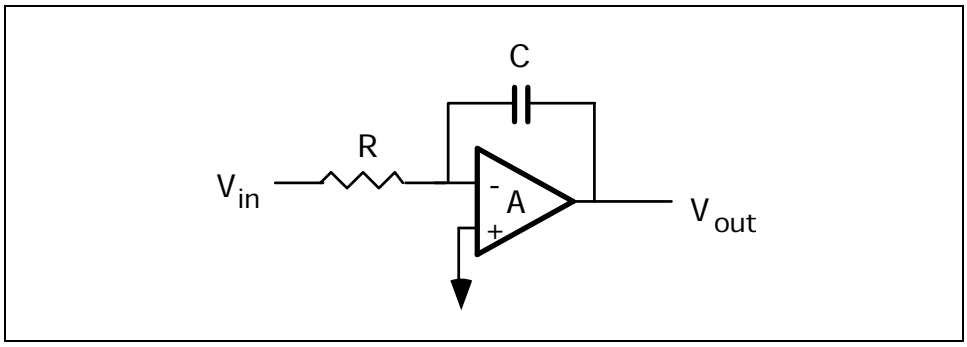


Figure 8-3. Op-Amp Integrator Circuit

From Lab 2, you will recall that the output voltage is the integral of the input voltage scaled by the charging time constant RC.

$$V_{out} = - 1/RC \int V_{in}\, dt \qquad (8\text{-}1)$$

If the input voltage is a constant and negative, the output voltage becomes a ramp increasing linearly until the output reaches the positive rail voltage. If you reverse the input voltage, the op-amp integrates downwards linearly until it reaches the negative rail. The ramp output is just

$$V_{out} = - (V_{in}/RC)\, t \qquad (8\text{-}2)$$

In order to simulate the operation of the V-F circuit, time is divided into time slices and the differential form of the above equation is used to calculate the output voltage V'_{out} at the end of each time slice:

$$V'_{out} = V_{out} - (V_{in}/RC)\, \Delta t \qquad (8\text{-}3)$$

where V_{out} is the voltage at the start of the time slice and Δt is the size of the time slice.

LabVIEW Demo 8.1: Operation of an Op-Amp Integrator

Load the program called Integrator1.vi from the chapter 8 library. This program simulates the dynamic operation of an op-amp using Equation 8-3. Each time you click on **Run**, a new output voltage is calculated and displayed on a chart (capacitor voltage versus time). Initially the output voltage is set to –3 volts. With V_{in} = –5 volts, R = 150 kΩ and C= 0.1µf, the incremental voltage will be 0.333 V for a 1 millisecond time slice. Click the run button a few times so you can see the ramping voltage. At any time, you can change V_{in}, R or C to modify the rate of change (the slope of the ramp). In fact if you change the sign of the input, the output voltage will ramp down.

Try clicking on the run button 10 times. Then change the sign of the input voltage. Again click the run button 10 times. What kind of waveform have you just generated?

LabVIEW Project A Real Op-amp Integrator

Use a real op-amp (741) to build the integrator circuit below.

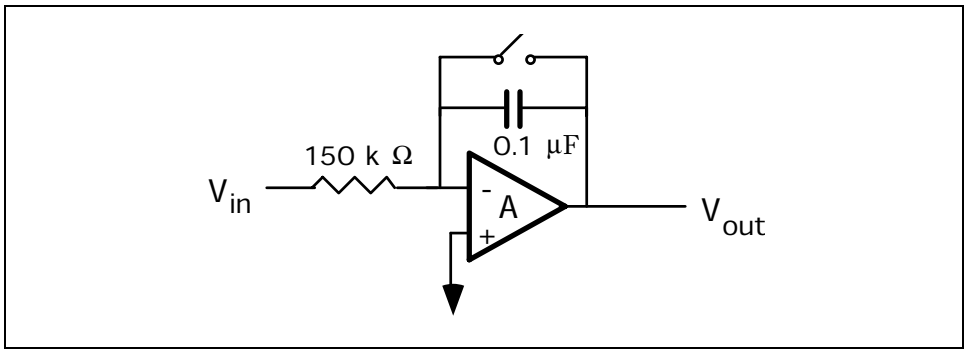


Figure 8-4. Op-Amp Intetgrator Circuit with Manual Reset

Apply a 2 volt P-P, 100 kHz square wave to the input. Observe both the input and output signals on a dual channel oscilloscope or DAQ card.

Observe what happens if the signal amplitude becomes too large.

What happens when the frequency becomes too small at a constant amplitude.

What do you think will happen if the input signal is a triangular waveform? Try it!

Block 2: Comparator

An op-amp with no input resistor and no feedback resistor becomes a comparator. If the signal on the summing input (–) is larger than the non-inverting input (+), then the output swings to the maximum negative voltage. If the signal at the summing input is smaller than the non-inverting input, then the output swings to the maximum positive voltage. The speed of the change from one rail to the other is related to the open loop gain and is called the slew rate. By connecting a reference voltage (V_{ref}) to one of the inputs, a trigger level can be defined at V_{ref} and a negative-going output will signal when the input voltage is larger than the reference voltage.

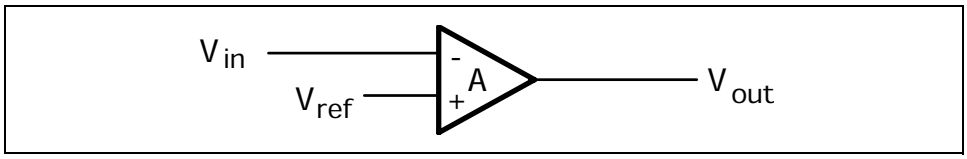


Figure 8-5. Op-Amp Comparator Circuit

LabVIEW Demo 8.2: Op-Amp Comparator in Action

Load the program called Comparator1.vi from the chapter 8 library. Run the program and observe that the comparator output can only be at one or the other rail voltage. In the V-F circuit, the reference voltage will be taken as zero volts. Modify the reference voltage to this value and run the VI again.

LabVIEW Demo 8.3: Op-Amp Integrator and Comparator in Series

Load the program called Integrator2.vi from the chapter 8 library. Each time you click on **Run**, a new output voltage is calculated. Initially the integrator voltage is set to –3 volts. As before with V_{in} = –5 volts, R = 150 kΩ and C= 0.1µf, the incremental voltage for each 1 millisecond time slice will be 0.333 V. Watch the comparator output when the integrator reaches the reference voltage at zero volts. Continue clicking until the integrator output reaches +3 volts. Now reverse the input voltage to +5 volts and integrate down through the reference voltage until the integrator output is –3 volts. Repeat the cycle (V_{in} = –5 v, 10 clicks; V_{in} = +5 v, 10 clicks) once more. The sign of the comparator output signals when the input is positive or negative. Notice that the comparator changes state or is toggled each time the integrator level crosses zero volts. In this mode, the comparator is a zero crossing detector. Notice the waveform produced on the integrator and comparator outputs.

In the V-F circuit, the reference voltage will be set to zero volts (upper limit) and the lower limit (initial voltage) will be set to some negative voltage.

Block 3: The Monostable

Recall from Lab 7 that when a 555 monostable is triggered, the output goes high for a period of time set by an external resistor and capacitor. The on-time is 1.1 RC. Setting R = 36 kΩ and C = 0.1 µf yields an on-time of 3.96 milliseconds. The monostable needs a falling edge to trigger the circuit, followed by a rising edge. This is accomplished in the real V-F circuit using a resistor-diode network consisting of a 1.5 kΩ resistor and two 1N914 diodes used to clamp the trigger voltage within the 555's input range.

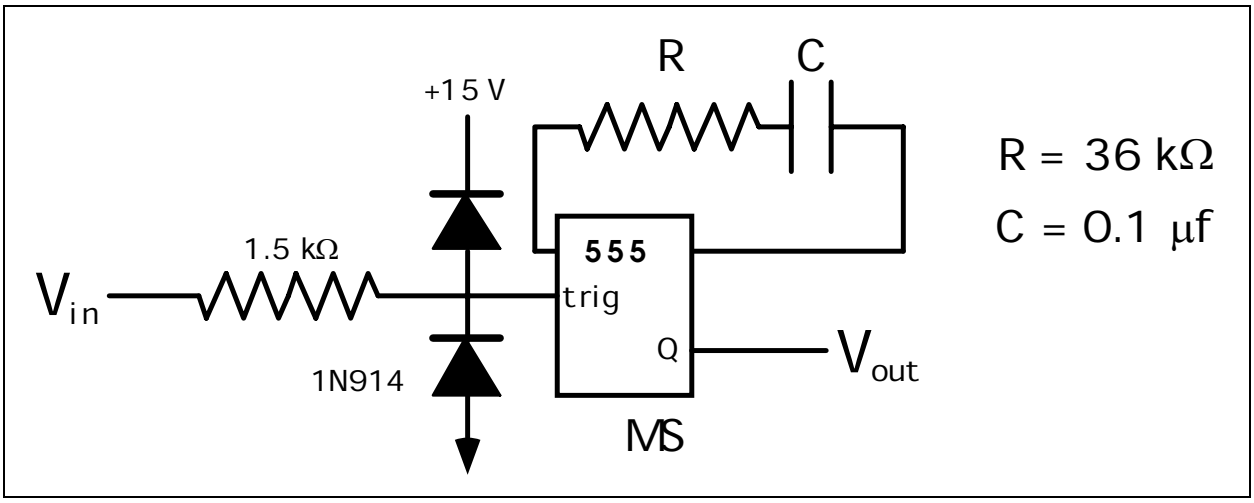


Figure 8-6. 555 Monostable Circuit

The monstable output when high will be close to the positive power supply voltage and will be used to forward bias a third 1N914 silicon diode shown in the next section. A high output allows current to pass through an 0.5 kΩ output resistor to ground. From Ohm's law, this current will be (15.0 V-0.6 V)/0.5 kΩ = 28.8 ma.

LabVIEW Demo 8.4: Monostable Operation

Load the program called Monostable.vi from the chapter 8 program library. Click on the **Run** button to activate the circuit and click on the trigger switch to trigger the monostable.

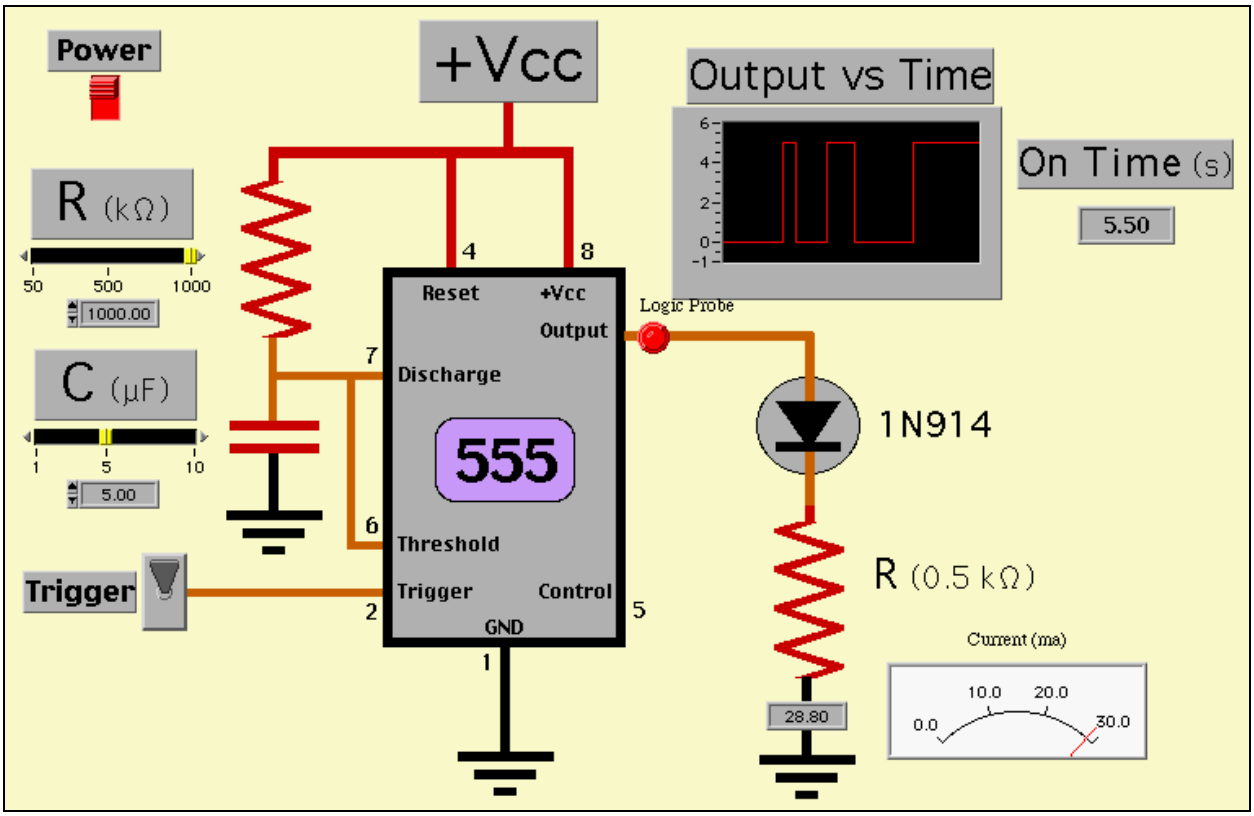


Figure 8-7. LabVIEW Simulation – Monostable Circuit as a Current Driver

Note that current only flows through the resistor when the monostable is triggered. The magnitude of the current can be adjusted with the choice of the resistor.

Question: Suppose the above current was used as the input to the integrator, how long would it take the integrator voltage to reach –10 volts assuming that the output voltage was initially 0 volts? The answer is contained in Equation 8-3.

$$V'_{out} = V_{out} - (i_{in}/C)\,\Delta t \quad \text{or} \quad -10 = 0 - (28.8\text{ ma}/0.1\mu\text{f})\,\Delta t \qquad (8\text{-}4)$$

$$\Delta t = 34.7 \text{ microseconds} \qquad (8\text{-}5)$$

Part 4: A Real V-F Converter

The output of the monostable is used in a real V-F converter circuit to generate a reset current for the integrator. When the diode is forward biased by the monostable, a reset current is applied to the summing point of the integrator. Since the monostable current is much larger than the input current, the summing point becomes the switch and the integrator is ramped down for a time defined by 1.1RC. When the monostable shuts off, the input current dominates and the integrator output ramps up to 0 volts. Here the

monstable resets and the cycle starts again. The period as seen on the monostable output is related to the input voltage level. When the monostable on-time is short, the frequency is directly proportional to the input voltage, a true V-F converter.

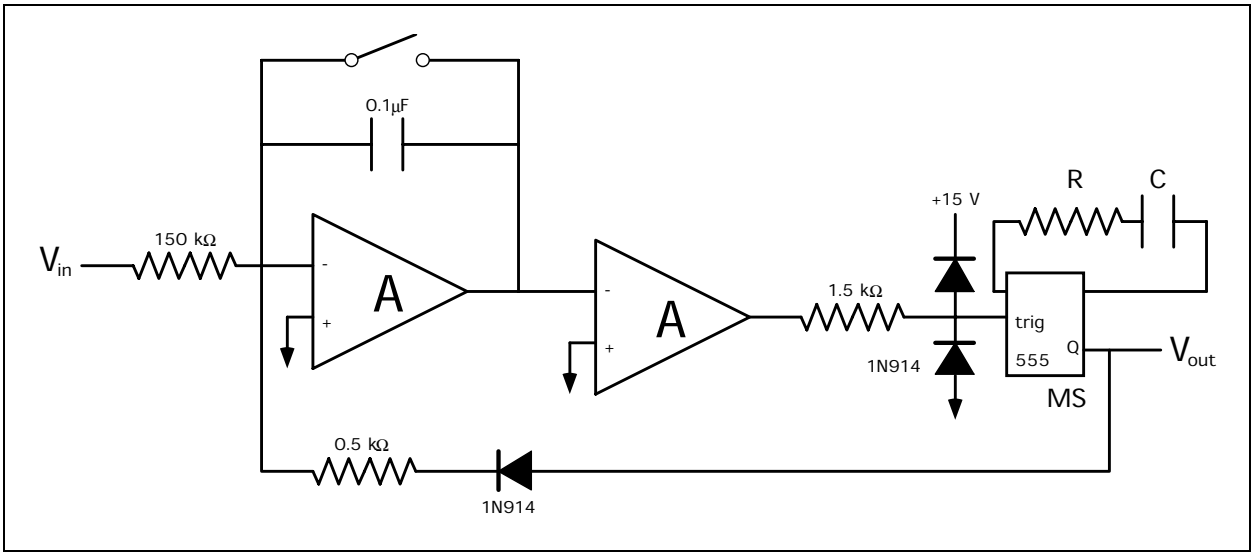


Figure 8-8. 555 Schematic Diagram for a Real V-F Converter Circuit

LabVIEW Demo 5: Operation of the V-F Circuit

Load the program called VF.vi from the chapter 8 program library, VF.llb. Click on the **Run** button and investigate the variation of the period with the input level. When the action is stopped, the magnifying cursor can be used to expand the time to see a close up of the complete timing diagram.

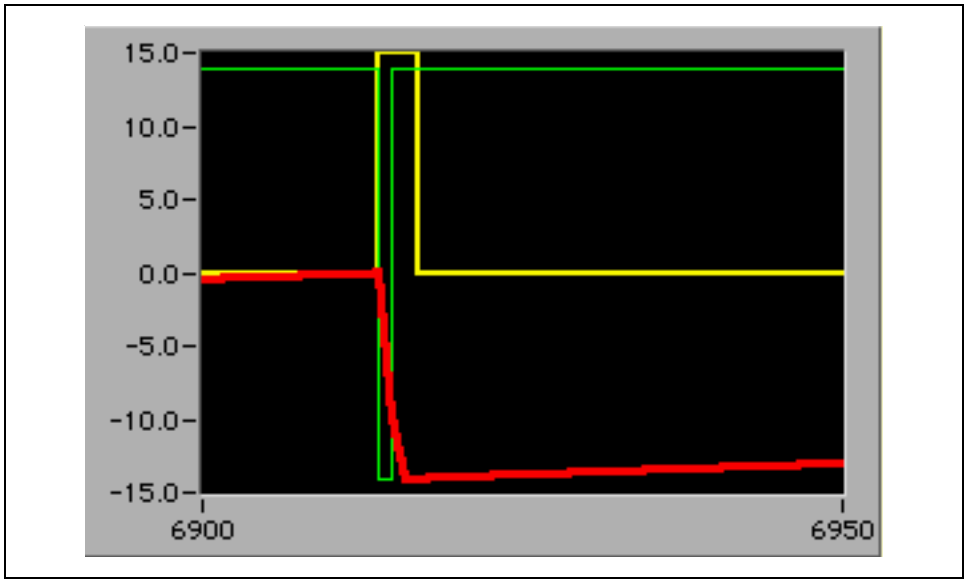


Figure 8-9. V-F Timing Diagrams

When the monostable is off, the input signal (red trace) ramps up until the integrator output reaches the comparator trigger level at zero volts. The comparator (green trace) flips to the opposite rail generating a trigger signal for the monostable which in turn (yellow trace) generates a reset current that is much larger than the input signal and of opposite sign. The integrator ramps down towards a negative voltage. As soon as the integrator voltage reaches zero volts, the comparator flips back to its initial state (+15 V). At the end of the monostable timing period, the reset current is returned to zero and the integrator ramps up again driven by the input signal level.

LabVIEW Exercise

Plot the output frequency versus input voltage.

eLab Project 8

Objective

To study the operation of a Voltage-to-Frequency converter circuit built from basic analog chips, the op-amp and the 555 timer.

Procedure

Build a voltage-to-frequency converter circuit using the schematic diagram of Figure 8-8. It requires four resistors, two capacitors, three silicon diodes, two op-amps and one 555 timer IC. The chip pinouts can be found in Lab 1, Figure 1-5 and Lab 6, Figure 6-1. The op-amps and timer chips are powered from +15 and –15 volt power supplies. The circuit requires that the integrator be in a known state (a negative or zero voltage on the input) for the feedback to work correctly. This is easily set be momentarily shorting the integrator capacitor. After started the circuit will run until the power is removed. The component layout is shown below.

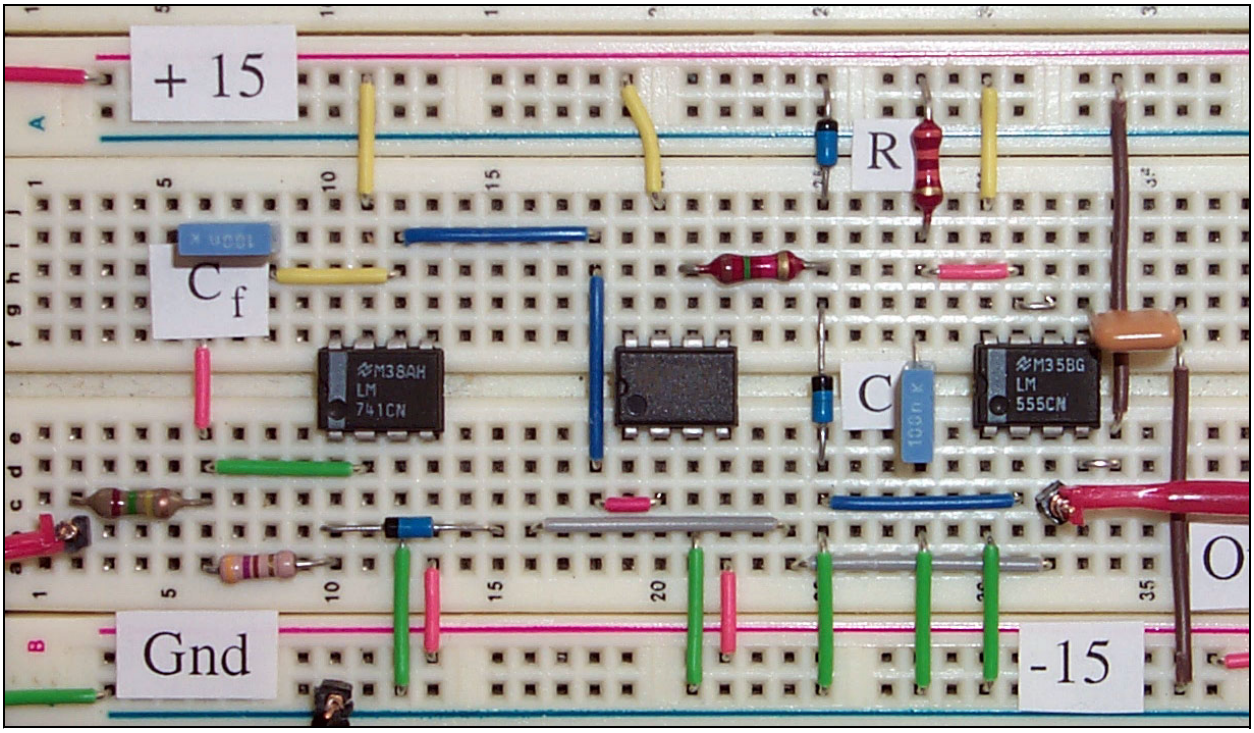


Figure 8-10. V-F Component Layout

Plot the V-F output frequency as the input voltage in varied from –5 to –0.5 volts.

Computer Automation 8: V-F Calibration Curve

Computer Automation implies the repetitive measurement of circuit parameters and the analysis and reporting of that data set. In this lab, a system test for the V-F circuit of eLab 8 is presented. The operator will be able to set a range of test input signals and the number of tests to be run. In operation, a test voltage is output on one pin of the DAQ card which is to be connect to the input of the V-F circuit. The V-F output can be connected to a counter input pin of the DAQ card and using Frequency.vi, introduce in computer automation lab 6, the frequency of the V-F circuit measured. After all data points are collected, the V-F calibration curve is displayed on the front panel.

LabVIEW Design

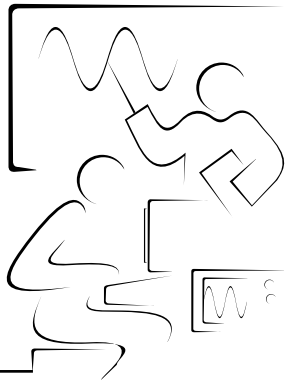
A starting design for a LabVIEW test program, called V-F Scan.vi is found in the program library. Launch this program and open up the diagram window. Notice that two subVIs, Write1pt.vi and Frequency.vi are used. Write1pt.vi is a subVI used in earlier labs to generate a test voltage on the analogout pin [device1/channel0]. Connect this pin to the input lead of the eLab V-F circuit. Frequency.vi is similar to FrequencyLow.vi introduced in

the computer automation lab 6, but modified so that it can be used as a subVI. Connect the output of the eLab V-F circuit to counter2 clk input pin of your DAQ card. Select the start and stop voltage levels and the number of points to acquire for the calibration curve. When all wires are connected and the V-F circuit is operating, click on **Run**. Each input voltage level and measured frequency will be displayed as it is measured. After all n points have been acquired, the calibration curve will appear in the graph display.

LabVIEW Enhancements

Design a LabVIEW program that fits a polynomial curve to your measured calibration curve and displays the polynomial coefficients.

Lab 9
Nonlinear Circuits: Log Amps



In the days before slide rules, calculators and computers, complex mathematical functions such as division, square roots and powers were solved using logarithmic tables. Two of the most common properties of logarithms reduced multiplication and division to addition and subtraction. These are

$$\text{Log}(AB) = \text{Log}(A) + \text{Log}(B)$$

$$\text{Log}(A/B) = \text{Log}(A) - \text{Log}(B).$$

We have already seen in Lab 2 how summing and difference op-amp circuits can add and subtract. Provided a log op-amp circuit exists, the above relationships can be used to build multiply and divide circuits.

The diode introduced in Lab 3 displays a nonlinear response in the current-voltage characteristic curve. When forward biased, the diode current is exponentially related to the voltage across the diode.

$$I_d = i_o \exp(V_d/a) \tag{9-1}$$

where i_o is the reverse bias diode current (a constant) and a= kT/e. Solving for V yields a natural logarithmic relationship between the voltage across the diode and the current passing through the diode.

$$V_d = a \log_e (I_d/I_o) \tag{9-2}$$

Log Op-Amp Circuit

A log op-amp circuit can be build by replacing the feedback resistor of the inverting amplifier with a diode.

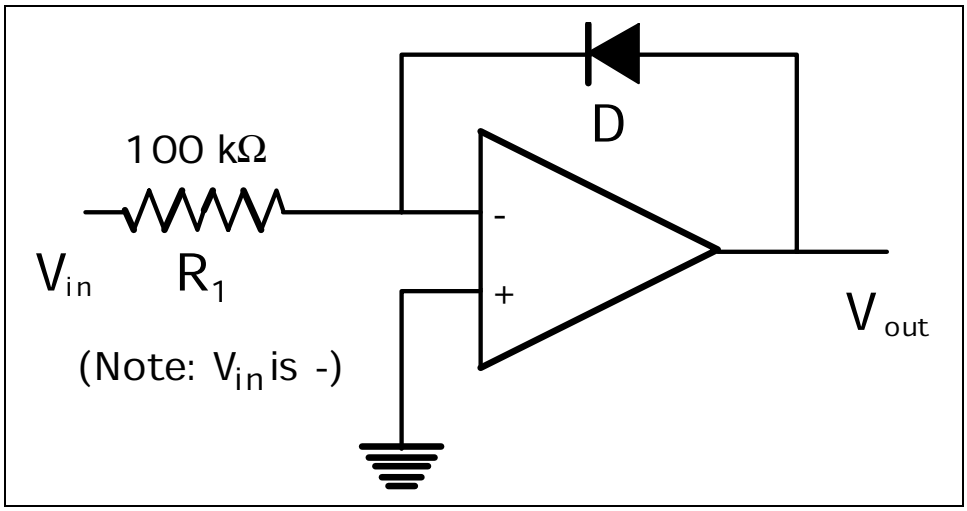


Figure 9-1. Schematic Diagram of an Op-Amp Logarithmic Circuit

For the input loop, the current is $i_1 = V_{in}/ R_1$ (9-3)

For the feedback loop, the current i_f is I_d and V_{out} is V_d (9-4)

At the summing point $i_1 = - I_f$ (9-5)

Together these equation are $i_1 = V_{in}/R_1 = - i_f = -I_d = - i_o \exp(V_{out}/a)$ (9-6)

and solving for V_{out} yields

$$V_{out} = a \log_e (V_{in}/ i_o R_1) \quad (9\text{-}7)$$

With the careful diode selection, this expression is valid over 5 - 6 decades. Diodes such as 1N914 and some common transistors (2N3900A) with the base and collector pins tied together, work well. The constant “a” is about 0.059 volts at room temperature and i_o is typically 10^{-11} amps.

LabVIEW Demo 9.1: Log OpAmp Circuit

Load the program called LogOpAmp.vi from the chapter 9 program library. Click on the **Run** button to activate the circuit. Investigate the output voltage as the input is varied over five decades of voltage levels.

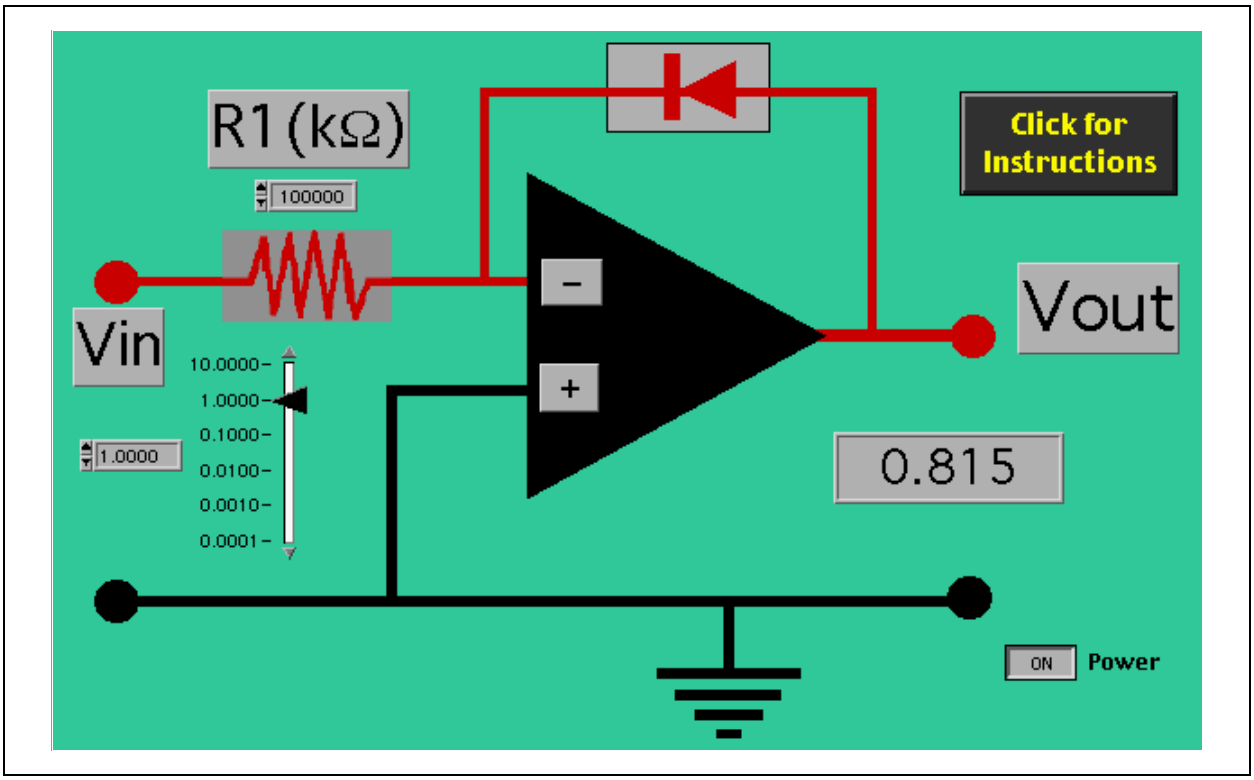


Figure 9-2. LabVIEW Simulation - Op-Amp Logarithmic Circuit

This VI is an exact op-amp simulation of Equation 9-7 using the typical values for "a" and i_o. To convert the input voltage (1.000 volt) into the natural logarithm for 'one', requires that the output be scaled so that $\log_e$ (1 volt) does in fact yield the numeric value ln(1) = 0.000. A second program entitled Ln.vi scales the output of LogOpAmp.vi to generate the correct natural logarithm values. A further scaling is required to convert the natural logarithm base(e) into the normal logarithm base(10). A third program LogN.vi further scales LogOpAmp.vi to demonstrate how a input voltage is converted into a numeric log value. The scaling and conversion factors are found in a subVI called Scaling.vi. It uses the op amp circuits for subtraction and multiplication by a constant found in Lab 2.

An Analog Decibel Calculator

Many analog measurements require that a signal be measured in decibels. Recall from Lab 4, a decibel is defined as $N(dB) = 20 \log(V_{out}/ V_0)$ where V_0 is a reference voltage. One can use two log amps and the difference circuit from Lab 2 to build an analog decibel conversion circuit.

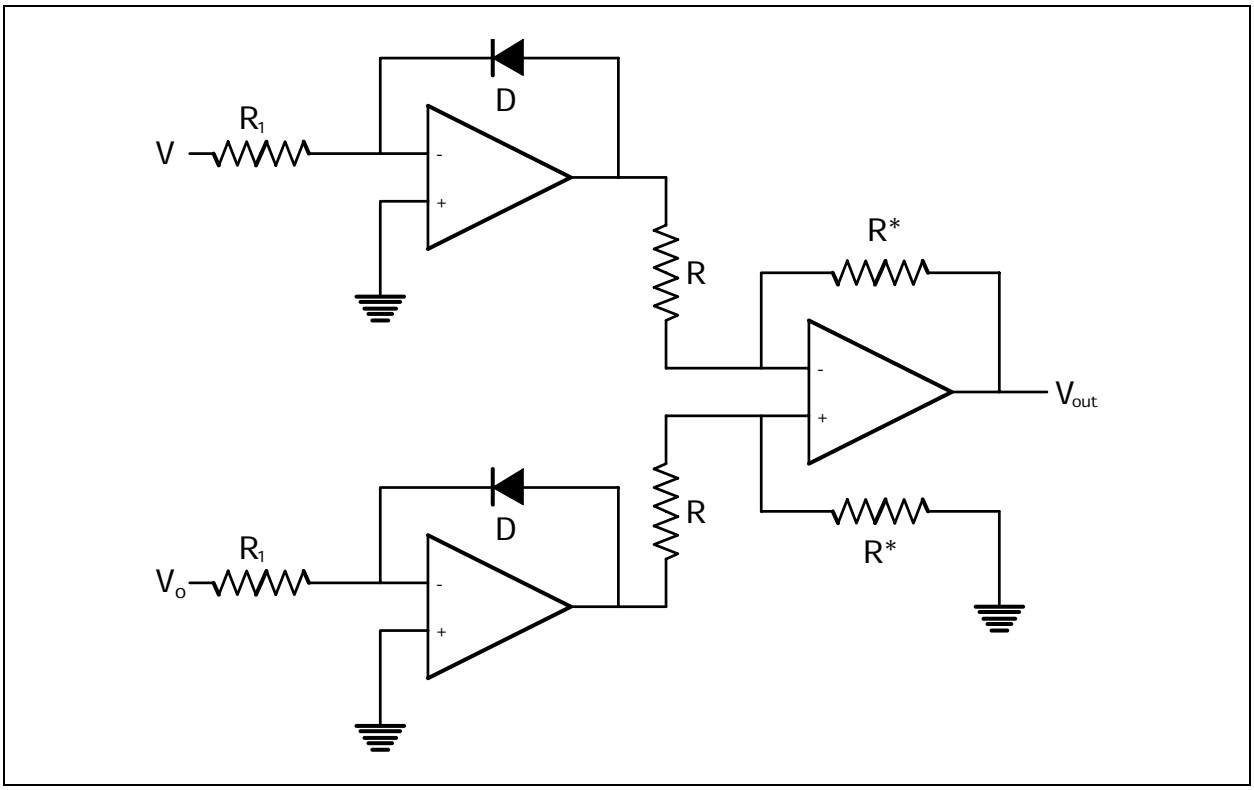


Figure 9-3. Schematic Diagram of an Op-Amp Decibel Analog Calculator

The op-amp circuit shown above calculates the logarithmic ratio log (V/V_o). This is a common calculation used in many applications especially in photometery. By replacing the resistor R* with 20 R, the above circuit calculates decibels. The equivalent LabVIEW simulation for the decibel calculator uses two log amps, a difference function and a multiplication by 20. The following figure shows the strong similarity of the LabVIEW simulation (diagram page) with the schematic diagram (Figure 9-3) for an op-amp decibel calculator.

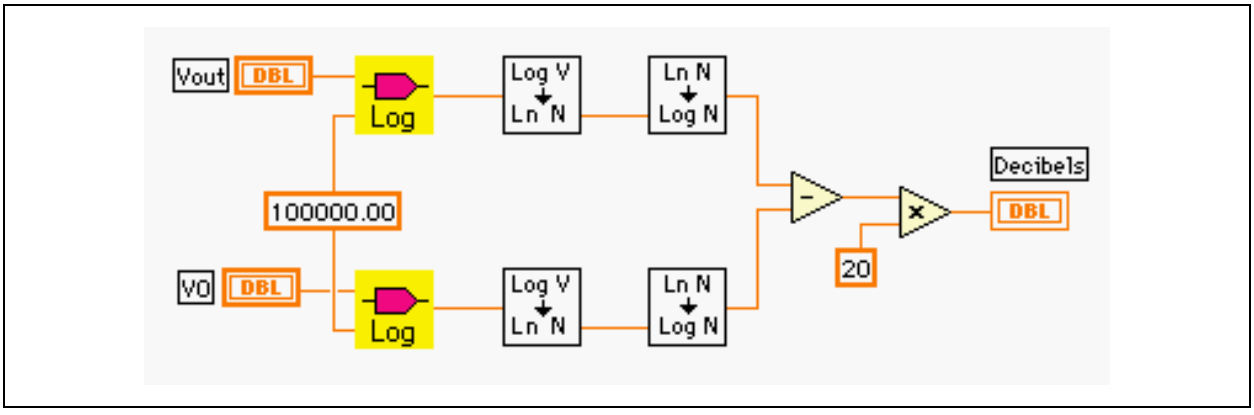


Figure 9-4. LabVIEW Diagram of an Op-Amp Decibel Calculator

LabVIEW Demo 9.2: Decibel Calculator

Load the program called Decibel.vi from the chapter 9 program library. Enter the output voltage and the reference voltage. Click on the **Run** button to execute a calculation. If V is the output voltage and V_0 is the reference voltage for this op-amp circuit, then the calculation gives the gain in decibels.

Exponential Op-Amp Circuit

You may have noticed the symmetry of interchanging a special component between the feedback and input loop. For example, a capacitor in the feedback loop yields an integrator while a capacitor in the input loop yields a differentiator. Diodes also have this symmetry property. A diode in the feedback loop yields a log amp circuit while a diode in the input loop yields an exponential circuit.

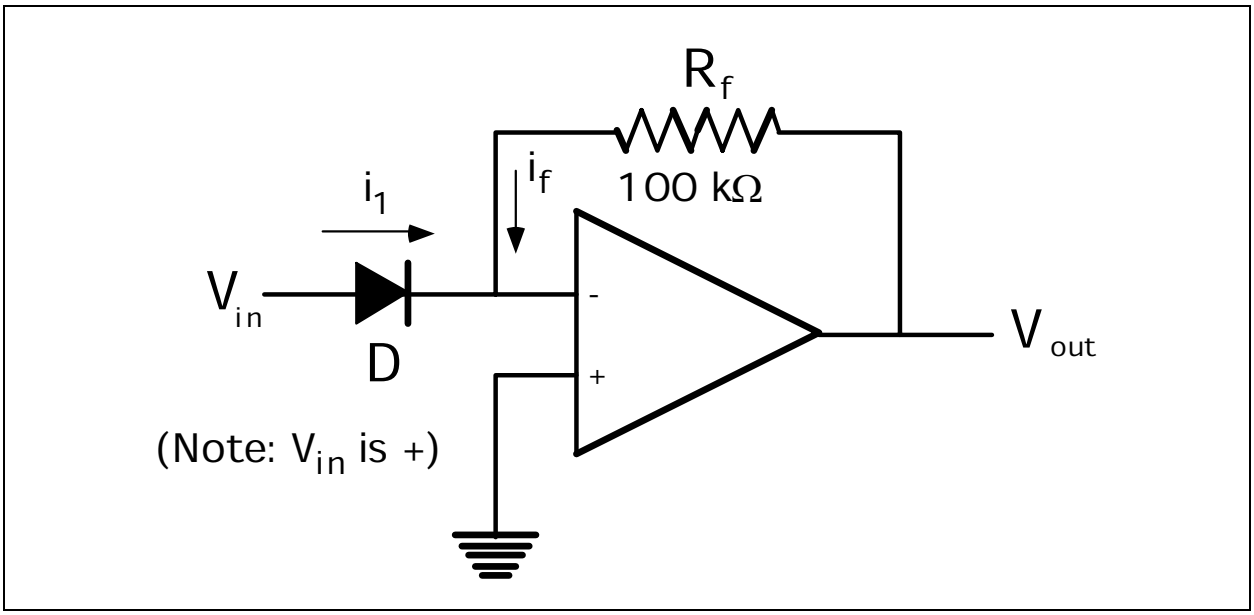


Figure 9-5. Schematic Diagram of an Op-Amp Exponential Circuit

For the input loop, the current is $i_1 = i_o \exp(V_{in}/a)$ (9-8)

For the feedback loop, the current is $V_{out} = - i_f R_f$ (9-9)

At the summing point $i_1 = - i_f$ (9-10)

Together these equations are $i_o \exp(V_{in}/a) = i_1 = -i_f = V_{out}/R_f$ (9-11)

$$V_{out} = i_o R_f \exp(V_{in}/a) \quad (9\text{-}12)$$

This circuit provides the antilog or exponential function which can be used to convert a log sum or difference back into simple numbers.

$$\text{AntiLog } \{\text{Log}(AB)\} = AB \tag{9-13}$$

$$\text{AntiLog } \{\text{Log}(A/B)\} = A/B \tag{9-14}$$

Notice that the previous op-amp circuits when followed by the antilog circuit provides the function multiply or divide.

Analog Multiplication of Two Variables

Multiplication of two variable signals X and Y can be accomplished with the help of Equation 9-13. It is expanded to read

$$\text{AntiLog } \{\text{Log}(X)+\text{Log}Y)\} = \text{AntiLog } \{\text{Log}(XY)\} = XY \tag{9-15}$$

First one calculates Log(X) and Log(Y) using the Log op-amp circuit. Then these are added together with the summing circuit from Lab 2. Finally the exponential of the resultant voltage is computed using an anti-log op-amp circuit. For noise reduction, the output is often reduced by a factor of ten. The schematic diagram for the circuit op-amp circuit follows

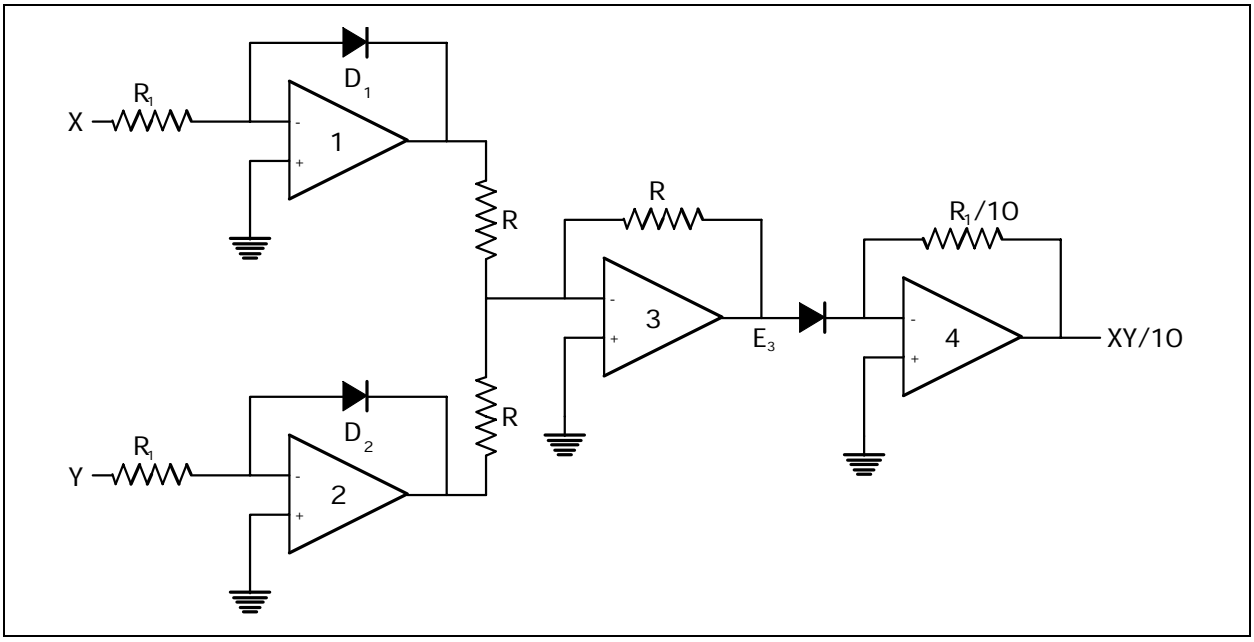


Figure 9-6. Op-Amp Circuit for the Multiplication of Two Variables

LabVIEW Challenge

Design a LabVIEW program that simulates the analog multiplication circuit shown above. Make good use of the Lab 9 program library and sub-VIs to produce a compact program.

Raising and Input Signal to a Power

Many physical laws follow a simple power law and a circuit that can raise an input signal to a specific power (possibily a fraction) is of great use. The op-amp circuit makes use of the logarithm property to raise an alog input X to a constant power y.

$$\text{AntiLog } \{y \text{ Log}(X)\} = \text{AntiLog } \{\text{Log}(X^y)\} = X^y \qquad (9\text{-}16)$$

First one calculates Log(X) using the log op-amp circuit. Its output is multiplied by the constant (y) using the inverting op-amp circuit from Lab 2. Finally the exponential of this voltage is computed using an anti-log op-amp circuit to give the final result X_y. The electronic schematic circuit for the power law follows.

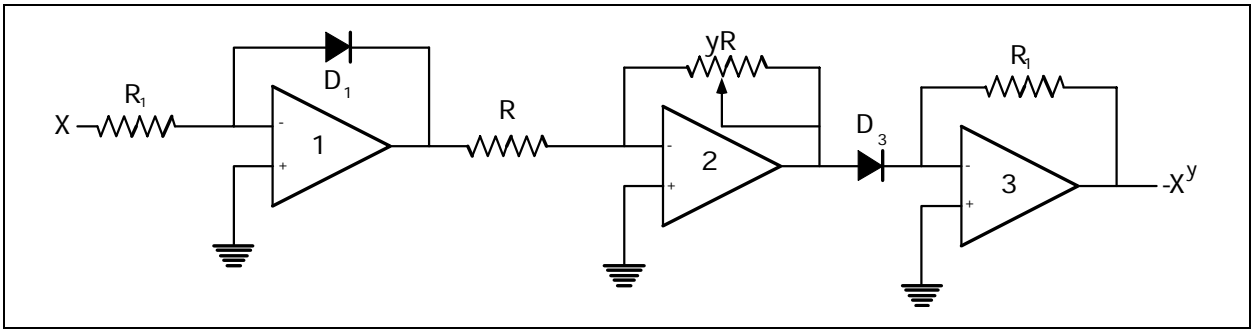


Figure 9-7. Schematic Diagram of an Op-Amp Power Law Circuit

Note that a resistance potentiometer with one lead shorted to the wiper lead is used to set the Gain of the second op-amp (G = yR/R) to y. The fraction y can be an integer number, half integer or any other fraction.

LabVIEW Challenge

Design a LabVIEW program that simulates the raise to a power op-amp circuit shown above. Make good use of the Lab 9 program library and sub-VIs to produce a compact program.

eLab Project 9

Objective

To study the operation of an op-amp logarithmic circuit.

Procedure

Build a log amplifier using the schematic diagram of Figure 9-1. It requires a resistor, a silicon diode, a 741 op-amp. If a silicon signal diode is not available, a transistor such as a 2N3900A with the base and emitter lead tied together works well. A small 0.001 µF capacitor is placed across the diode

to suppress noise. The op-amp is powered from a +15 and a –15 volt power supply. The component layout is shown below.

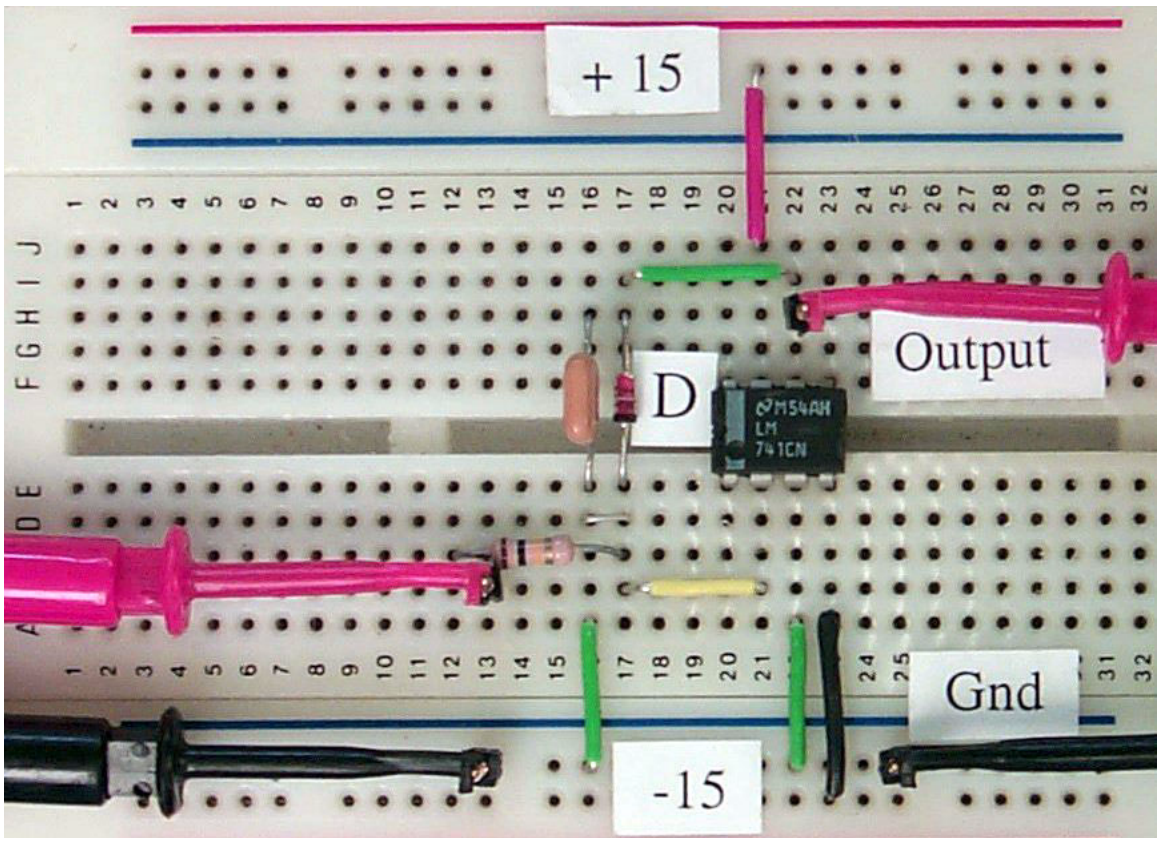


Figure 9-8. Component Layout for Op-Amp Log Amplifier

Investigate the operation of the logarithmic op-amp circuit by applying a variable amplitude DC voltage to the input pin 2. Plot the output voltage as a function of the input voltage on both a linear and logarithmic plot. How well does the circuit work for AC input signal levels?

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